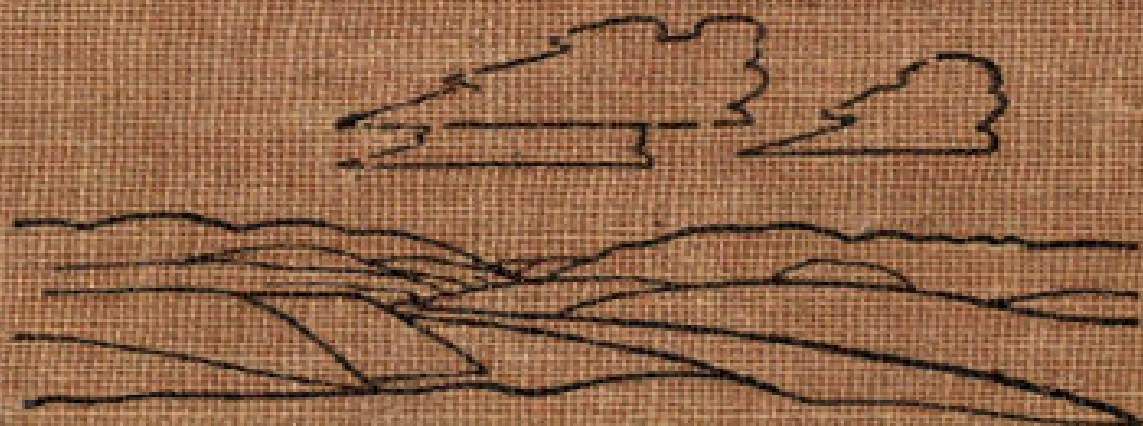




THE FLOW OF TIME
IN THE
CONNECTICUT VALLEY



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geological imprints

Author: George W. Bain
Howard A. Meyerhoff

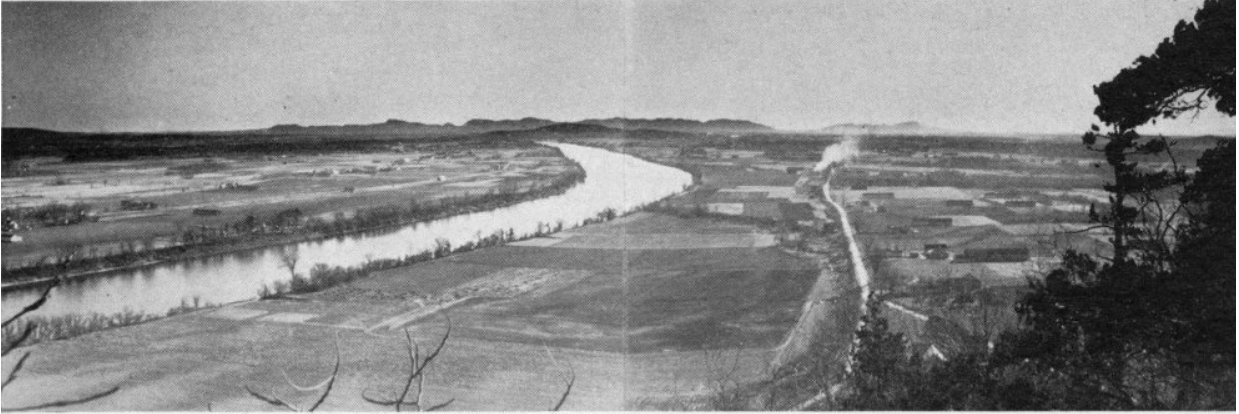
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PL. 1. *The Connecticut Valley as it is seen from Mount Sugarloaf.*

The western highland shows through the pine boughs at the extreme right. The eastern highland balances it on the far left. The Holyoke Range hems the basin on the south except at the gap where the river escapes to the Springfield area.

The Flow of Time IN THE CONNECTICUT VALLEY

Geological Imprints

by
GEORGE W. BAIN
and
HOWARD A. MEYERHOFF

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Introduction

In every region there is an evening drive which lures the city dweller from the cramped vistas of the office, the home, and the dingy streets to the limitless expanse of hills and valleys, where mental tension relaxes and vision broadens as the physical horizon expands and acquires depth. In less favored localities, the drive may be long and the relaxation short, but not so in the Connecticut Valley. Half an hour of travel, either to the east or to the west from any large community, provides an escape to the hills, where people, cars, houses, and all the minutiae of urban civilization are blurred on the canvas of upland and lowland.

Local pride and personal prejudice may proclaim one view superior to another; but the praise so liberally bestowed upon the heights beyond Westfield, the Mount Tom Reservation, the land called Goshen, Shelburne Summit, and many another site, merely bespeaks the rivalry of equally favored vantage points. Perhaps the trail to Pelham would not be singled out for special mention by the indiscriminating enthusiast, but the connoisseur of New England's scenic beauty returns and follows it again and again. A good road may take some credit for its popularity, but there is a deeper cause than this which brings him back; for, if there is drama in scenery, he finds it here. The road leads out of Northampton, and from the graceful arch of the Coolidge Memorial Bridge he views the flood-scarred lowlands that border the river, and across the flat plain into Hadley he sees visible reminders that river and farmer periodically struggle over ownership of the land. Then a rise in the road constricts the view but offers a promise of something different. Ahead, rolling fields stretch to the beckoning hills beyond Amherst, but the hills appear and disappear in tantalizing cadence as the car tops each rise and drops into the ensuing hollow. Soon West Pelham comes into view, and the rise to the highland begins. Beside the road a brook tumbles into the valley; and as the car climbs the heights to Pelham, and miles of wooded land are suddenly spread before the eye, the wayfarer realizes that here is the dramatic climax to his trip and to the murmured story of the brook. But the long ridges reaching out to the north and to the south,

the deep valleys between them, and the sky which meets the farthest ridge do not enclose the panorama. It has a fourth dimension—time—a dimension as limitless as the horizon.

With just a dash of imagination, the wayfarer may journey backward through time; through scenes of infinite variety; through countless years of unceasing change; through situations so different that he would scarcely have recognized his New England. The scarred plain of the river, the brook, the soil, the rocks, the upland and the valley,—all tell a fascinating and a logical, if surprising, geological tale. A detour down this fourth dimension promises as much interest as a journey through the other three.

Today and Yesterday

From the Coolidge Memorial Bridge the broad lowland seems to reach out in all directions towards the encircling hills. Far down the river, the distant bank rises a sheer thirty feet from the water and is high enough to surmount even the worst of floods. Yet each year this bank recedes as the unconsolidated sediment at its base is sapped by the stream and is carried away. Three times the river road has been moved back from the insatiable Connecticut, and today the main Hockanum highway takes the long route far from the water's edge.

The River Works

Nearer the bridge the land is lower, and it shows the effects of frequent inundation, but not of scour. A great sand bar lies in the curve of the stream, and the low parallel ridges suggest that they, too, were awash in the Connecticut before its eastern bank encroached so far upon the town of Hadley. The tongue of land which serves as Northampton's airport is a succession of bars and abandoned channels which record the migration of the river away from its old bank along Bridge Street. The Connecticut is robbing Hadley to pay Northampton, but there was a time when Northampton was pilfered, too.

Swales line the landscape as far as Hadley; and each year, at the time of high water, they must now be content with the meager overflow, where once they sped the entire stream upon its southward course. But even now, in flood, their original function may be restored. For the swale just west of Hadley was a roaring torrent in 1938, 1936, and 1896. Indeed, it threatened to appropriate the entire stream, and each of the great curving hollows that furrow the lowland are scour-channels which were made at other times.

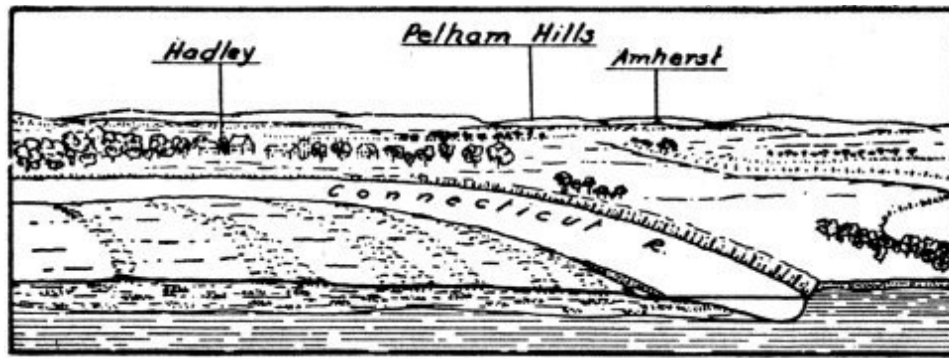


FIG. 1. *The Connecticut River undercuts the Hadley bank at Hockanum.*

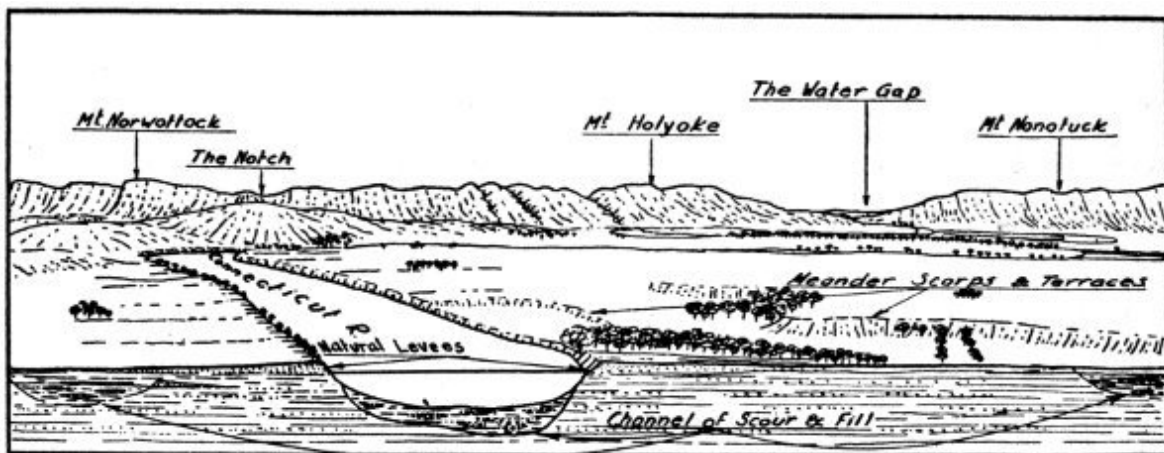


FIG. 2. *Natural levees border the Connecticut River south of the Sunderland Bridge.*

The river has moved at will from one side of its alluvial plain to the other, and its threats to change its course are not to be taken lightly. 3
 Until 1830 it flowed past Northampton, around the great ox-bow to Easthampton and then back to the watergap between Mount Tom and Mount Holyoke. It served as the main line of communication to the Atlantic seaboard and was a much travelled route. In the spring of that year high water breached the narrow neck of land between the two ends of the meander loop, and practically overnight the route to New London was shortened by three miles. Although the event was not a source of rejoicing to the landowners, Northampton declared a day of thanksgiving because they were now, thanks be to Providence, three miles nearer the sea. How often the river has changed its course may never be determined, but the floodplain is grooved with swampy or silt-filled ox-bow lakes, not only near

Northampton, but all the way from Brattleboro, Vermont, to Middletown, Connecticut. They tell of older shifts in the course of a river which still displays its brute power within the limits of its alluvial plain.

The inundation of 1936 did more than scour the river's floodplain; it left thick deposits of sand and silt upon many of the fields. Each preceding flood has done the same sort of thing, dropping coarse sand in greatest abundance on the banks where the river flowed straightest. Flood by flood, the deposit has risen higher on these favored sites, where the swift main current slackens as it spreads over the broad, flat plain. Today the banks form natural levees sloping away from the river at many points southward from the Sunderland Bridge.

Just when the river started to shift back and forth across its alluvial plain is not revealed, but it was long before the white man penetrated the country. Indian graves and campsites have been laid bare as the high water of each new flood has removed the silt left during earlier inundations. The sites rarely yield any implement brought by the Europeans; they record long years of Indian occupation in the land called Norwottock, a land in which the 4 red man found a river which temperamentally shifted its course in response to periodic floods.

The floodplain ends at a rise in the road not far east of Hadley. The rise is a scalloped embankment, reminiscent of the high bank on the river bend downstream from Hadley; even the long narrow swamp at the base looks like a filled ox-bow, and the scallops look like bites which the hungry river took from its banks. This embankment continues northward past Mount Warner, following the present channel closely through North Hadley, and it passes just east of Sunderland village. Corresponding banks are present on the west side of the stream in South Deerfield and Hatfield. Within the confines of those terraces the Connecticut has had free play, but its course has never strayed east or west of these well defined boundaries.

Wave-like hills of sand cap the embankments in several localities north of Hatfield and North Hadley. Some, perched on the terrace edge, were partly cut away when the river was establishing the limits of its floodplain. Wherever the pine trees are cut down, or the grass plowed under, the sand within these hills begins to drift. They look and act like those hills of the

desert, the sand dunes, and they record the drift of wind-whipped sand across a naked land, before the river had established a floodplain within its present confines.

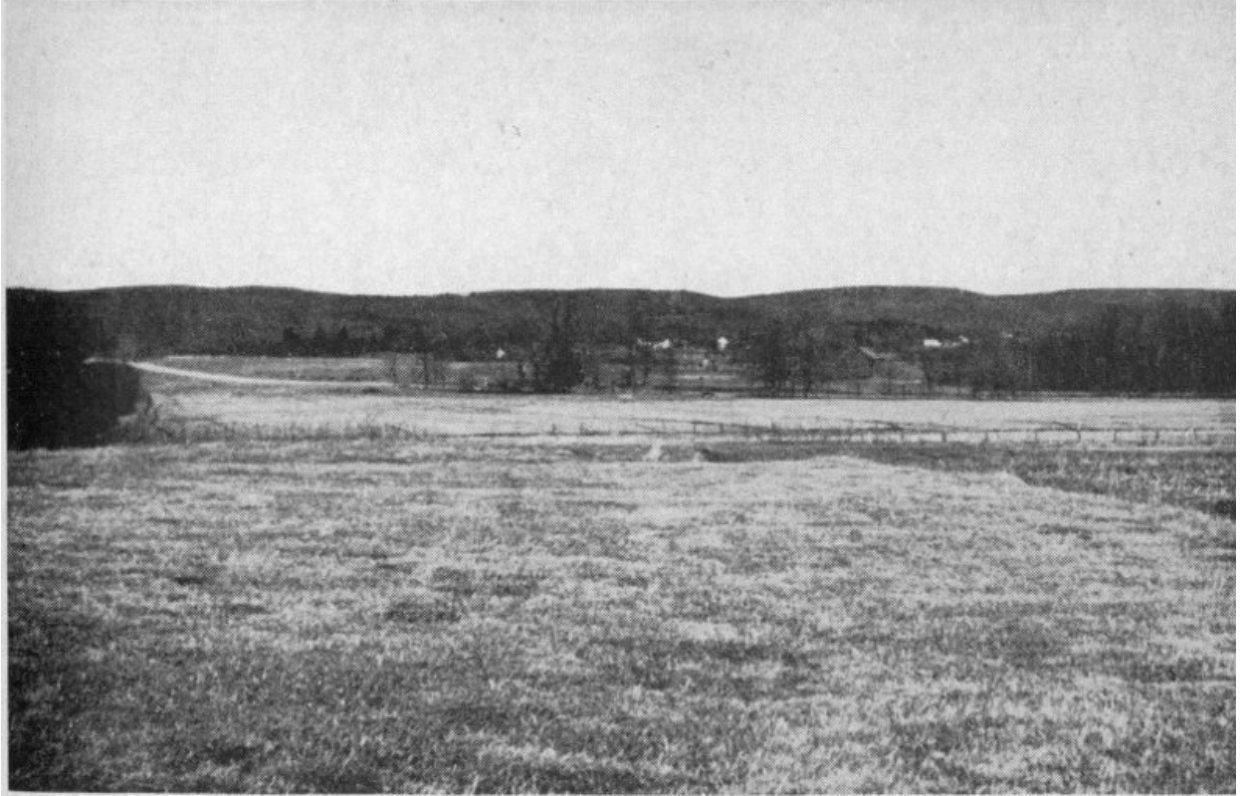
The Landscape Changes

Fine sand, silt, or clay is found beneath the windblown sand wherever the river banks undercut the dunes. The clays are especially widespread, for each of the numerous local brickyards has its clay pit, and there are many more clay banks which have no brickyard. The clays are rhythmically banded. One band, composed of very fine material which settles from suspension only after weeks of absolute quiet, retains moisture tenaciously; adjacent bands dry more rapidly, are somewhat sandy, and settle from suspension in less than a week. A large body of quiet water in which so much fine clay could settle must have occupied the valley before the river was there, and the only type of water body which could have provided the proper environment is a fresh-water lake, free from agitation during the long winter months when its surface was frozen over. These thin clay bands are deposits of a winter season, when streams are low and their load light. Then, even the finest particles can settle, during the many weeks of quiet water, as a paper-thin layer upon the lake bottom. The coarser sandy layer just above the finest clay records the spring break-up, the melting of the ice, and resuscitated streams flowing from the hills with a vigor that can be acquired only when the melt-water from the winter snow combines with the normal run-off. The sand which these freshets bring to the lake diminishes as the spring floods subside, and the sediment becomes progressively finer until next spring comes around.

PL. 2. Features of the landscape which originated during comparatively recent time.



a. Air view of the ox-bow lake between Northampton and Mt. Tom.



b. Roches moutonnées of the Pelham Hills seen from Hadley.

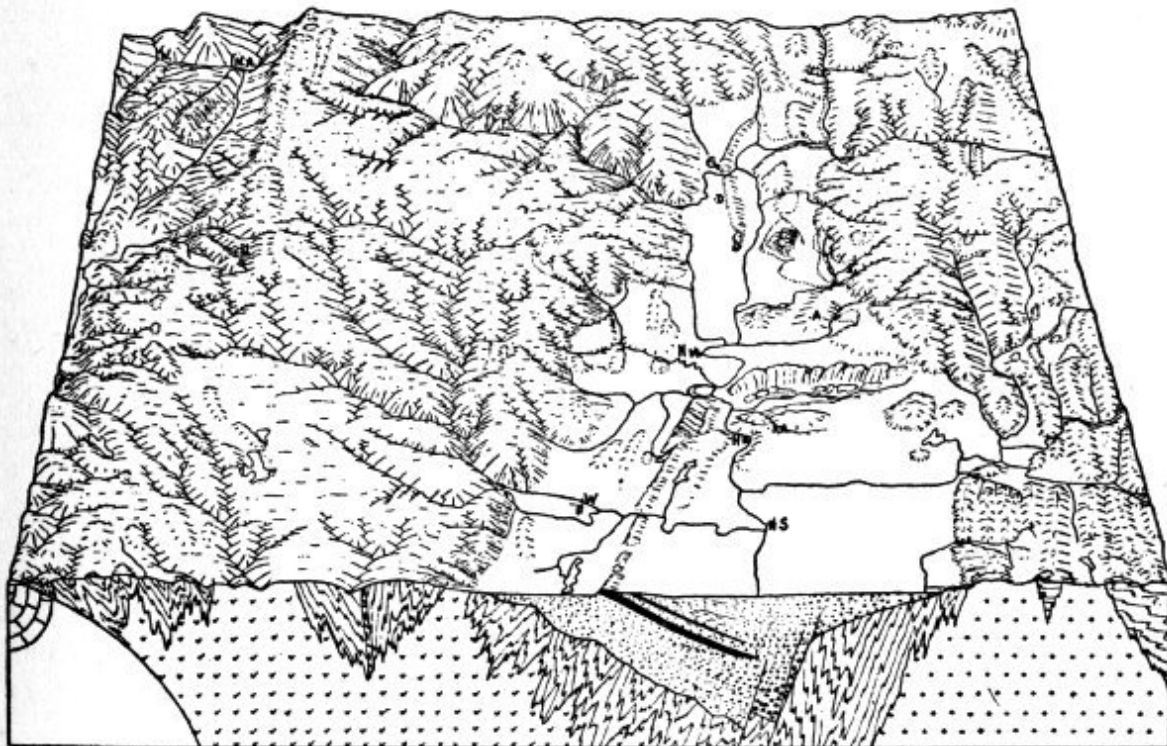


FIG. 3. *Block diagram showing the main features of central Massachusetts at the present time.*

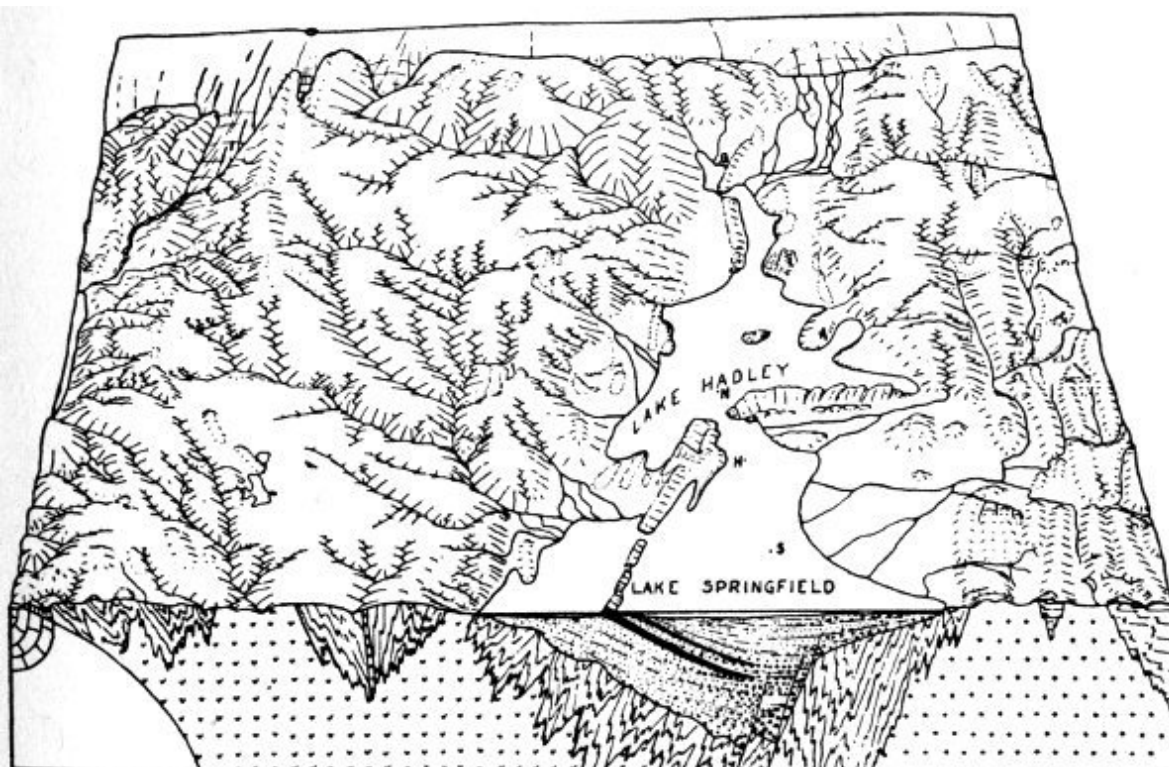


FIG. 4. Block diagram showing the main features of central Massachusetts during the recession of the Ice Sheet.

Each sandy layer is a spring; each clay band, a winter; and the two 6 together mark the passage of a year. High spring floods are rarely local; floods on the Connecticut are usually matched by floods in the Merrimack watershed to the east and along Housatonic to the west. Floods of the past were much the same, and many of them can be identified readily in the banded clays of the Connecticut Valley. Each one can be traced in contemporaneous deposits which were formed in other parts of the lowland and in neighboring lake basins.

Some of the winter bands, together with the layers below them, are torn and folded, and the tops of the folds have been sheared off. Covering them invariably is the sand layer of the spring break-up. Plain from these features is a winter episode of freezing to the lake bottom, and of ice contorting the clays as it expanded and contracted in response to fluctuations in the surface temperatures. The normal cyclic repetition of sand and clay was resumed when these particularly hard winters came to an end.

At South Hadley Falls the lake clays rest upon a gravel bed, and the bottom layer records the lake's first year of life in that locality. The overlying 7 bands provide the evidence of a characteristic climatic sequence which can also be recognized in the clays at Chicopee and at other points still farther south in the lowland. But at Chicopee there are many layers which are older than the bottom layer at South Hadley Falls; and at Springfield many layers appear that are older than the basal band at Chicopee. From the sediment deposited in its waters, the story of the lake is not difficult to decipher. It existed at Springfield years before it appeared at South Hadley Falls; in fact, it flooded the meadows near Middletown, Connecticut, for nearly 6,000 years before its waters existed near Northampton.

These beds of clay hold the moisture close to the surface throughout the lowland, making it available to the fields of vegetables and tobacco. Towards the valley margins these crops disappear because the fine sediments end against the rocky shores of the adjacent hills which pass into and beneath sloping terraces of sand and gravel. In the numerous terraces which fringe

the hills, the horizontal beds of gravel lie above lakeward-dipping beds of coarse sand; they underlie broad flats furrowed by channel-like depressions which radiate from the valleys at the apex of each flat. On these terraces one can easily picture sand-laden waters coursing through the channels and building deltas outward into the lake.

Deltas were built wherever streams from the highlands entered the valley, and they mark the ancient level of the lake. Strangely, their elevation drops from 315 feet at Montague to 300 feet at Amherst, and is only 268 feet at South Hadley. The changing elevation shows either that the lake surface sloped southward—and indeed this would be unique—or that the shoreline was raised in the north and that the lake drained southward. The latter surmise is plainly the more plausible.

Most deltas on the east side of the valley are pitted by numerous conical depressions. In a depression on a delta plain near Montague, an excavation, made to obtain road fill, disclosed a mass of disordered gravel which must originally have been deposited in the horizontal top-set beds of the delta, but which now lies in the bottom of a depression mingled with the fine sand of the underlying fore-set beds. The top-set beds seem to have been supported for some time and then collapsed as if the underpinnings were removed. The crudely circular or elliptical outlines of the depressions suggest that stray icebergs drifted upon the delta slopes, where they were anchored or buried by the sandy outwash. The buried ice-cakes survived until the lake was drained, and the baselevel of the streams was lowered, for the depressions have no outwash within them. They collapsed soon after the lake vanished, because water soaking through the delta sands melted the ice, much as it thaws the ground for dredging in the Yukon. Even today this gravelly ground, particularly the beach of the ancient lake, is well drained, and it forms the best land for the apple orchards of the valley.

Glaciers Came

The delta deposits and the clays form a thin veneer over a bouldery soil that comes to light along the delta-top margins and in gulches cut down through the gravel and sand. Some of the boulders are huge, attaining diameters of twenty feet; and all are strangers to their present resting places. Some are set

upon a bare rock floor, scratched as though by sandpaper, and they teeter to the weight of a child; most are embedded in soil. These “erratics” seem to have been left like unwanted objects, picked up and carried for a time, and then dropped when the bearer wearied of their weight. The scratches on the rock floor are parallel grooves, all of which trend southward. They are unmistakable tracks left by glaciers, and the boulders are like the stones perched on glacial ice for a ride to the terminal moraine.

The land above the old lake shore is bare scratched rock or rocky soil called boulder till. Every hill farm has been cleared of more stones than trees, and it is only with the vogue of the rock garden that these erratics have found any merit in man’s estimation. It has been said with a considerable element 9 of truth that the lake margin can be identified by the stone fences heaped up by exasperated farmers at the line where the water once lapped the slopes of the glaciated hills. Striations and erratics decorate the tops of Mount Tom and Mount Holyoke, and those who visit Mount Monadnock or Mount Washington will find they must inscribe their initials over the signature of the great ice sheet.

The stranger rocks or erratics, stranded promiscuously over the countryside, can be traced to hills farther north. Clearly the ice sheet was moving southward, picking up debris and abrading the countryside like a great sanding machine. Northern slopes were worn to long gentle inclines and the southern slopes kept their original forms or were steepened as the ice plucked fractured blocks from their moorings. One imaginative writer likened the glaciated rock hills to the wigs of sheep’s wool worn by the jurists of his day; the name stuck, and they are still known as *roches moutonnées*. Look at the Pelham Hills from the Coolidge Memorial Bridge and you will see the top of Jeffrey Lord Amherst’s wig facing towards Canada.

Within the Connecticut Lowland the moving ice often picked up a load of debris more cumbersome than it could drag along. It handled the situation most satisfactorily by dropping the load and streamlining it, and these piles of glacial debris with blunt north slopes and gentle southerly sides are drumlins. When next you pass the apple orchards of South Amherst, recall that the smooth elliptical hill east of the road to South Hadley is a drumlin, a relic of an overloaded glacier.

Just Before the Ice Age

The glacier advanced as far as Long Island and Martha's Vineyard, and the lakes of the Connecticut Valley formed along the ice margin and spread northward as the ice front receded. The distinct layers, or varves, of clay mark off 25,000 years since the recession began, but for a million years before its final retreat, the ice covered all New England intermittently. 10

This length of time transcends human comprehension unless one considers years in terms of what has been done. A million years is not too long for a sand-laden ice sheet, moving only a few feet each year, to grind tens of feet of solid rock off the north sides of the "everlasting" hills. To those who study the earth, "Before the Ice Age" has about the same significance as "Before the Hurricane" has to the average citizen of New England. It is in such terms that geologic time must be considered.

The ice sheet simply modified the pre-glacial topography; it changed symmetrical hills to asymmetric *roches moutonnées* and left boulder till spread over much of the bedrock floor. The greatest changes were effected in the White Mountains, where the steep-walled river valleys were changed to troughs with a U cross-section, as in the scenic notches; or with steep headwalls like that in Tuckerman Ravine, a typical alpine cirque. Within the lowlands boulder till was left as a blanket, concealing the irregularities which were made in the rock floor at an earlier geologic date. These irregularities may pass unnoticed unless some construction project happens to reveal them. Bedrock is rarely over seventy feet down at any point in the lowland, but work at the Sunderland Bridge and the Coolidge Memorial Bridge encountered masses of glacial debris in a deep fluvial channel more than three hundred feet below the river surface and at least two hundred feet below the present level of the sea. This deep trough is not over one hundred yards wide, and if it were fully exposed to view, it would look like a miniature Saguenay gorge. Similar trenches in every part of eastern North America, from Hudson Bay to Cape Hatteras, show that the land once stood higher than it does now, and that the main rivers flowed in deep, narrow canyons, although the upland surface between the rivers had its present characteristics. Thus, in Pliocene time, while primitive members of the human race were entering old England, New England rose high above sea level, and its lowlands were trenched by quickened streams.

The narrow gorges are an eloquent, if mute, record of rivers suddenly rejuvenated, their current accelerated and the exuberant waters cutting into freshly elevated rock. Massachusetts and the neighboring states along the Atlantic seaboard formed a plateau-like upland, perhaps one thousand feet higher than today, and the coastline lay fifty to one hundred miles out under the present waters of the Atlantic. 11

The Pliocene episode of stream incision was of short duration. The gorges are not wide, and only near the sea do they cut deep into the coherent crystalline rock which gives New England its solid foundation. Nowhere did the land remain elevated long enough to permit the rivers to widen their canyons through the plateau-like country and to modify the essential features of the landscape. The latter were acquired in an earlier geologic epoch called the Miocene, and the scenic pattern carved by running water in that relatively remote division of time still dominates the region's topographic form.

Rivers Carried Off the Everlasting Hills

Every stream has its load of sediment, as the silt- and sand-filled reservoirs along the edges of the valley so effectively testify. Each sandy river bed is an aggregate of rolling grains, moving with the current, slow where it is slow and faster where the current is accelerated, but travelling always towards the sea. Every grain is a piece of the countryside lost to the land and soon to become a part of the ocean floor. Very little of this sand comes from the lowland itself, for the Connecticut may cut the bank below Hadley, but it leaves almost as much sand as it acquires on the opposite shore. The river's burden is brought to it by swift tributaries—the brook at West Pelham and hundreds more like it. Their sides are cut-banks, but no extensive sand bars are built to balance their erosive work; what they pick up they carry to the lowland, and what they bring to the lowland is soon transported to the sea.

The contribution which the tributaries make to the lowland rivers was demonstrated only too conspicuously by the great fans of coarse debris spread across the valley of the Deerfield River and the West River during the floods that accompanied the torrential rains of the hurricane. Parts of the village of Townshend, Vermont, nestling in the flat floor of the West River 12

valley, were buried in gravel wash, and the hillside roads above were gullied ten feet deep. One harassed traveler aptly remarked that the original road level could be recognized from the few concordant remnants of pavement beside the trout brook.

The hill slopes at Townshend rise and end near Jamaica, about one thousand feet higher in elevation. Here the roads are in good condition. There are no signs of erosion, and the rolling uplands extend for miles with no signs of gullying or wash by the heavy rains.

The debris handled by the West River now and for ages past has come from the steep hill slopes along the main valley. Each load of sand has cut these slopes back from the main stream and has widened the lowland floor. So, for millions of years, the tributaries of the Connecticut have pushed the valley walls farther from the main river, and their tributaries in turn have pushed their hill slopes back, while the valley floors have steadily widened. The Connecticut Lowland was broadened in this way, and the tributary Deerfield has developed its valley in similar fashion but to a lesser degree. Today streams near the headwaters acquire sediment, not from the upland across which they flow to reach the deeply entrenched valleys, but from the steep slopes in the most remote recesses of the upland on which they rise.

Flat valley floors are broadened in coherent rocks as well as in unconsolidated sand—less rapidly, indeed, but just as surely; and every region is worn down to the grade of the streams which drain it, except for those rare masses of resistant rock which defy decay and yield reluctantly to their inevitable fate. The rocks of the Mount Holyoke and Mount Tom ranges, Mount Warner, the Pocumtuck Hills and the highlands on both sides of the Connecticut Valley are made of tougher ingredients than the lowland, and even millions of years of incessant onslaught by running water did not suffice to level them by Miocene time, when the lowland was excavated.

PL. 3. Erosion remnants or monadnocks surmounting base levelled surfaces.



a. Mt. Sugarloaf, a remnant of Triassic rocks disappearing grain by grain down the Connecticut River.



b. Mt. Monadnock, a hill surmounting the New England peneplain, seen from Mt. Lincoln.

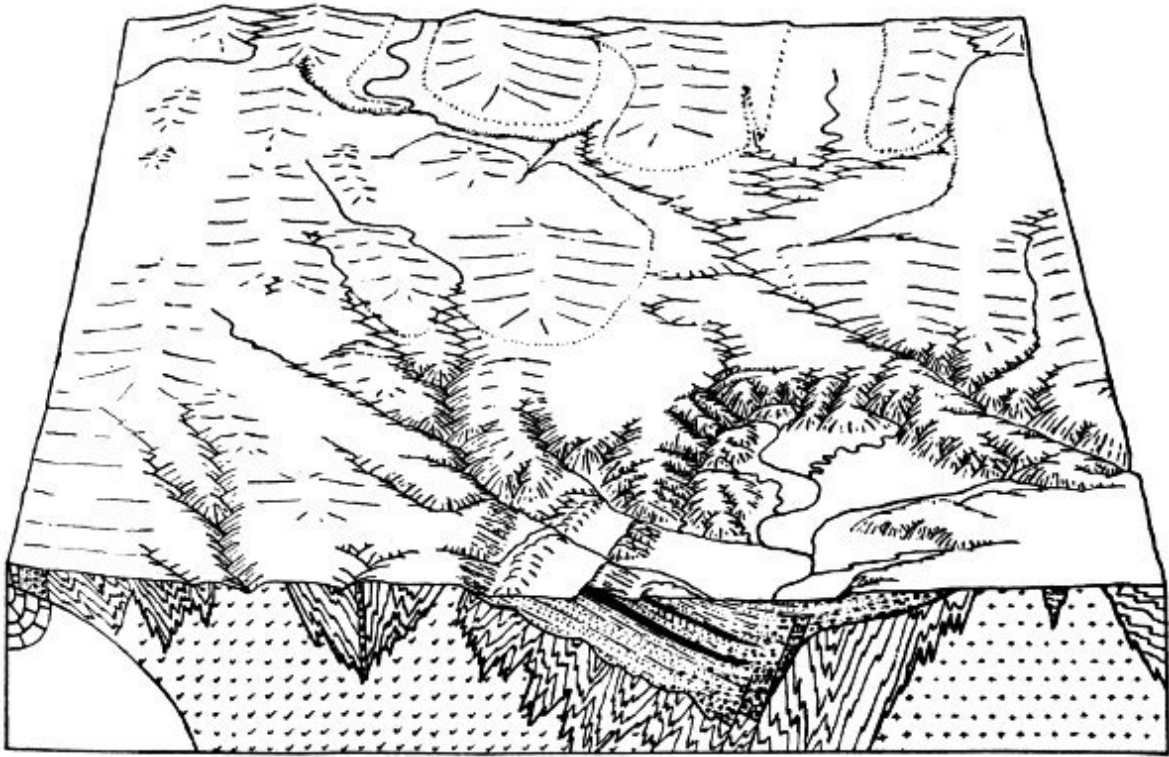


FIG. 5. *Block diagram showing the main features of central Massachusetts during the excavation of the lowland.*

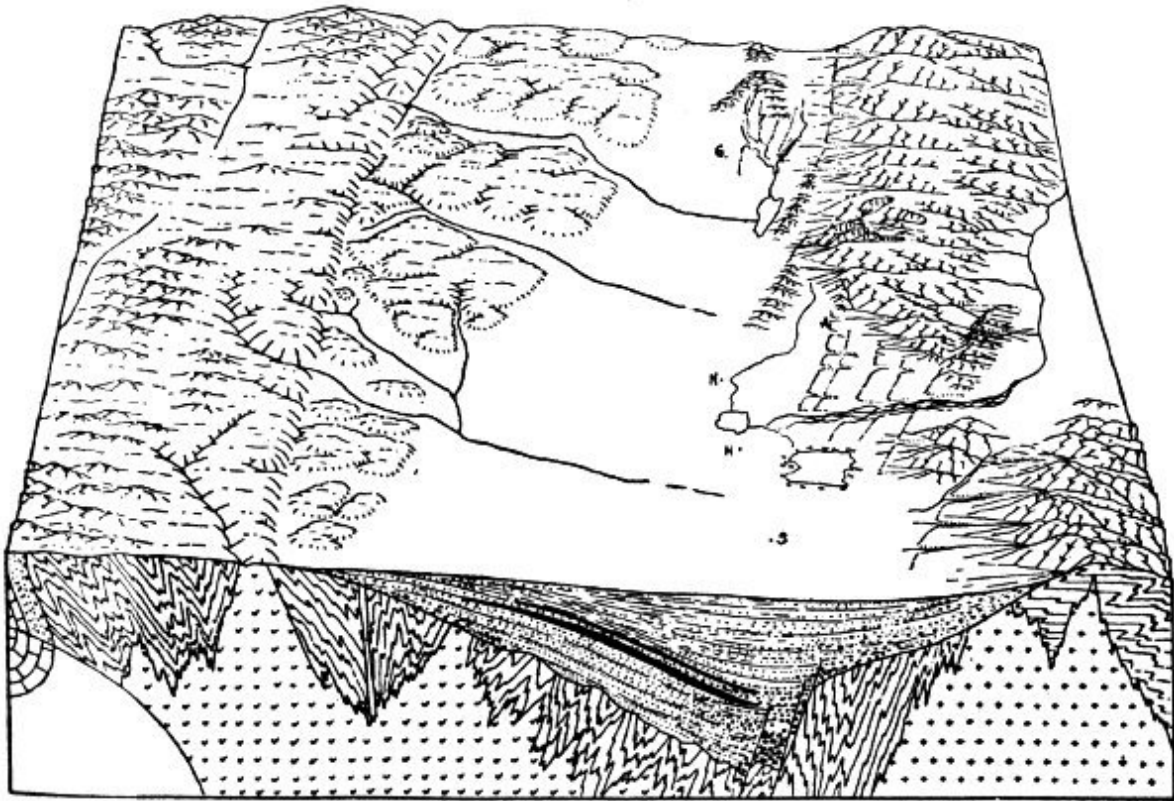


FIG. 6. Block diagram showing main features of central Massachusetts after the Triassic basins were filled.

The lowland extends beyond our immediate region. It continues southward with diminishing elevation to New Haven, where it joins another broad depression, now flooded by the waters of Long Island Sound. 14

Before the Rivers Cut the Valleys

Those who would see the land as it was before the rivers carved the lowlands must put back every grain of sand the waters carried away; they must fill in these valleys to the level of the Jamaica upland. Then only will the country be as it was before the streams were rejuvenated and started to cut deep trenches and to widen them as the Deerfield has done at Charlemont.

Broad, open valley flats or straths surmount the steep V-shaped notches of both the Deerfield and Westfield Rivers. Surely, everyone who has paused at the lookout on the east summit of the Mohawk Trail has seen the upland

sloping gently towards the Deerfield and then breaking sharply at the top of the present canyon. The same view confronts the motorist who drives from Adams to Cummington, just after he leaves the village of Plainfield. Here the shallow bowl in front of him holds no hint of the deep notch in which the Westfield flows. The gentle contour of the land suggests only the slow but methodical sort of change which comes with maturity. Those who favor air travel will see, as they fly over Mount Tom, a similar but more dissected strath reaching into the hills northwestward from Northampton. Aeroplanes flying the Boston-New York route pass over straths which have been trenched by the Connecticut along its course from Middletown to New London.

The straths are part of a mature, but ancient drainage system, which 15 was graded a thousand feet above the level of the present streams and only a few hundred feet below the main upland. Certain broad depressions through the highlands east of the Connecticut Lowland suggest that this drainage pursued a southeastward course to the Atlantic, and that the river did not establish its modern course until the straths were elevated and notched.

The land level above the strath-margins is a still older surface from which the rock-benches were cut. The higher surface stretches to the horizon at Pelham, but Mount Monadnock and Wachusett stand conspicuously above it. And on the Mohawk Trail one must ascend the tower at the eastern summit before any higher land comes into view. Greylock's summit and the long chain of the Green Mountains attain greater elevations. The West River and Deerfield basins are graded to the level of this higher and older erosion surface, but farther north a chain of peaks including Stratton and Okemo swing eastward towards Ascutney. They appear to have formed a divide on this ancient land, as they do today; and beyond their crests rivers have run to the Saint Lawrence and Hudson basins from a time which antedated any of the familiar features of the New England landscape.

Although this flat upland surface is more complex than it appears to the eye, it dominates all of southern New England, and ramifying arms of it penetrate northward into the White Mountains of New Hampshire and Maine. Another great arm passes west of Mount Greylock and spreads out between the Catskill Mountains and the Adirondacks. During the long period of erosion

when it was formed, New England was reduced nearer to the grade of the main rivers than at any other time either before or since, and only rocks which have effectively resisted all later assaults by the geologic processes of destruction surmount the surface. To the eye, the region appears so nearly planed that it has been called the New England peneplane.

The upland continues southward through the Berkshire and Litchfield Hills, descending in a series of almost imperceptible steps towards Long Island Sound and the Atlantic. A few miles south of Litchfield, Connecticut, its low angle of declivity increases abruptly, and the more steeply inclined surface passes beneath the waters of Long Island Sound. The sudden change in dip suggests that two erosional planes are present and that each was formed under somewhat different circumstances and in different periods of geologic time. The soundness of this surmise can be demonstrated in Long Island where sediments laid in a Cretaceous sea rest upon the older and more sharply inclined erosional plane and rise approximately to the level of the New England upland. The deposits form a wedge between the two planes, and their Cretaceous age supplies a series of dates that would otherwise be difficult to establish in New England's geological history. Erosion fashioned the New England upland in the early and middle epochs of the Tertiary period, immediately following the deposition of sediments in the Cretaceous sea. And the southward sloping plane upon which those sediments rest records an even earlier episode of denudation—an episode lost in the shuffle of later events in Massachusetts but preserved in fragmentary form in Connecticut, thanks to the protection afforded by the sedimentary cover.

Had we lived in central New England when erosion of the upland and of the younger straths was in progress, we would have noted that the valley forms were well defined in the headwaters and lower reaches of the streams, which made their way through a country of light-colored or gray clayey soil. In the middle reaches the valley boundaries were blurred and indistinct, and the country through which they flowed was surfaced by red and sandy soil. The middle region is now the lowland, but even then it formed a depression athwart the topographic and hydrographic features of the country; and its distinctive red soil resembled alluvial wash or fill in a long basin. Its low relief would have been as impressive in early Tertiary time as its higher relief is today, for then it had little topographic competition anywhere

between the present sites of New Haven, Connecticut, and Northfield, Massachusetts. 17

The land had one dominant characteristic—a relatively flat or faintly terraced surface. But this surface concealed a mosaic made of an infinite variety of rocks, each responding to the attack of weather in its own particular way. Erosion has brought out the pattern of the mosaic, and we have retraced the steps in its development. Viewing the evolution of the countryside in retrospect, we see its features take form much as a worker on an inlaid bronze might watch the design come out when it is etched. The creation of the mosaic or inlay is another part of the history, and the relief of the land now permits closer scrutiny of the pattern than would have been possible in Cretaceous time.

The Mosaic of Central Massachusetts

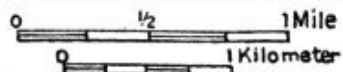
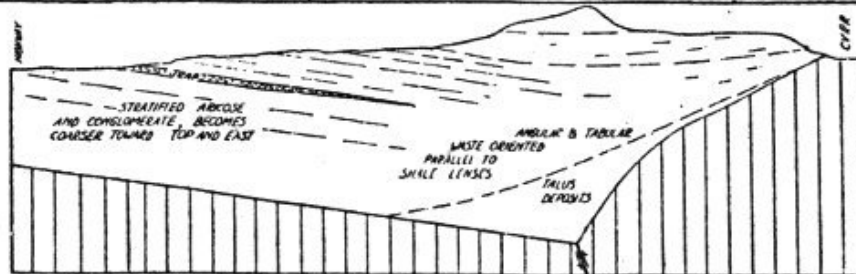
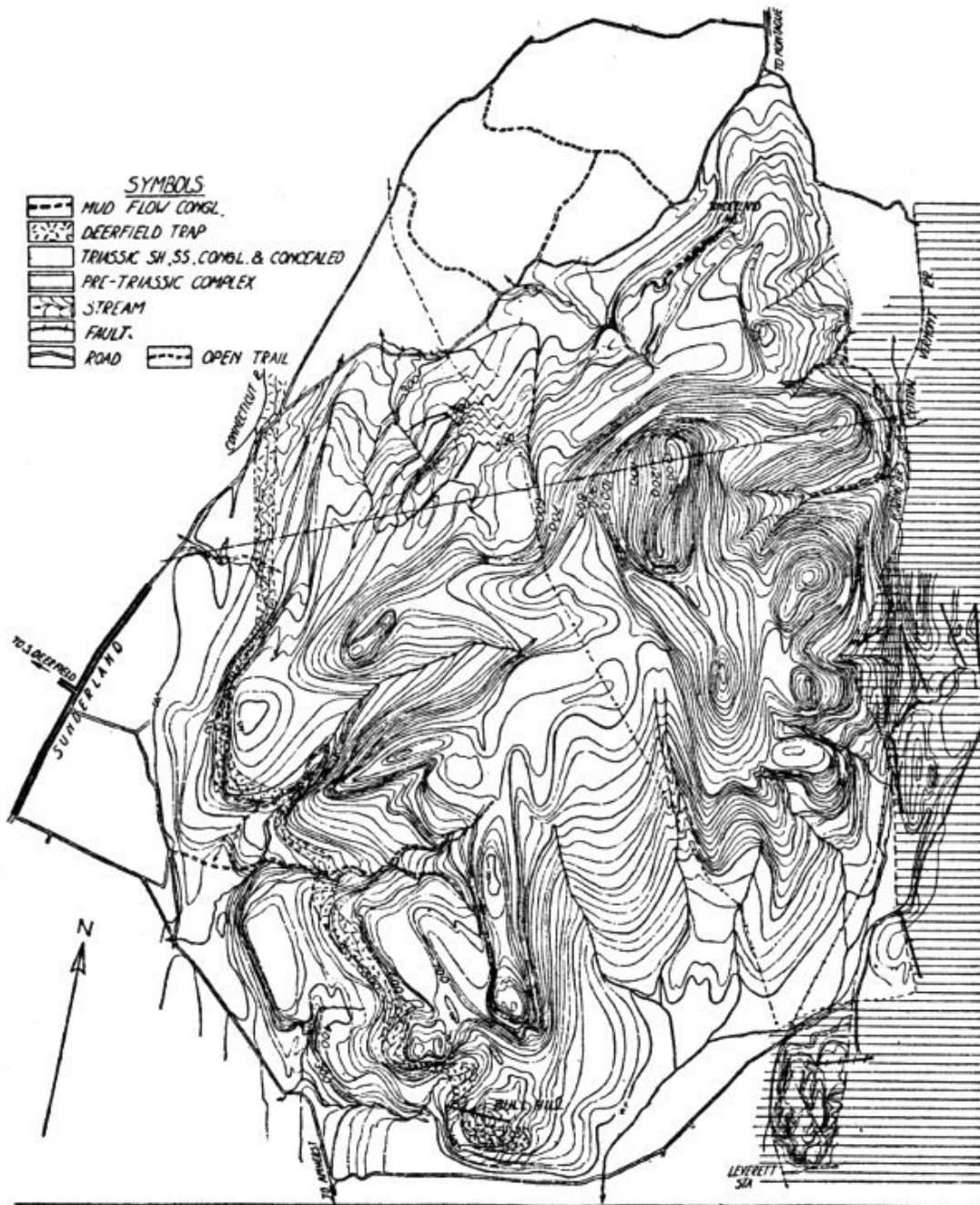
The great artisan incorporated three main features in the mosaic beneath the New England upland, and from them erosion developed the major pattern of the present landscape. The three units of the pattern comprise a somewhat heterogeneous but durable foreground in the east, a weak inlaid design in the center, and a moderately homogeneous and durable background in the west. The foreground and background are simply a suitable base for the younger, central feature of the design—an inlay which was completed in Triassic time, while the mighty dinosaurs were beginning to gain confidence as the new rulers of the earth. Skillful artistry and complicated technique were expended on the Triassic inlay, for in part it was rolled in, partly melted in, and some of it was cut in amid the tougher materials now found on either side.

The Red Rock Basin

The youngest ingredients which were incorporated in the inlay are a series of fine-grained red sandstones and consolidated clays or shales. They are horizontal layers, turned up slightly at the edges of the lowland, but elsewhere they lie in almost horizontal beds that extend from South Hadley through Chicopee (Chicopee shale), Springfield, and Longmeadow (Longmeadow sandstone) to a point just south of Hartford. Near the hills which form the eastern boundary of the lowland these fine-grained sediments locally give way to coarse tabular deposits of angular gravel, which appear along the base of the Wilbraham Mountains and again in Mount Toby and northward. The deposits are isolated or detached masses which resemble fans emerging from mountains, not unlike the more modern sands and gravels which the Westfield River left where it emerged from the western hills. But the Triassic gravels are red, and they are firmly 19 cemented into conglomerate; yet it is plain that this part of the inlay was made by washing and rolling the red muds, sands, and cobbles into a depressed basin waiting to receive them.

The southern part of the basin was deepened, and the highlands were rejuvenated spasmodically from Springfield to New Haven. The sinking of the lowland on the west and the rising of the highlands on the east took place along a fracture plane, commonly called “the eastern border fault,” near the eastern limits of the red sediments throughout that part of the valley. The rocks composing the alluvial fans are flexed sharply downward east of Portland, Connecticut, like compressed pages in a book, where the great eastern mountain block pushed obliquely against them. In this way the mountain range was renewed as erosion wore it away, and the basin was deepened periodically as the wash from the highlands filled it. The intermittent uplift sustained the growth of the fans along the edge of the lowland, but the frequent recurrence of movement never permitted these graded accumulations of waste to extend far out from their mountain sources.

The great fracture, which sharply delimits rocks of different origins, and the deformation in the strata near Portland record, as surely as the writings of any human historian, a tale of periodic rock compression and paroxysmal release that must have been accompanied by violent tremors. Connecticut and Massachusetts had their earthquakes and had them as violent as any now originating in the western ranges of the United States and Mexico; but happily they shook a land which was overrun by the dinosaurs, and which was not yet ready for human habitation.



CONTOUR INTERVAL 20 FEET
DATUM PLANE 40 FEET ABOVE SEA LEVEL

FIG. 7. *Map of Mount Toby showing gorges filled with conglomerate.*

Near the northern terminus of the Triassic basin the eastern boundary 21 was not subject to intermittent and violent movements during the later stages of sedimentation, as it was in the south. Instead, the youngest part of the red-rock inlay consists, in some places, of unfractured boulder beds which were washed far out towards the center of the lowland; elsewhere, landslides brought masses of rock debris upon soft red and gray shales, which may have accumulated in shallow lakes; in still other localities, long stringers of red sediment reach far back into the eastern highlands. Many boulders in the conglomerate at the south end of Mount Toby are eight feet in diameter, and torrential mountain streams brought them to their resting place. A few are scratched and grooved, much like the boulders in the till left by the ice sheet; perhaps they signify the presence of snow fields and glaciers in the mountain range, but the scratches may have been acquired by avalanching. The landslide masses buried in the shales at the Sunderland caves show that the mountain front was steep, and the ancient talus or slide rock near the Central Vermont Railroad south of Roaring Brook shows plainly that the mountain front was a precipitous cliff of granite. The stringers of conglomerate extending eastward into the granite upland south of Montague, north of Leverett station, at Amherst, and again near Granby, are alluvial fill in ancient mountain gorges.

This old mountain mass stood out as a long, straight range extending from a point east of New Haven northward into New Hampshire. It was of moderate height in Connecticut, but it became higher and more rugged to the north; glaciers may have nestled around its crest east of Deerfield, and its front was an impressive slope of slide rock. Granite gorges with tapering gravel plains, dry one day and raging torrents the next, fingered eastward into the mountain block. At that time the Connecticut Valley was much like the land east of the Sierra Nevada in California, where greater contrasts in heights and depths are to be found than in any other part of the United States.

A Dinosaur Diary

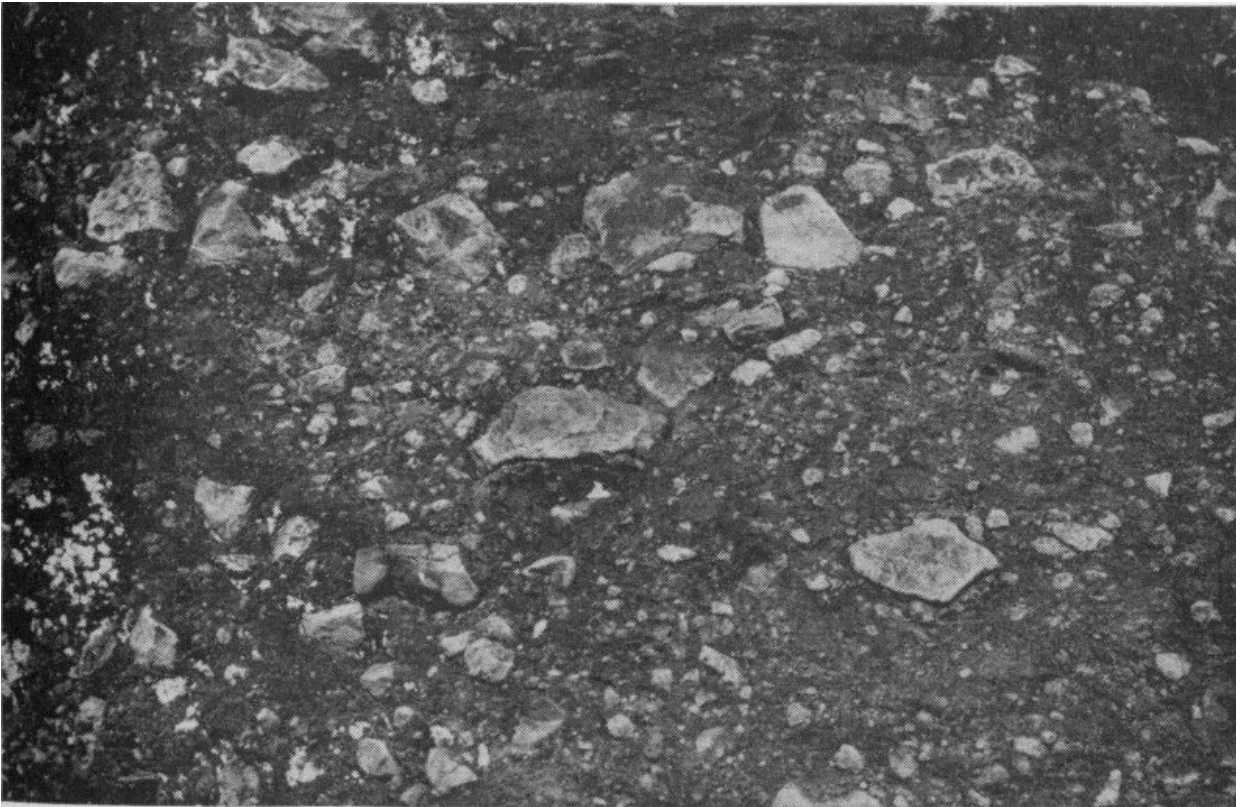
Like the valley east of the Sierras, the depression in central Massachusetts contained playa lakes and intermittent streams. Sand brought by the mountain torrents clogged the channels and spread into broad alluvial plains, while silt accumulated in muddy lake basins. Black sandy shales now 22 mark the sites of the lake beds, and their black color comes from the coaly remnants of Triassic plants. Some swampy lake margins supported peat bogs, which have been preserved in coal seams two to three inches thick between Granby and South Hadley. Many of the lakes lasted long enough to become stocked with half a dozen species of fish. But the fish led a precarious existence, and their skeletons were buried in great numbers in the upper lacustrine layers when the lakes dried up, and dust and sand drifted over the parched basins at Durham, Connecticut, and at Sunderland, Massachusetts. The remains were interred even more effectively when cloudbursts in the hills brought thick layers of gravel out over the ancient lake beds.

Most of the lakes and ponds were ephemeral, but the fact that their presence was more than a mirage in a Triassic desert is clear from the ripple-marks retained on their sun-hardened surfaces, and from the impressions of objects which touched them while they were still soft. Stray series of parallel furrows record the passing of drifting shrubs, and the abrupt disappearance of rain-drop imprints at a well defined line in the hardened mud marks the exact position of the water level in a few of these Triassic water bodies. Footprints register the activities carried on by a bizarre animal population. Beside the road to the French King Bridge and in the river bed at Turners Falls the ripple-marked surfaces contain the impressions of many feet, and the dinosaur tracks at “the Riffles” beside the Northampton-Holyoke highway are known throughout the country. In Connecticut, Middletown and Durham are famed for their tracks, and the impressions left in the playa beds by muddy feet are so widely distributed throughout the lowland that it must have taken a lot of walking by many generations of dinosaurs to leave such an ample record.

PL. 4. *Rocks of the Triassic basin and their record.*



a. A dinosaur walked from the raindrop marked surface at the right to a shallow pond at the left.



b. Volcanoes ejected much ash and many bombs to form the Granby tuff.

Some of these three-toed animals were like the modern lizards and walked on all four feet; but the great majority walked on two feet and, like the kangaroos, used their tails to balance their bodies, and their short fore limbs to support them when they crouched. In any single playa deposit, variations in the sizes and kinds of footprints reveal that many individuals made them; yet strangely, most of the tracks at any one place are headed in a single direction. Apparently the herd instinct must have been strong in these reptiles, as it is in kangaroos or in a flock of turkeys, all following a leader, with only an occasional individual going off to one side or back-tracking in a display of independence. And so the dinosaurs dominated the life in the early Connecticut basin, as it sank and trembled, and as mountains rose to the east; on dry days and days of cloudburst, on hot days and days when frost crystals formed in the mud, they roamed the plain, as the lowland settled nearly two miles and filled to the brim with red sands, muds, and marginal gravels. 23

Volcanoes

Red is the predominant color in the central inlay of the New England design, but greens and blue-black lines have been worked into the pattern. The dinosaur-ridden basin has a rim south of Middletown in Connecticut, and another north of Holyoke in Massachusetts; it lies just west of the dinosaur-track ledge near Holyoke, and the tracks themselves are only thirty feet above the bottom of the basin. The rim is an odd ensemble—now red and now green; here solid and hard and black, there soft and fragmental and crumbly. The fragments may be angular or round; sandy or glassy; dense and solid, or full of bubble holes like molasses taffy. The whole looks like the spread-out ash dump from a giant power plant. And not only does it resemble an ash heap—it *is* the ash heap of a volcano; and the hard black layers within it are lava flows interspersed with the heavy falls of ash.

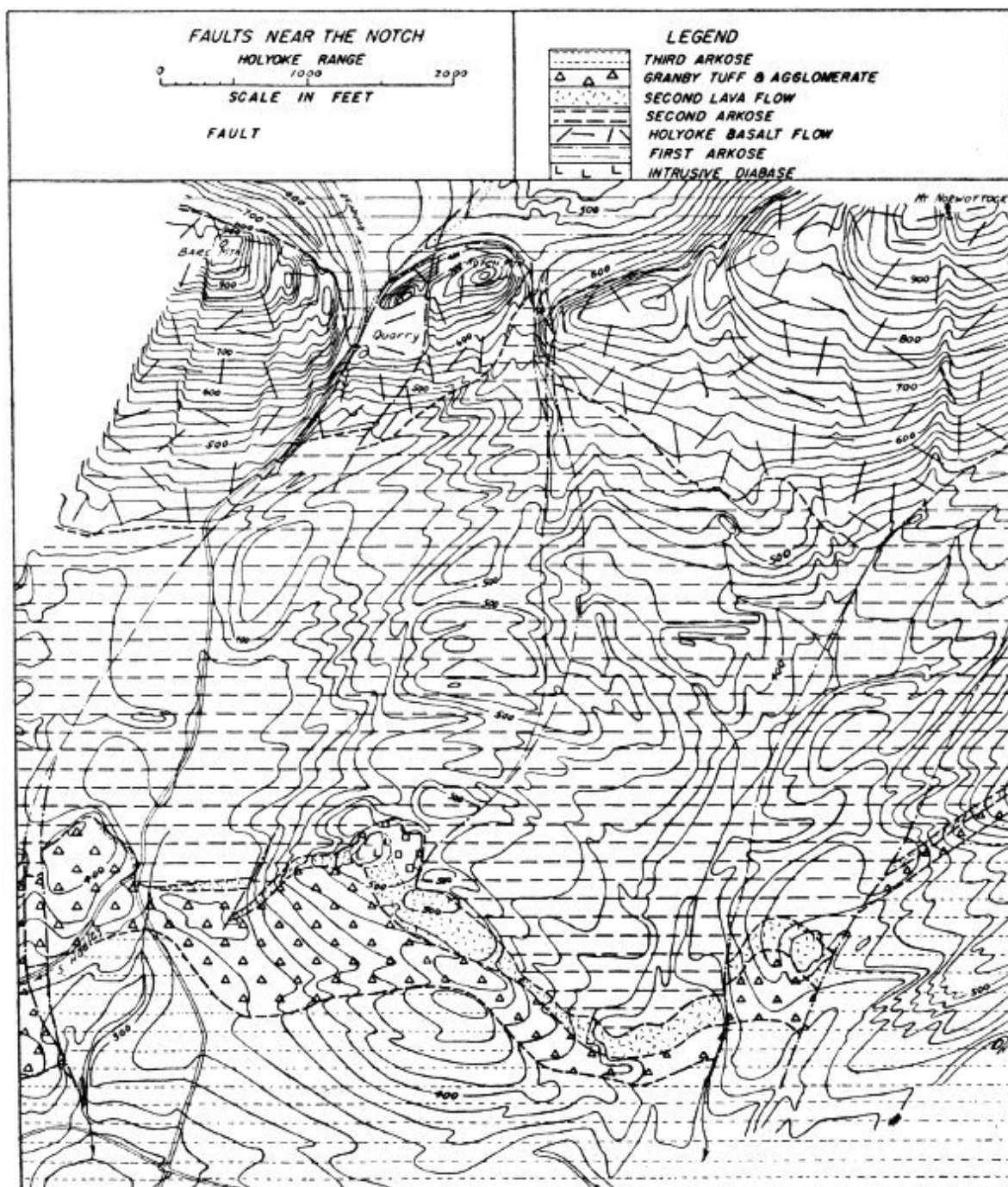


FIG. 8. Map showing agglomerate burying a fault scarp on the power line through the east gap of the Notch.

The ancient ash heap grows thicker east of the Connecticut River, and it is more than 3,000 feet from top to bottom around a series of massive blue-black rock-columns southeast of the Mount Holyoke Hotel.

These are the lava-filled necks of craters which became quiescent with the dawn of the dinosaur days. The ash deposit, called the Granby tuff, grows thinner eastward away from the craters and disappears completely northeast of Granby, where a stream deploying from a valley in the eastern mountains washed it away as fast as it fell and left coarse gravel in the form of a huge fan.

The floor on which the ash came to rest was not everywhere the same. Where now it crosses the Northampton-Holyoke highway and the Amherst-South Hadley road it was a lava flow; but north of Granby and at numerous places between the Hockanum and Amherst-South Hadley roads the ash lies on conglomerate. Along the Amherst-Springfield power line, a block of the conglomerate floor was pushed up five hundred feet above the same beds farther west, forming a small block mountain which was entirely buried beneath the ash. Similar block mountains can be observed under the blanket of ash, especially on the south side of the Holyoke Range; and renewed movement subsequently affected many of the blocks north of Granby, where the ash deposit and even some of the sediments laid down in the earlier days of the dinosaurs were fractured and displaced. As a rule, along any one fault, the block on the east was pushed up and moved southward; and the block on the west was pressed down: as a group, the fractures may form the beginning of the great eastern border fault which bounds the basin farther south.

The volcanoes which made the Granby tuff or ash bed erupted intermittently for a long period of time. Usually, the river which emerged from the eastern mountain range brought so much fluvial debris that ash is not in evidence except in the immediate vicinity of the craters located between the Notch and the summit of Mount Holyoke. Even though alluvial sands and gravels supplant the tuff here and there, the river did not succeed in closing or quenching those fiery vents. The rocks now present recount a struggle in which, at times, the river encroached upon the cinder cones; at others, the ashes choked the stream and buried its alluvial wash.

While the volcanoes rumbled and erupted, earth forces intermittently thrust the eastern mountain range southward and upward, dragging the eastern margin of the lowland with it and upturning the sedimentary fill, much as a plow might upend a layer of snow at the roadside before shearing it off and pushing it out of the way. The relentless movement caused the

entire eastern floor of the basin to be broken into blocks; the easterly ones were piled against the westerly, and their eastern edges were pushed down into the basin floor and the western borders rode up on their neighbors. Through all this tremendous disturbance the great stream pouring out of the mountain pass kept the elevated blocks cut down and the small basins filled in. Earthquakes, erupting volcanoes, and shifting rivers made life for the dinosaurs troubled and a bit uncertain.

Only once did the volcanoes dominate the situation in the valley, and that was very early in their history. A group of vents, localized along a southward trending zone about a mile west of the Notch, and another group along the present course of the Connecticut River from Turners Falls to Sunderland poured out billions of cubic feet of black basaltic lava into the center of the lowland. Eruptions followed in such rapid succession that the rivers never scoured the surface of the earlier flows. Lava piled up 400 feet thick in the center of the basin east of the Mount Tom Range; it moved eastward in a flow which thinned against the fans of rivers issuing from the eastern mountain, and it ended in a formidable wall of scoria confronting the mountain streams. Lava buried the northern basin from Sunderland to Turners Falls and beyond, while the southern basin filled from Northampton to New Haven. But lava dominance was short-lived, and even before its bubbly surface reddened to the weather, streams had covered it with gravel.

The lava flows are the most resistant materials used in the lowland design. They form the ridge east of Greenfield in the northern basin. The Holyoke and Mount Tom Ranges are remnants of these flows, tilted at moderate but varying angles by the recurrent movements which enlivened the epoch of dinosaurs and volcanoes.

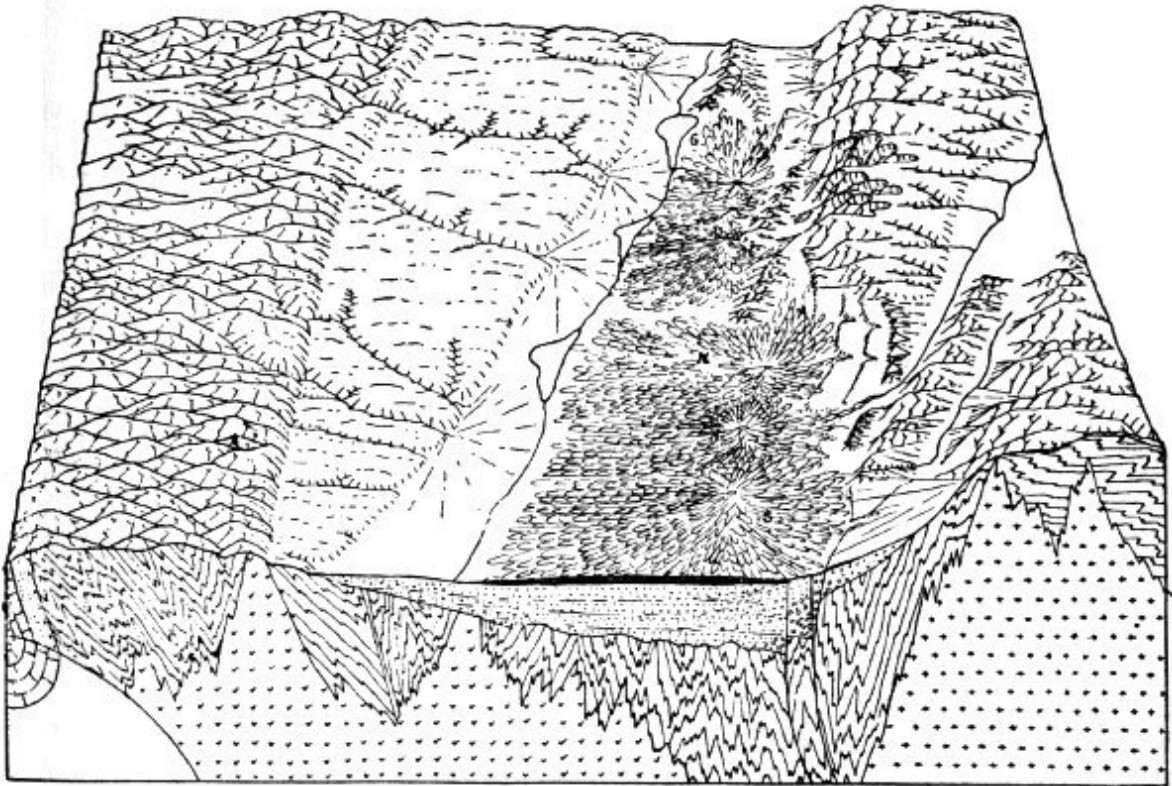


FIG. 9. *Block diagram showing the main features of central Massachusetts during volcanic stage.*

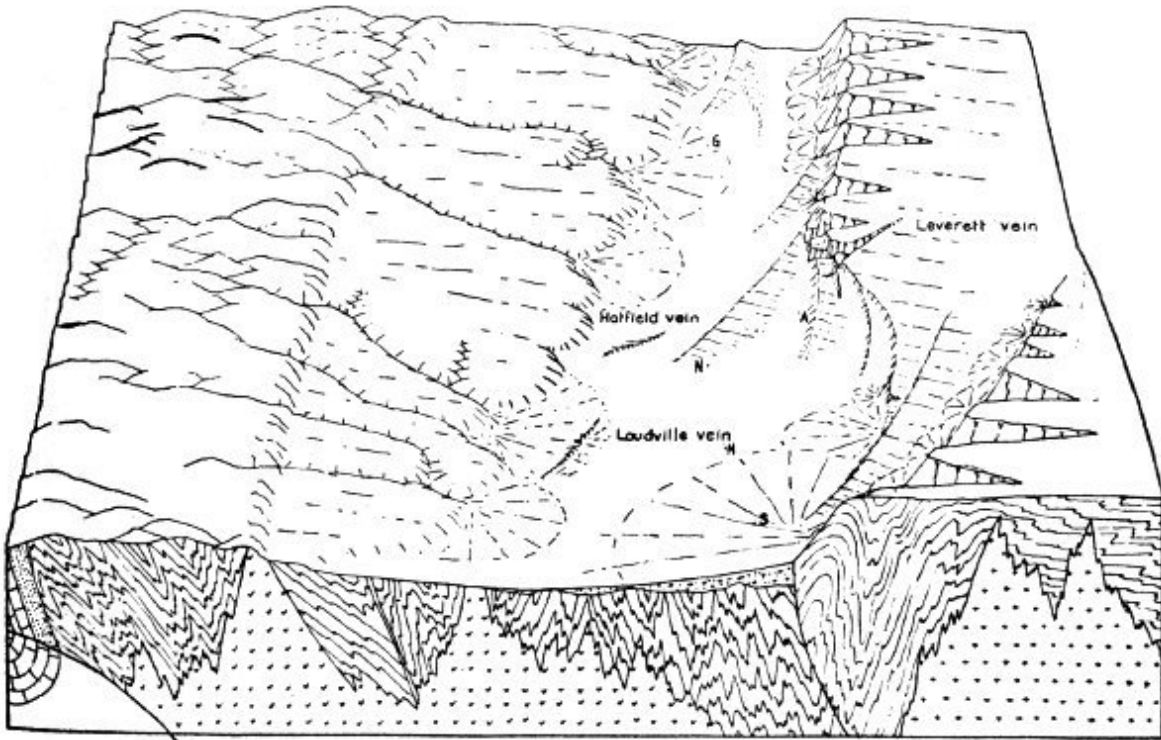


FIG. 10. Block diagram showing the Triassic basins of central Massachusetts.

The most spectacular episode of lava extrusion was localized in a small volcanic center situated about one mile west of the Notch in the Holyoke Range. All flows in the range moved away from this center, and before the great outpouring took place, minor explosive outbursts had built cones of ash with bases up to a mile in diameter. Small lava tongues are interspersed with the ash beds, and mixtures of sand and lava tell of breaks through the 1,500 feet of sandy fill which was rapidly accumulating in the basin. Throughout this early period of volcanic activity the streams brought out so much wash from the eastern mountains that they soon dominated the scene in Massachusetts, and in Connecticut volcanoes gained ascendancy for one brief moment of geologic time, when an early flow covered much of the valley from Hartford south.

28

The Original Valley

The first and oldest ingredients in the central design are entirely red. The materials are fragments of older rocks—granite and gneiss, schist and pegmatite, feldspar and quartz. They are invariably coarse, for every layer of inwashed sediment has pieces over an eighth of an inch in diameter, and only the coarser particles were smoothed. The finer particles were not moved about enough to have their sharp corners worn away. The pebbles and clay in the thick layers of conglomerate at the French King Bridge were dropped by rushing, overloaded torrents deploying on a lowland—a situation not unlike the one at Townshend, Vermont, during the hurricane deluge. Only fine debris was transported across the fans to the far side of the basin. The western hills made small contributions of sediment; but their streams brought particles which never exceeded an inch in diameter, and in quantities so moderate that the fragments underwent some sorting and sizing as they were spread over the lowland. From the very start the valley was deeper near the east wall than the west; and the eastern mountain block was greatly elevated, whereas the western block was simply a hilly upland.

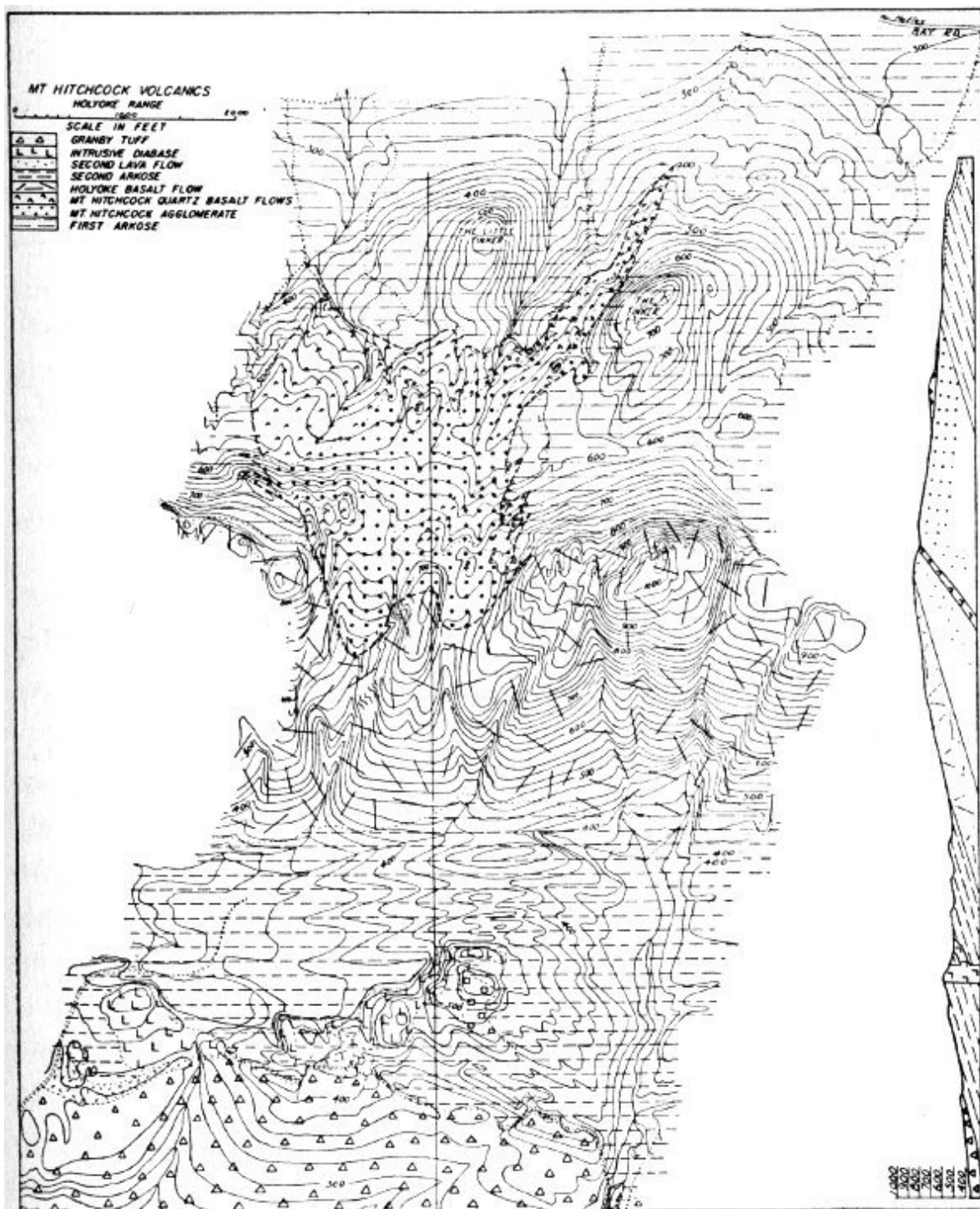


FIG. 11. Map of the old volcanic region near Mount Hitchcock and west of the Notch.

The edge of the eastern mountain mass is located at the French King Bridge and passes half a mile west of Montague. Its location beneath

the younger fill is known less perfectly farther south, but it seems to extend through Amherst, certainly west of South Amherst and Granby and probably east of the Notch. At least two mountains rose above the ancient lowland floor; the northern one is a long ridge of schist between Bernardston and Mount Hermon, and the other is Mount Warner. Mount Warner is an island of highland rocks in a sea of red sandstone fill. The Bernardston ridge resembles a peninsula in somewhat analogous sedimentary surroundings. The two eminences reveal the form of the valley floor and the western hills at the dawn of the Triassic period, for they were spared from destruction by burial, until deep erosion exposed them again in Miocene time.

Hot Springs in Central Massachusetts

Hot springs the world over register their presence by leaving deposits of unusual minerals, and they have left this sort of record at Loudville. Here the coarse sandstones of the lowland rest upon gneiss, and at the south end of the Loudville lead vein barium sulphate crystals, called barite, formed in the sand before it was cemented into solid rock. The crystals are the product of highly charged mineral water, rising through the sands from a subjacent fissure. The fissure itself is also filled with barite, and with galena and quartz as well. It is the vein which was worked in the old Loudville lead mine. There are other veins in the western and eastern highlands at Hatfield, at the Northampton reservoir near Whately, and at Leverett. All are in fractures which were still partly open when the valley first took form.

The Marginal Uplands

The rocks which formed the high eastern mountain range of Triassic time and the rocks which made the old western hills and underlay the basin floor comprise essentially a single group characterized by its complexity. At one place the rock resembles sandstone, but the layers stand on end; at another, it looks like shale, but the stone breaks across the color banding instead of parallel to it; and at a third place a fissure seems to have opened and had a crystallizing melt poured into it. These tabular, filled fissures can be found nearly everywhere, coursing in every direction and at all conceivable

inclinations to form a network that binds the older rocks into a firmly knit whole. The fillings, or dikes, are like reinforcing rods, holding the rocks together and withstanding the agents of destruction. Thus, the story of the highlands has three distinctive phases,—a relatively young phase when the interlacing reinforcements were poured into fractures; a somewhat more remote stage, when the bedded rocks were crumpled into their inclined positions; and an earlier stage, when the bedded rocks were deposited. The geologic dates of these three events may vary from one locality to another, and they certainly are different in the Eastern Upland as compared with the Western Upland; but the events always occurred in this sequence and constitute the broader aspects of the story at all places.

31

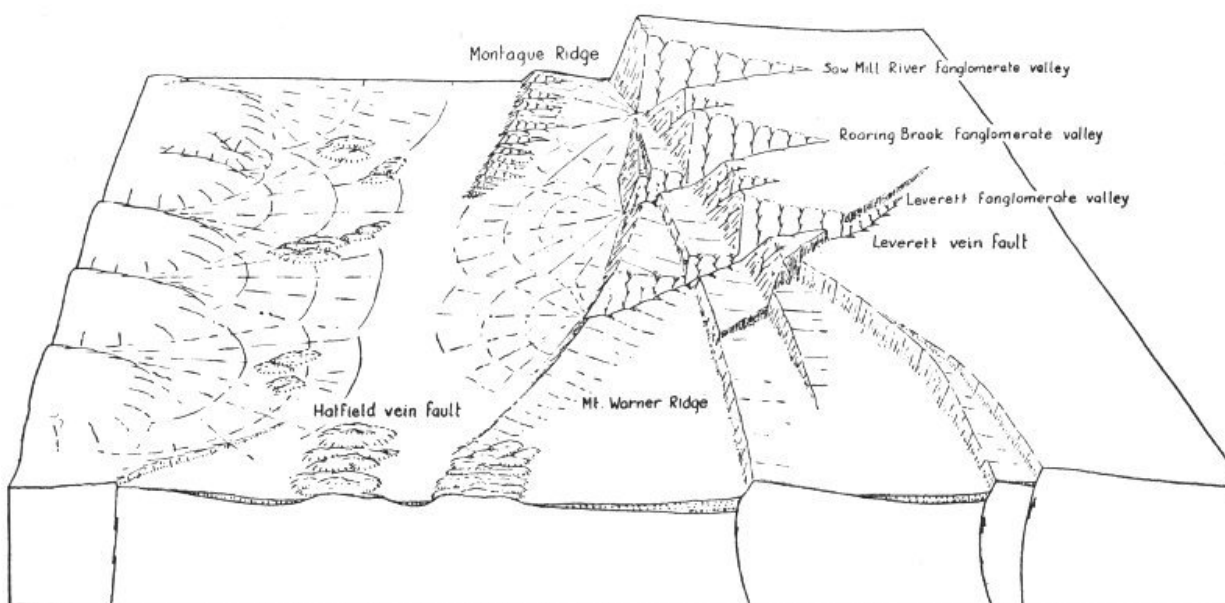


FIG. 12. Block diagram showing topography during formation of the lead veins.

The Eastern Upland

32

The Eastern Upland includes the land between the Connecticut Valley and the Atlantic Ocean. At present, it has the general form of a broad rolling highland with ridges and valleys that have a north-south trend. Close inspection shows that the rocks in the ridges are different from those in the long valleys. Also the layering of the materials ranges from a vertical

attitude, as at Ware and Brimfield, to undulating and almost horizontal positions, as at Spencer and Worcester.

Through vertical and horizontal beds alike run those reinforcing sheets—some tabular and vertical, called dikes; others also tabular but horizontal, called sills; and some are just huge, irregular masses without visible bottoms, called stocks and batholiths. Some of them, composed of uniform, small, light-colored minerals, are granite; others are made entirely of large minerals over an inch across and are called pegmatite; a few, with cuneiform intergrowths of a dark mineral in a light one, arranged like Arabic writing, are known as graphic granite.

PL. 5. *Intrusive and extrusive igneous rocks.*



a. *Columnar lava rests upon red sandstone in the cliffs at Greenfield.*



b. Fissures were filled with liquid rock that became solid and bonded wall to wall at the Windsor Dam.

Every one of these masses flowed into the rocks along fractures and other zones of weakness, crystallizing as they lost their heat and solvents to the hot springs of that ancient time. They are all invaders, or intrusives, which inserted themselves into the older beds. Whether they were squeezed into the fissures by the pressures that crumpled the original beds into their upturned positions, or whether they, like the liquid in a hydraulic press, transferred pressure from a deep reservoir to the walls of the fissures and so pushed the beds into their distorted forms, is unknown. Two features are clear; the distortion of the beds and intrusion of the liquid bodies were almost simultaneous, and the hot springs associated with them were still active at the dawn of Triassic time. These profound disturbances transformed the land into a series of elevated, wave-like folds, and rains promptly began to tear away at the summits of the newly raised mountains. From them was carved a serrate and rugged landscape, part of which was later buried beneath the Triassic fill.

Coal Swamps in Massachusetts and Rhode Island

Among the strata of the Eastern Upland which were folded, intruded, baked hard, and stewed in hot spring water, one group stands pre-eminent. It forms a broad band starting north of Worcester and reaching to Providence and beyond. Nearly everywhere it carries coaly material or impressions of plants which are now extinct, but which flourished in the Coal Age or Carboniferous period. Some of the coal seams were mined in the Providence basin, but they had been so heated by intrusive granite that they are partly graphitic and proved difficult to burn. The great extent of some of the coal seams suggests a panorama of immense swamps, and of land so flat 34 that, for long periods, streams brought no sediment, and the trees and water-loving plants furnished the only fill. At other times sluggish rivers, flowing from the northwest, laid thick layers of sandy mud over the surface of the bogs. The alternating muds and coal seams are thousands of feet thick, and they record the story of a basin which sank as fast as it filled—a depression which was never built high enough to be a well drained plain, yet never subsided sufficiently to be inundated by the sea. The Carboniferous peat bogs and mud flats may have extended westward almost to the Connecticut Valley; and farther to the northwest they were bounded by a chain of rolling hills.

The rock floor of the coal basin contains a variety of ancient materials. Some rocks were river deposits, some were marine limestones, a few were lava and volcanic ash, and many were granite and gneiss which crystallized at great depths and became exposed only after streams had stripped away the thick overburden. The basin floor thus holds a complex story, in which land and sea, vulcanism and quiet, erosion and deposition, all played their respective roles. Only in the west, along the margin of the Connecticut Valley, is the involved story at all clear. And in the Western Upland across the red-rock inlay, it is possible to see some of the land as it was before trees took root in the swamps, and rivers brought sands and muds from the vegetated hills that hemmed in the coal basin.

The Western Upland

Many years ago, when transportation facilities were not what they are now, New England settlers mined iron ore from the hill north of Bernardston and smelted it in local charcoal furnaces. The rocks containing the iron are creased into sharp, close folds, and they came into such close contact with a hot granite intrusive that their minerals were changed by its action. This granite, however, is older than the one which is associated with the disturbed Carboniferous beds, for it was intruded when the Devonian sediments from Gaspé to Connecticut were deformed. It was this profound disturbance 35 that turned the red rocks of Roche Percé from a horizontal to a vertical position and raised a mountain range which stretched through all of northern New Brunswick, Maine, the lowland section of New Hampshire, and a belt extending for some miles east and west of the Connecticut Valley. The eroded remnants of these Shickshock Mountains formed the backdrop for the great Carboniferous coal swamps in Rhode Island, Massachusetts, and Acadia.

The iron ore was a hot spring replacement of a limestone containing shells of sea organisms which lived when chordate animals first became abundant. This was the Devonian period in geologic history—the time when a backbone appeared essential in every really high-grade animal. The limestone rests upon a beach gravel, now consolidated into a quartzite conglomerate. The gravel consisted of small white quartz pebbles which came from the many veins in the steeply inclined slates of the adjacent coast.

Marine deposits of Devonian age are found as far south as Leverett, and scattered outcrops indicate that the old seaway reached northward up the Connecticut, entering Canada east of Lake Memphremagog. Thence it spread eastward to Gaspé and westward to Montreal, and around the north and west side of the Adirondack Mountains into New York State. A low rolling land where the Green Mountains stand today formed the western shore of the Devonian sea for many miles northward into Quebec. The Adirondack and Taconic Mountains were a fused aggregate of undulating uplands which limited the seaway on the south along the International Boundary. Its eastern shore lay far beyond the horizon of the region described in this brief account.

The rocks of the old Devonian coast in Massachusetts were chiefly slates, cut by many quartz veins. They are exposed along the Mohawk Trail in the

ascent from Greenfield to Shelburne Summit, and they continue northward in an almost unbroken band through Bernardston, Brattleboro, and Northfield (Vermont) to Lake Memphremagog. They contain casts of planktonic life which inhabited the Ordovician seas in these northern latitudes, and the Ordovician strata, together with still older Cambrian sediments found below them, meet the Devonian beach deposits at a sharp angle, just as the slates along the coast of Maine meet the modern beach sands and gravels. Like the slates of Maine, they were eroded deeply before the beach existed, and their slaty structure and their steeply inclined attitudes were acquired in a still more ancient epoch of deformation. 36

The folded rocks of Ordovician age flanked the highland area which now constitutes the axis of the Green Mountains. West of the Green Mountains they make the Taconic Range, and to the east they appear in ranges that go under a variety of names, including the Northfield and the Lowell Mountains. In the Taconics the folds have the shape of waves advancing westward from the center of disturbance in the Green Mountain axis; within the Connecticut basin the Ordovician folds have wave-fronts which advance from the same axis eastward across the Memphremagog sea. Along the eastern margin of the old land a series of dark green intrusives called peridotite welled up from the depths of the earth, and they now cut through the rocks extending from Chester, Massachusetts, to Thetford Mines, Quebec; they are like giant boundary posts marking the ancient line of demarcation between sea and land in Cambro-Ordovician time.

Originally the folded strata in the Taconic region were deposited in clear marine waters, where calcium carbonate accumulated rapidly. But the sediments of the same age east of the Green Mountain land represent an unbroken succession of hardened muds, which rest on sandy muds, and on fine and coarse products of violent volcanic eruptions—tuffs and agglomerates—and lava flows. No lime-secreting animals could thrive in this sea, although they numbered billions in the western waters; for only floating plankton could escape the interminable mud, and they drifted up and down the coast from Quebec to Connecticut. One or two straits may have connected the clear waters of the west with the muddy waters of the east, for some of the planktonic organisms have been found in the muddier sediments of the westerly waterbody. 37

The Cambro-Ordovician sea lapped even older rocks, contorted and cut by intrusives which bonded them precisely as much younger invading liquid rock bonded the younger sediments of the Eastern and Western Uplands. The older rocks were also laid in a sea—a sea so much more ancient than the Cambrian and Ordovician seaways that its shoreline and even its form and extent are at best conjectural. And when we study these oldest marine beds, we find that their ingredients were in part derived from still more ancient sedimentary rocks, which accumulated in the sea, and that these old beds were elevated into the land that supplied the waste now found in the oldest coherent section of rocks in western New England. Indeed, the dawn of the Cambrian period, when life first became abundant, was merely a half-way mark through geologic time. Although half a billion years have elapsed from the Cambrian to the present, another half a billion years reach still farther back towards the beginnings of earth history, beyond which science has not yet peered successfully. These billion years are but a finite segment of history, bounded by the infinite past and the infinity of the future.

It seems appropriate, therefore, to end our journey down the fourth dimension at this point, and as we retrace our steps, we can profitably survey the chronologic succession of events and scenes which followed each other from Cambrian time to the Twentieth Century A.D.

The Story of Central Massachusetts

The protracted story of central Massachusetts might be that of many another section of eastern North America, except for minor details. In Cambrian time an inland sea, well stocked with simple marine organisms, washed the shores of an archipelago which extended north and south through the Berkshire Hills, the Green Mountains, and the Notre Dame Mountains. Composed of rocks which themselves had had a long and involved geological past, the islands rose intermittently as streams and waves wore them away. Clear water and sandy beaches stretched along their western shore, and the original Adirondack Mountains were just visible from the summits of the higher islands. Swift streams raced down their eastern slopes, carrying gravels, sands, and silts into the eastern arm of the sea, and only free-swimming animals could survive in its turbid waters. For a time, volcanoes erupted and fumed along the entire eastern coast from Thetford Mines, Quebec, to Plainfield, Massachusetts, but their activity was short-lived. Only the streams which drained the broad islands endured, and they never ceased to pour mud into the eastern ocean. Gaps in the island chain permitted some of the free-swimming organisms to migrate to the western sea, where bottom-living plants and animals were actively secreting the limy shells and skeletons which helped build thick deposits of Cambrian limestone.

These conditions continued into the ensuing Ordovician period of geologic time, but gradually the situation changed. Again the volcanoes renewed their activity, and masses of dark peridotite were intruded along the eastern shore; the island chain rose rapidly, and the straits closed. The elevated land began to expand outward, and folds spread eastward on the east and westward on the west, like waves from a center of disturbance. So great was the pressure that portions of the old land were sheared outward over the folded sediments. The Taconic disturbance was on from the city of Quebec to the city of Washington; and the streams, like ants, kept at their endless task of carrying sand and gravel into any and every depression they could find. They piled up great thicknesses of Silurian sandstone in Maine and New York, and so effectively did they tear down the Taconic Mountains that the

Silurian sea was ultimately able to penetrate the region from Thetford Mines, Quebec, almost to White River Junction on the Connecticut River.

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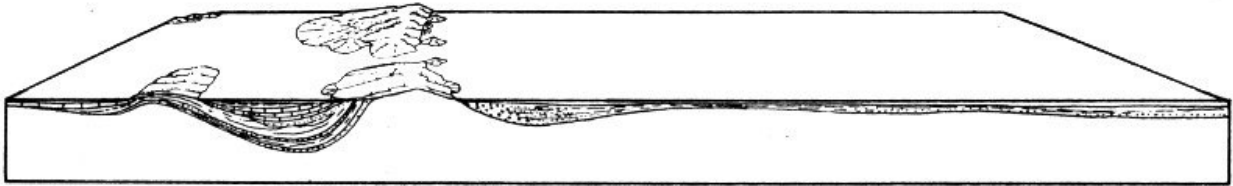


FIG. 13. Block diagram showing main features of central New England during middle Ordovician time.

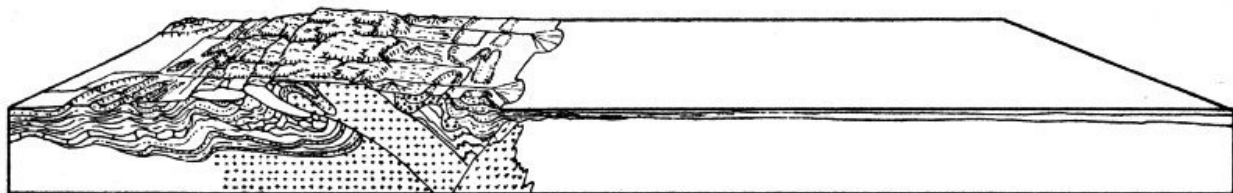


FIG. 14. Block diagram showing main features of central New England at the end of Ordovician time.

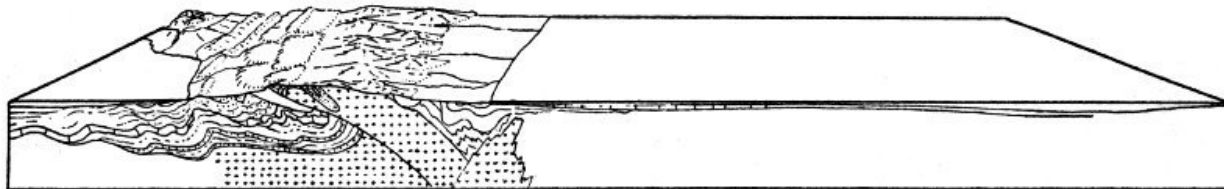


FIG. 15. Block diagram showing main features of central New England during the Devonian period.

One period later a Devonian sea followed in the wake of the Silurian sea, but its waters penetrated even farther south to Leverett, Massachusetts. The quartz gravels of its advancing beach covered the worn flanks of the Taconic folds. Sea animals left their shells to form a bed of limestone which may be seen today at Bernardston. But again the sea was shouldered aside by the restive land, which rose from Gaspé to Virginia. Much of the region affected by the Taconic disturbance was elevated again, and a broad band of Devonian sediments was folded closely through

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northern New Brunswick, southern Quebec, northern Maine, northern and central New Hampshire, and central Massachusetts. Granites welled up into the sediments, and dikes filled all the fissures. The baking, stewing, and reinforcing they gave to the older sediments made them so firm that they are still one of the most coherent and resistant series of rocks in New England and maritime Canada. This was the Shickshock or Acadian disturbance. Meanwhile the first forests took root on the long piedmont plains that spread from the rising mountains westward into the Catskill Plateau of New York State (Catskill sandstone) and eastward to the coast of Maine (Perry formation).

The margins of the piedmont plain sank. Vast, luxuriant swamps succeeded the old forests in Pennsylvania on the western piedmont, and in Rhode Island, Massachusetts, and Acadia on the eastern piedmont. The swamp vegetation later became the coal seams of eastern North America, and well does this time merit its name—the Carboniferous period. The Shickshock Mountains remained in the hinterland forming highlands from Spencer, Massachusetts, westward into New York State; but they were shorn of their crags, and only on rare occasions were the streams swift enough to carry silt into the swamps and to bury the accumulated peat.

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FIG. 16. Block diagram showing the main features of central New England during the Carboniferous period.

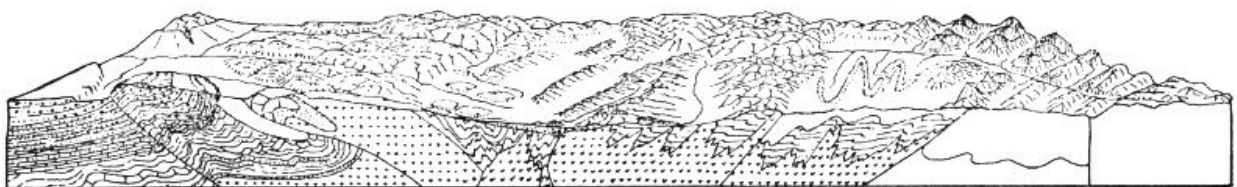


FIG. 17. Block diagram showing the main features of central New England in early Triassic time.

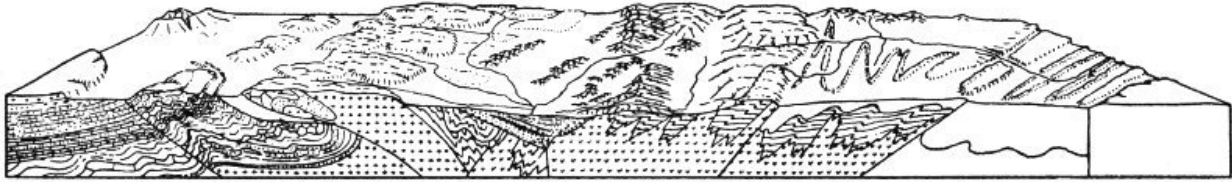


FIG. 18. Block diagram showing the main features of central New England during late Triassic time.

Torn and twisted as New England had been by the two previous disturbances, it was to suffer yet again. The entire northern section of the eastern coal swamps began to rise, and the movement spread southward through New Jersey, eastern Pennsylvania, Maryland, Virginia, the Carolinas, and Georgia. Granites insinuated themselves once more into fissures in the elevated landmass; the rocks were pushed outward from the raised block; and the sediments of the coal fields were thrown into folds which diminished in magnitude towards Ohio on one side and Cape Breton Island on the other. This was the Appalachian Revolution. When it was over, even the youngest sediments were interlaced with granite sheets and dikes; they were cooked hard in hot spring waters; and they were crumpled into close, long north-south folds. The landscape was changed completely: mountains had replaced the peat swamps; and from their summits alpine glaciers were plucking rock fragments which they dumped into the Boston basin. Streams, too, cut deeply into the mountainous upland, but there were no other local basins in which the fluvial debris could come to rest. 42

This was, in brief, the course of events which transpired in that era of geologic time called the Paleozoic. Twice as long as all ensuing time, the era was one of kaleidoscopic change, with placid seas, eruptive volcanoes, swift streams, and towering mountains competing for the lead roles in three rather similar historical cycles. When the Paleozoic era was over, the matrix of tough, resistant rocks was ready for the delicate inlaid design which was imposed upon it in the Triassic period.

There was nothing tranquil about Triassic time. While hot springs, born in the cooling granites, still oozed from rents in the mountainsides, a tremendous 100-mile-long rift tore through the east margin of the old Shickshock Mountain foundation. The rift was a clean break at some places, but elsewhere it was splintered and offset. Each northern sector of the break 43

invariably ended west of the beginning of a southern one, and the intervening rock is characterized by multiple fissures with more or less displacement of their walls.

The block east of the rift moved south and rose, while that to the west was depressed into a tilted and asymmetric basin. Mountain streams flowing eastward to the Atlantic were caught at the base of the rift, and a new set of torrents dashed down the west-facing scarp of the elevated block. After every cloudburst these new streams left their contributions of boulders in screes along the east side of the basin. The gravels steadily increased in thickness, covering the hills and valleys that furrowed the lowland floor. Much of the ancient relief still lies buried beneath the fill, but some of the eminences were exhumed one hundred and fifty million years later and have received man-given names like Mount Warner and Bernardston Ridge. As the basin subsided vertically for more than a mile, the mountain streams spread fans westward across most of its floor, restricting the contributions of the western rivers to a zone which is now less than two miles wide. The largest of the eastern rivers wore a valley three miles wide where it entered the lowland northeast of Granby.

Then volcanoes broke loose in the basin floor. Lava oozed through the sand west of the Notch in the Holyoke Range, and it frothed out of the openings or was blown violently from them. But by sheer persistence the rivers still dominated the scene as volcanic activity waxed and waned, and 1,500 feet of alluvial wash piled up around the volcanic cones. The energy of the volcanoes was ultimately spent, but for some time lava poured out of craters along a line extending southward from the main eruptive center, and from a second center which approximates the course of the Connecticut River from Sunderland to Turners Falls. It flowed westward into the middle of the 44 basin in a series of sheets until it was 400 feet deep; it pressed upward against the sand plains along the western hills; it surged east up the fan slopes where it ended in a frothy wall; and it spread southward from these two centers and from others to New Haven. The lava, now tilted, gives substance to the Greenfield Ridge, the Mount Holyoke and Mount Tom Ranges, and the long line of hills that pass through Hartford and Meriden.

Spectacular was this outburst in its time, and profound was its influence upon later scenery, but short was its duration. Before weather could redden

the lava surface, streams washed gravel over it; and only at the main centers between the Mount Holyoke Hotel and the Amherst-South Hadley road were the volcanoes able to hold out against the relentless activity of running water.

The block east of the rift continued to move southward and to rise, while the streams draining it entrenched themselves in an effort to remain at grade with the basin floor. The moving mountain mass pushed the lava flow up on end and twisted its eastern edge around, dragging it along to the south. The rock splinters which were formed in the process sliced the basin sediments into small blocks, some of which can be seen north of Turners Falls and also at the Holyoke Range. Ultimately the upward and southwestward movement along the rift piled the eastern blocks against the more westerly ones, pushing the west side of each eastern block up on the east side of the adjacent western one, and depressing its eastern side more deeply into the basin floor. The many fractures which were made weakened the basalt lava sheet along certain zones where, in recent time, the elements have worn the notches in the Holyoke Range.

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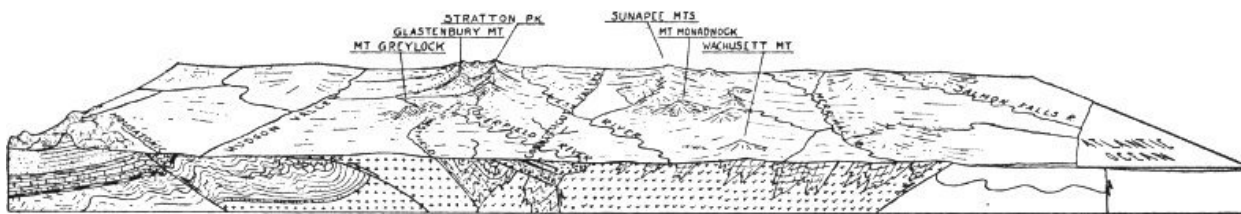


FIG. 19. Block diagram showing the main features of central New England at the opening of the Cenozoic era.

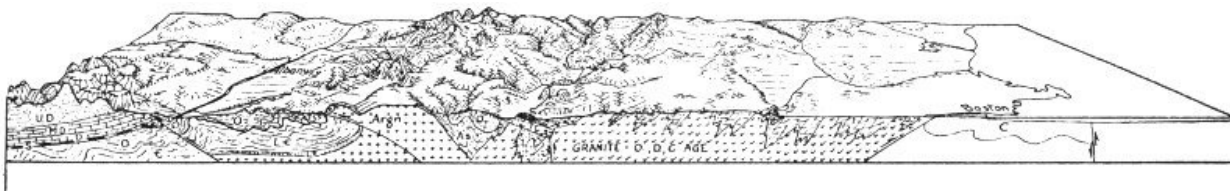


FIG. 20. Block diagram showing the main features of central New England at the present time.

Streams from the eastern highland stubbornly filled up the holes and planed off the raised blocks during the entire period of intermittent movement. In the midst of the tussle between earth forces and fluvial agents the volcanoes again broke into explosive eruptions, and volcanic ash filled many of the block-like depressions all the way from Granby to localities south of Holyoke. Then the fiery vents cooled, and the earth movements diminished in their vigor. But they left a mountain front so steep that talus and landslide deposits accumulated at its base near Mount Toby; and the block mountain range was so high that glaciers may have wreathed its summit. The mountain mass descended southward, and it was penetrated by at least one low pass northeast of Granby. 46

In the basin itself, alluvial fans encroached from the eastern mountain front, but out in the middle of the valley ephemeral playas and shifting lakes were numerous. Rushes fringed the lake shores; fish stocked their waters; and dinosaurs lumbered over the adjacent flats. The region was one of violent rains and seasonal droughts, of hot days and frosty nights—a semi-desert country lying in the lee of the Appalachian ranges, somewhat as the intermontane valleys of the West lie in the rain shadow of bordering mountains. Eight thousand feet of sediments poured into the Triassic trough while these conditions lasted, but the situation altered slowly as the Jurassic period dawned.

Throughout earth history, vulcanism and mountain-making have been spasmodic events; but so long as rain has fallen and water has run downhill to the sea, the unspectacular rivers have never relinquished their task of reducing the lands to the lowest grade on which water will flow. During all of the Jurassic and Cretaceous periods, and even into the Eocene epoch of the Tertiary, New England's rivers worked towards this end, and they came as close to attaining their goal as the restless earth has ever permitted them to do. The region from the Atlantic to the bases of the Green Mountains and the White Mountains was reduced to a broad, faintly terraced erosional plain. Not all of it was leveled, for Mount Wachusett, Mount Monadnock, the summits of Mount Greylock and Mount Ascutney resisted the wear and tear of the weather and of running water, and retained some of their original stature. At the headwaters of the streams the Green Mountain chain and the White Mountains also withstood reduction to the common level, forming the divide between St. Lawrence and Atlantic drainage. Such rivers 47

as the Merrimack, the West, the Deerfield, and the Farmington followed somewhat different courses than they do today, for some of the drainage heading in the Western Upland of New England flowed straight across the red-rock valley to the sea.

During Tertiary time the entire region rose vertically as a unit. The rise was intermittent, punctuated by long stillstands of the upland with respect to the sea. One of the earlier uplifts carried the land some 200 feet higher; and although the rivers maintained their courses, they deepened their valleys and ultimately widened them into broad, open plains far back towards their headwater reaches. In the resistant rocks on either side of the red-rock basin the valleys were sharp and well defined, but in the soft Triassic sediments the rivers cut wide swaths, nearly eliminating the low divides which kept them in their independent courses.

In Middle Tertiary time renewed uplifts occurred, and ultimately the strathed surface was elevated 1,800 feet inland at the Green Mountain divide. Once more the rivers started busily cutting down; but in a protracted stillstand, while the New England upland still lacked 700 feet of its present elevation, the Atlantic Ocean planed off the hills in southern Connecticut as far north as Middletown, and the Farmington River adopted a more direct route across the marine plain to the sea. Before the West, Deerfield, and Westfield Rivers could lower their channels to grade in the reinforced rocks of the Eastern Upland, a tributary of the Farmington worked headward along the poorly consolidated red rocks of the basin and diverted the waters of the northern streams into its own channel. This was the birth of the Connecticut River, and in late Tertiary time, the energies of the new-born stream were effectively expended widening the whole of the Triassic basin. Even some of its larger tributaries developed wide valley floors with steep walls in the hard crystalline rocks of the uplands. Only the lava flows and the buried 48 old-rock mountains withstood planation in the red-rock basin. The flows form such trap ridges as Greenfield Ridge, the Mount Holyoke Range, the Mount Tom Range, the Hanging Hills of Meriden. Exhumed mountains are typified by Mount Warner.

All of northeastern North America was raised to great heights in late Pliocene time, and the Atlantic Ocean withdrew at least fifty miles southeastward from the present shoreline. The rejuvenated rivers deepened

their valleys, forming narrow, sharply incised canyons like the gorges of the Hudson and the Saguenay; and the Connecticut made a deep groove in the lowland floor, cutting to depths which have been partly disclosed by drilling at the Calvin Coolidge Memorial Bridge and the Sunderland Bridge.

While the land stood in this high position, one winter's snow in the White Mountains failed to melt before the next began to fall. Snowfall accumulated upon snowfall, covering not only the White Mountains, but all of Canada and New England; and the Ice Age was here to stay more or less continuously for a million years. The ice piled up against the highest mountains and ultimately rose so far above them that it slid over their tops without attempting to detour around them. Its surface may have been 13,000 feet above sea level in northern New Hampshire, and its surface slope, which is estimated at 150 feet per mile, would give a thickness of 10,000 feet at Northampton. The continent yielded slowly under this great load, and it sank until all of the elevation gained in the Pliocene movement was wiped out, and more besides. The ice radiated from the centers of maximum accumulation—at first from the White Mountains, and then from northern Ontario, and finally from Labrador. The continental glacier crept southward to Long Island and Martha's Vineyard, where its front melted in the waters of the Atlantic as fast as new ice came up behind. It dragged and pushed and carried debris, only to dump it in a hummocky ridge, like a rampart, to mark its farthest advance.

At last the glaciers started to melt even faster than new masses moved down from the north, and the ice front began to recede 400 to 700 feet per year. The sea followed it, up the Hudson, up the St. Lawrence, in over the coastal lowlands for a short distance; and everywhere pounding waves made beaches at the water line. And in the path of its slow, deliberate retreat, the glacier left rock debris—boulders on the hills and in the valleys, boulders everywhere; all the landscape was marred and desolate.

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The ice had weighed the pre-glacial valleys down more deeply in the north than in the south. One such valley was the Connecticut Lowland, in which water gathered to overflow-height at Middletown. Thus Lake Springfield came into being, and it spread northward as the ice front receded. North of the Holyoke Range another lake formed, and this northern body of water has been named Lake Hadley. Streams flowed off the ice, off the hills—flowed

with unimpeded vigor, for there were no trees or grass to retard the run-off. Deltas grew out from the shores, and annual layers of clay settled on the lake bed.

The ice grew thinner, its area smaller, and its load lighter; and as Mother Earth lost her heavy burden, the land rose, more in the north than in the south. The differential rise decanted the water southward out of the lake basins, and the seas retired from the coastal lowlands. Old shores and sea beaches remained as flat terraces sloping gently southward. The rivers raced down the steep beach slopes to the old lake floors and sea bottom. They cut their channels deeply into the unconsolidated deltas and meandered back and forth over the flat, ungraded lacustrine plains, as if uncertain where to flow. They flooded the lands in the spring, leaving loose sand and silt for the winds to blow when the water was low. Sand dunes rose near the river banks at North Hadley, Sunderland, Hatfield, and South Deerfield; but the march of the dunes was arrested as post-glacial vegetation repossessed the land. It was at this point in the story that man found and settled the Connecticut Valley, becoming a witness to the geologic work of the river and an aid to the work of the wind as his plow bared the fertile soil to the elements.

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Interesting Places

Books and periodicals supply dinner menus for the hostess and list theatrical offerings for the habitué. Surely suggestions of places for a picnic or an evening drive are equally in order. Experience, some of it painful, soon reduces the number of pleasant picnic sites: poison ivy or a deceptive bog may linger in the memory and automatically eliminate some otherwise delightful spot. But places suitable to every taste lie within the Connecticut Valley or along its fringing uplands. Some are near the highways and others are on woodland trails; a few are interesting for their immediate surroundings and many because of their expansive view. Here is a landscape which can be appreciated without leaving or stopping the car; but there is a sight which can be relished only from a trail, or from a pinnacle accessible to the agile climber. Drives satisfy some tastes; but places to stop, meditate, and conjure up the past appeal to others. The Valley and its environs have something for every temperament and every mood.

Mount Lincoln in Pelham

Mount Lincoln is remote enough from highways to offer some measure of retreat, yet it is not discouragingly inaccessible. The summit rises about 300 feet above the nearest road, which lies a mile away by woodland trail. It is Pelham's highest eminence, and its height is enhanced by a fire tower which affords a magnificent view in every compass direction.

The gently undulating New England upland stretches off to the north and east for miles. The innumerable hills which compose it integrate to form a horizontal skyline, which suggests a flat erosional plane, originally formed at, or near, the level of the sea. To the northeast Mount Monadnock in 52 New Hampshire rises prominently above the general level, for its extremely resistant rock withstood reduction by weather and water more effectively than the weaker bedrock on every side.

The valley lowland begins but three miles to the southwest. The range of hills stretching away like beads on a string is the Holyoke Range. Mount Toby, Mount Sugarloaf, and the Pocumtuck Hills are the prominences in the lowland to the northwest. The lowland was eroded out of the New England upland after the land was elevated far back in Tertiary time, and the disintegrating rock was carried to the sea by the rivers. The hills in the lowland were left where the rocks resisted destruction more successfully than elsewhere, but they only approximate the level of the upland of which they were once a part.

Mount Lincoln and the surrounding hills are strewn with boulders. Every slope is dotted with large irregularly shaped rocks, many of which have smoothed facets marred by minute scratches. The boulders were left by the Great Ice Sheet when it melted off New England, and the scratches were made when the ice dragged the boulders over hard rock surfaces. These stones came down from the north, and among them you may recognize types which you have seen in the ledges around Orange and Northfield. Early Pelham settlers found the boulders as much in their way as the trees; so they burned or used the trees, and they piled the stones in long rows to fence their fields. Stone fences characterize all glaciated regions, and here they follow the roadsides for miles, reaching to the edge of the deposits in glacial Lake Hadley.

Mount Toby

“Let’s go to Mount Toby” usually means to go to the camp ground along Roaring Brook at the east base of the mountain, or to one of the sugar camps on the west slope, or to the Sunderland Caves at the north end. All of these places are worth knowing, but the view from the mountain top deserves at least one trip, and the wood road from Roaring Brook is replete with interesting sights.



PL. 6. *View of the Holyoke Range from Mt. Lincoln.*



FIG. 21. Map showing location of interesting places.

1. Davis pyrite mine
2. Plainfield manganese mine
3. Lithia spodumene pegmatite
4. Chesterfield tourmaline locality
5. Westfield marble quarry
6. Williamsburg galena vein
7. Hatfield lead mine
8. West Farms lead mine
9. Loudville lead mine
10. Westfield trap quarry
11. Bernardston magnetite mine
12. Gill dinosaur track quarry
13. Mount Toby

14. Sunderland Caves
15. Roaring brook
16. Whittemore's Ferry fish quarry
17. Mt. Sugarloaf
18. Leverett lead vein
19. Notch quarry
20. Northampton granite quarry
21. Titan's Piazza
22. Titan's Pier
23. Ox-bow lake
24. Smith's ferry dinosaur tracks
25. Varved clay pits
26. Mt. Grace
27. French King bridge
28. Mt. Lincoln
29. Pelham asbestos mine

The side road to Roaring Brook leaves the highway east of Mount 54 Toby just north of the old cemetery, and the camp site is on the west side of the Central Vermont Railway tracks. The gray rocks east of the tracks are part of the ancient mountains of Triassic time. Their lofty summits have been worn away by the unceasing activity of weather and running water, and they are now lower than the fans of waste which was discharged from the ancient valleys. Roaring Brook is continuing the work of erosion as it tumbles down from Mount Toby, and frost has loosened the great boulders that lie on the mountainside.

The rock along Roaring Brook is very different from that east of the railroad. It looks a great deal like concrete, with a large assortment of aggregate materials mixed in with the cement. The rock is conglomerate, a mass of coarse stones washed out of the ancient Triassic mountains, deposited at their base and in contemporary stream valleys, and then cemented during the ensuing ages. Many of the pebbles in the conglomerate cannot be found in the old rocks east of the railroad tracks. Actually these rocks change in character at different levels in the uplands of today, and still higher changes which were present in this mountain group during Triassic time have been destroyed, though the record of their presence has been retained in the fragments which compose the conglomerate.

The woodland trail starts up the mountain about 100 yards north of the picnic grounds. The rock beside it is red granite, and the streams of Triassic time flowed over it as they carried the gravel which now makes the Mount

Toby conglomerate. The latter first appears about 100 feet uphill, and it is virtually the only rock exposed from this point to the summit. Interspersed sandstone beds disintegrate easily and form quiet pools and basins in the adjacent brook; the pools end a few feet upstream where the water cascades over the edge of the next higher conglomerate stratum.

Mount Toby's summit rises above any other eminence in central Massachusetts east of Ashfield and south of Mount Grace near Northfield. From it the entire country to the south appears low and flat, except for the teeth of the Mount Holyoke Range and the long ridge extending southward from Mount Tom. A slope rises westward from the lowland to meet the edge of the flat New England upland along a line that passes through Shelburne, Conway, Goshen, and Granville. East of Toby this same upland comes so close that it seems but a step across to it.

55

Many peaks may be seen rising above the New England upland. The one far to the east is Wachusett. Up there to the north-northeast are Monadnock and Mount Grace. Over in the northwest are Stratton and Glastenbury in Vermont, and much nearer and lower is Bald Mountain at Shelburne Falls. Mount Greylock, the highest point in Massachusetts, is almost due west.

The lowland was excavated after the New England upland was elevated, and the main features which distinguish the present landscape were carved out before the end of the Miocene epoch of Tertiary time. The high points which surmount the upland are monadnocks which, like their prototype Mount Monadnock, successfully resisted the ravages of time and New England's changing but rigorous climate.

The Sunderland Caves

The Sunderland Caves are on the northwest side of Mount Toby, just a short walk and an easy climb from State Highway 63. They penetrate a cliff made of conglomerate overlying a shale which accumulated in a Triassic lake. The shale makes the floor of the cave. Joints, forming a right angle with the cliff, break the conglomerate into giant blocks. Frost, smooth shale surfaces, and gravity have caused the two end pieces to creep away from the other

conglomerate blocks. The second block from the end has fallen against the end block, forming a high-roofed cave about 100 feet long.

Directly southwest of the lower entrance to the cave, the shale beds are highly distorted along the borders of a trough-like mass of angular conglomerate or breccia, in which boulders up to six feet in diameter are numerous. It is believed to be the record of a Triassic landslide, which avalanched down the mountain front immediately to the east, and into the old lake at the mountain base. It plowed up the clays in the lake bed, carried some of them away, and furrowed the others into the crumpled forms that are clearly visible along the path to the caves. 56

Mount Sugarloaf

Mount Sugarloaf does not offer Mount Lincoln's retreat from crowds nor Mount Toby's expansive landscape, but it is accessible, and it provides an unrivaled view of the valley between South Deerfield and the Holyoke Range. Its red sandstones and conglomerates rise almost sheer for 500 feet above the Sunderland-South Deerfield road. On the northwest and southeast sides the cliffs are determined by nearly vertical joint planes. During the Ice Age, the southward-moving glacier plucked away the loosely attached blocks facing the South Deerfield and Sunderland sections of the lowland, leaving Sugarloaf as a remnant between the joint surfaces.

The great bites which the meandering Connecticut River has taken out of the lowland are visible east of Sunderland village and south towards Hatfield. Each arc in the edge of the scalloped flood plain is the extremity of a meander loop which the wandering river carved in its bank and then abandoned by breaking through the narrow base or tongue, as it did at the Northampton ox-bow.

An area of low, rolling, sandy hills extends through the pine woods for a mile southward from South Deerfield. The hills are dunes which formed when the Connecticut was picking its channel across the newly exposed and barren bed of glacial Lake Hadley.

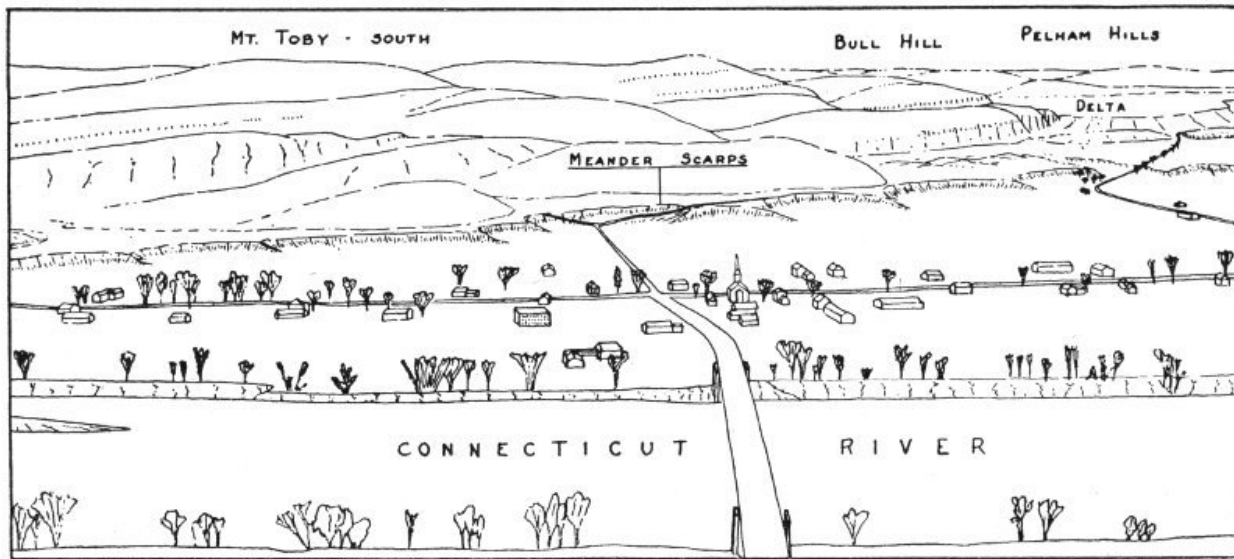


FIG. 22. Meander scarps form a margin to the Connecticut River flood plain at Sunderland.

The panorama from the west side of Mount Sugarloaf centers about the deep gorge of the Deerfield River. The top of the gorge widens out into a broad strath and affords a glimpse of the more remote upland. The river, emerging from this canyon during post-glacial time, built a huge delta into glacial Lake Hadley, and much of the delta still remains in the terrace which is utilized by the Boston and Maine Railroad as it descends into Greenfield.

Turners Falls

Rushing water has a fascination which was frankly recognized by the highway engineers who made the parking place facing the Connecticut where Route 2 passes along the north side of Turners Falls. Here the river drops over a series of sandstone ledges into a deep and narrow channel at the east base of the trap ridge. Waterfalls are not common in rivers flowing through lowlands; they indicate disturbances of normal stream development and sometimes change in course.

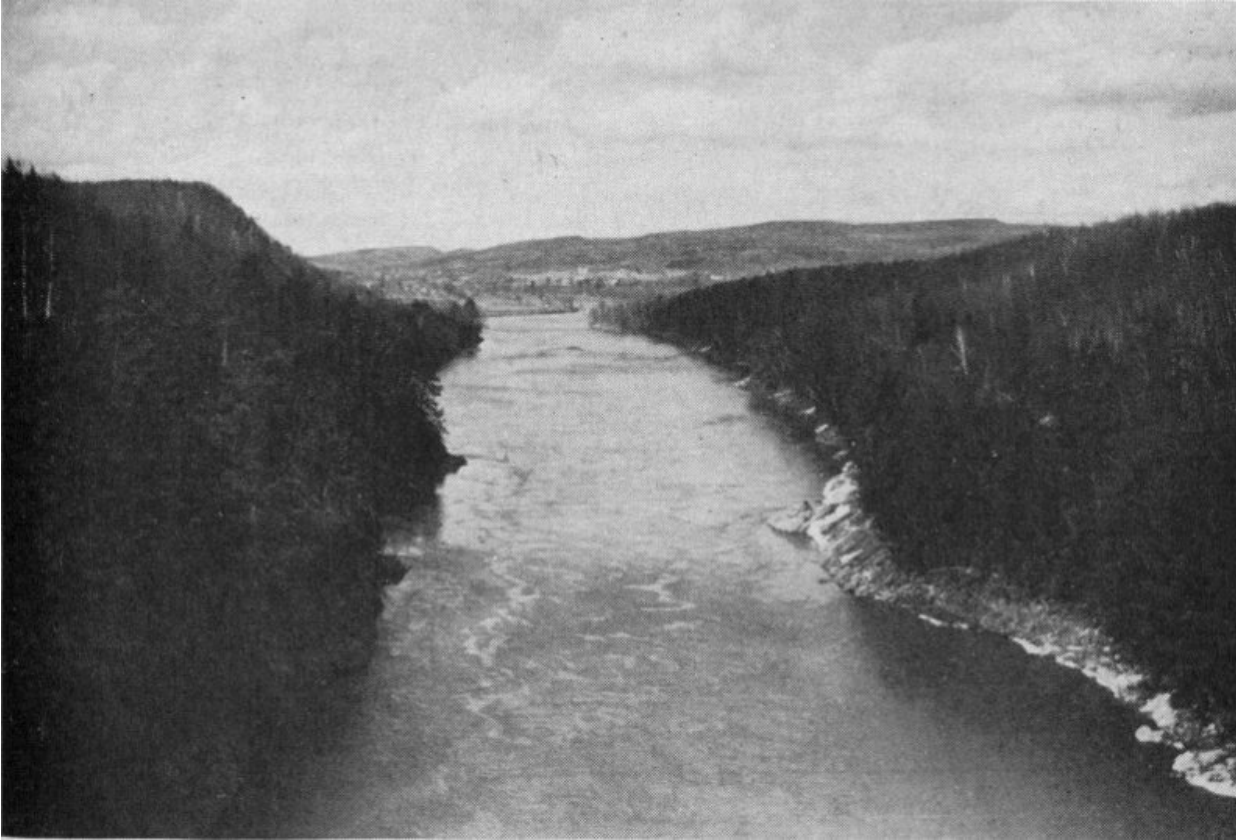
The Connecticut Lowland is old, but its ancient drainage lines were buried by the deposits left in glacial Lake Hadley. The river's present course was

established upon these lacustrine sediments, and the inner valley plain is excavated in them. Before entrenchment took place, the south-flowing reach of the river above Millers Falls was deflected westward across the lake plain by the delta of Millers River. It was turned southward once again by the trap ridge near Turners Falls. The river soon cut through the unconsolidated lake beds and found that it was out of its pre-glacial channel. The delta of Millers River had diverted the water from the old rock valley beneath the Montague sand plain, across a rock divide, and into the pre-glacial valley of Falls River. The lake-fill in Falls River has been almost completely removed, and Turners Falls now mark the spot where the Connecticut pours over the bank and into the channel of its pre-glacial tributary. The falls have receded upstream several hundred feet and have cut a deep gash in the Triassic rocks.

PL. 7. *Gorges, in highland and lowland alike, were formed when the rivers were superimposed on coherent rock.*



a. *View of the Deerfield River gorge emerging on valley lowland as seen from Mt. Sugarloaf.*



b. View of the French King gorge as seen from the bridge.

Turners Falls are the product of a series of coincidences. First, the ice sheet and Lake Hadley buried all established drainage lines and forced the streams to adopt new routes over the bared lake bottom. While the lake existed, Millers River threw a weak obstruction in the path of the Connecticut, diverting it to that part of the lowland where one of its pre-glacial tributaries had excavated a slender rock gorge along a fault plane. The river washed the lake deposits out of the gorge, exposed the old bank of Falls River, and was busily cutting a new gorge back into this bank when the dam was constructed and its erosive activities were suddenly arrested.

59

The French King Bridge

The highway from Greenfield to Athol and Fitchburg passes Turners Falls and crosses the Connecticut River near Millers Falls by way of the French King Bridge. Here the roadway is more than 130 feet above the water level.

A picnic ground and parking place at the west end of the bridge make it a particularly attractive place to stop and enjoy the view upstream towards Northfield.

The river occupies a narrow rock gorge for a mile north of the bridge, but at that point the valley widens out. This entire section of the river's course was established on the old bed of glacial Lake Hadley; but after the unconsolidated deposits were washed away, the stream found itself flowing along the weak contact between the Triassic conglomerate on the west bank and the metamorphic rocks of the highlands on the east bank. The river deepened its channel on the weak contact zone and made the scenic cut over which the bridge was built.

The pre-glacial valley lies beneath the sand plain east of the river. Millers River crosses this old valley between Millers Falls and its confluence with the Connecticut, at the east end of the bridge. The rapids at the junction can be traced to the ridge of crystalline rock between the east bank of the present Connecticut and the west bank of the pre-glacial Connecticut. The resistant ledge forms a barrier which Millers River has not yet eroded to its grade.

The conglomerate beds on the west wall of the gorge dip steeply eastward towards the river and end against the crystallines. The beds were originally laid down with a gentle westward inclination. They were tilted steeply in the opposite direction against the crystallines by faulting, which elevated the ranges and pressed down the adjacent basin during Triassic time. 60

Titan's Piazza and Titan's Pier

Not so long ago, giants and the devil received the credit or the blame for such oddities in nature as rock-masses broken into six-sided columns. Ireland has its Giant's Causeway, and Yellowstone National Park its Devil's Post-pile. Titan's Piazza and Titan's Pier were likewise attributed to activities of the leader of fallen angels and were given names appropriate to such an origin by the early settlers. Dr. Hitchcock, in characteristic fashion, undertook the task of correcting the errors of puritanical psychology by renaming these places during one of the early Mountain Day trips from

Amherst College. The entire college body sojourned to the west end of the Holyoke Range to hear the cliffs renamed and their true nature explained.

Devil or no devil, those huge columns had a hot origin. The dark rock in them is part of the main lava sheet which stretches across the valley in the Holyoke Range and swings southward in the Nonotuck—Mount Tom Range. The lava poured out of a series of volcanoes which were strung out along a fissure about three miles to the east, and the molten mass had a temperature of 1200° to 1300° C. The hot lava radiated its heat to the sandstone below and to the air above; and, as it cooled, it contracted like any other substance. The shrinkage was so great that series of cracks formed in regular pattern, with each crack perpendicular to the cooling surface. The stresses producing the fissures were equal in all directions and would have made circular cracks and cylindrical columns; but cylinders have non-cylindrical spaces between them, and the pattern in which the columns are most nearly cylindrical and yet completely occupy all space is hexagonal. So contraction broke the lava into hexagonal columns perpendicular to the cooling surface. The columns are parallel where the lava floor is regular but are curved or radial where the floor is rolling.

PL. 8. *Trap ridges, near and far.*



a. View of Titan's Piazza at Hockanum showing the columns resting upon the gently inclined sandstone.



b. View of the Springfield lowland from the Westfield marble quarry. The Wilbraham Mts. appear in the distance. The trap ridge extends through the middle and is breached by the Westfield River.

The columns on Titan's Pier lie across the river from the Northampton-Holyoke road in the narrow gap at Mount Tom station. The basalt flow is inclined 15° southeastward, and the columns stand perpendicular to the surface—hence they are inclined with respect to the water level. Doubtless the devil docked his boat on the gently inclined rock surface of the cove on the downstream side of the pier. 61

Titan's Piazza is situated east of the road to the Mount Holyoke House. It is an extremely narrow ledge backed by a stockade of columns. The front of the piazza is literally strewn with wreckage from the house, for a slope over 100 feet high is covered with angular pieces of basalt which have fallen from the back wall. The lower ends of the columns break off into shallow hexagonal saucers with the concave sides up. Many have slid down the slope, to the delight of the birds that bathe in them. Higher up the cliff, the

saucers become deeper, and towards the top the columns scale on into bullet shaped masses.

Westfield Marble Quarry

Anyone who drives westward on the Jacob's Ladder route from Springfield passes first through the open, rolling country of the Connecticut Lowland. Hills are in sight, but they seem remote until he leaves Westfield, and there the upland rises before him like a 900-foot wall. The road uses the gateway cut in the wall by the Westfield River, and the drive westward towards the headwaters of the river is one of the best known scenic attractions in western Massachusetts. But a greater treat awaits the person who will venture southward on the road along the Little Westfield River. It follows the canyon brink about 500 feet above the stream. Near the hilltop, a side road 62 turns north to the Westfield Marble quarry, which provides a vantage point overlooking fifty miles of country to the north, east and south.

The Westfield River meanders eastward across the flat lowland. Its banks are terraced, each level cut in the lake beds or in the delta which the river built in glacial Lake Springfield. The scalloped margins of the terraces are the extremities of meander loops which developed when the river was not entrenched as deeply in the unconsolidated deposits as it is today.

The flatness of the twenty-mile strip of lowland is impressive, for it ends only at the Wilbraham Mountains, eight miles east of Springfield. Beneath the lowland lie soft and gently dipping sandstones and sandy shales, capped by a thin veneer of lake clays and river sands. The shales are the youngest Triassic beds remaining in the region, and they outcrop between Thompsonville and Windsor Locks, Connecticut. Younger shales above them succumbed to Tertiary erosion.

The Wilbraham Mountains are granite and gneiss which formed the roots of the ancient Triassic ranges. Their present accordant summits are a tribute to the leveling activities which running water performed on a quiescent land, whereas the deep V-shaped valleys incised in the level summits record uplift and quickened erosion in Tertiary and glacial time. Indeed, the lowland itself

owes its existence to the power of rejuvenated streams working on non-resisting rocks.

The Holyoke and Mount Tom ranges are visible far to the northeast, and a chain of low hills connects Tom with the ridges between Hartford and Avon, Connecticut. These linear hills surmount the lowland because they are made of basaltic lava, which is better able to resist the rain and the weather than the sandstones and shales above and below. Scattered flat-topped hills between Southwick and Granby are sheets of basalt-like rock called diabase, which was inserted between a sandstone roof and floor. Nowhere can one better appreciate the highly individualized imprint which each geological element has made upon the central New England landscape. 63

The Old Lead Mines

The colonial period in our nation's history was characterized by an ignorance of its mineral wealth and a dependence upon Europe for most raw materials, especially essential metals. During the War for Independence, European supplies were cut off, and Yankee ingenuity had to make the most of local deposits of metallic minerals. It was not long before mines were in operation on several lead veins in the Connecticut Valley, yielding a supply of lead for the duration of the war. But the mines were small, and most of them were soon abandoned, remaining only as historical sites, or as collecting localities for the mineralogist. Five of these old deposits are still accessible: four lie west of the valley at Loudville, West Farms, Hatfield, and Williamsburg; an important one is situated east of the valley at Leverett. All are very similar in geology and mineralogy, yet each possesses its own individuality.

The Loudville vein was worked intermittently as late as 1861. It follows a fault fracture between walls of gneiss, but at the southwest end of the vein some of the minerals are disseminated through the Triassic sandstone and conglomerate. This feature indicates that the sediments were unconsolidated at the time of mineralization. The fault zone resembles many analogous fissures which give forth hot mineral-bearing waters in the Basin and Range region of Nevada, for the charged waters have impregnated the sands which cover the fissures.

The Loudville vein contains numerous well-formed crystals. Barite was the first mineral deposited, and it is readily recognized as a heavy, easily scratched substance with one set of cleavage planes at right angles to two others. Gray metallic galena and resinous cleavable sphalerite or zinc blende occupy much of the space between the barite plates. Hard hexagonal crystals or white masses of quartz coat and even replace the barite plates. 64 Spike-shaped crystals of calcite and siderite line many of the cavities and coat the quartz. A patient search will be rewarded by the finding of other minerals, including pyrite, chalcopyrite, pyromorphite, wulfenite, malachite and azurite.

The old shaft has been closed and the tunnel at the river level has collapsed, hence the only exposures are in the open cuts. The most interesting is the one at the south end, where the barite plates are disseminated through the sandstone.

Another series of pits can be found easily about 100 yards west of the road to West Farms and about one mile north of the Loudville deposit. The vein attains a maximum width of three feet between walls of gneiss, and it occupies a fault fracture which seems to be continuous with the Loudville zone. Included in the vein are many fragments of a black phyllite resembling the Leyden argillite, as well as pieces of gneiss. The minerals are identical with those found in the Loudville deposit, but the specimens of quartz, galena and sphalerite are more spectacular.

The Hatfield vein occurs in a rock of igneous origin, known as the Williamsburg granodiorite. It is exposed at the west edge of the valley, about 200 feet from Federal Highway 5, at the northern limit of the settlement called West Hatfield. The workings are full of water, and the very thorough mining activities carried on by mineral collectors and by Smith College and Amherst classes have reduced the waste pile to negligible proportions. Early collections and records reveal that the vein is essentially like those farther south. At Hatfield, West Farms and Loudville the fractures do not parallel the systems in the Triassic sediments and lavas.

A galena-bearing vein outcrops near the Whately-Williamsburg town line at the north end of the Northampton reservoir. Leyden argillite forms the walls of a fault fissure. Barite is absent from this vein, but fine quartz, pyrite and

chalcopryite coat the walls. Coarse comb quartz encrusts the older minerals, together with breccia fragments and cubes of galena. The vein is remote from the valley and differs in mineralogy and texture from those within the valley. Other deposits like it have been found in the nearby hills.

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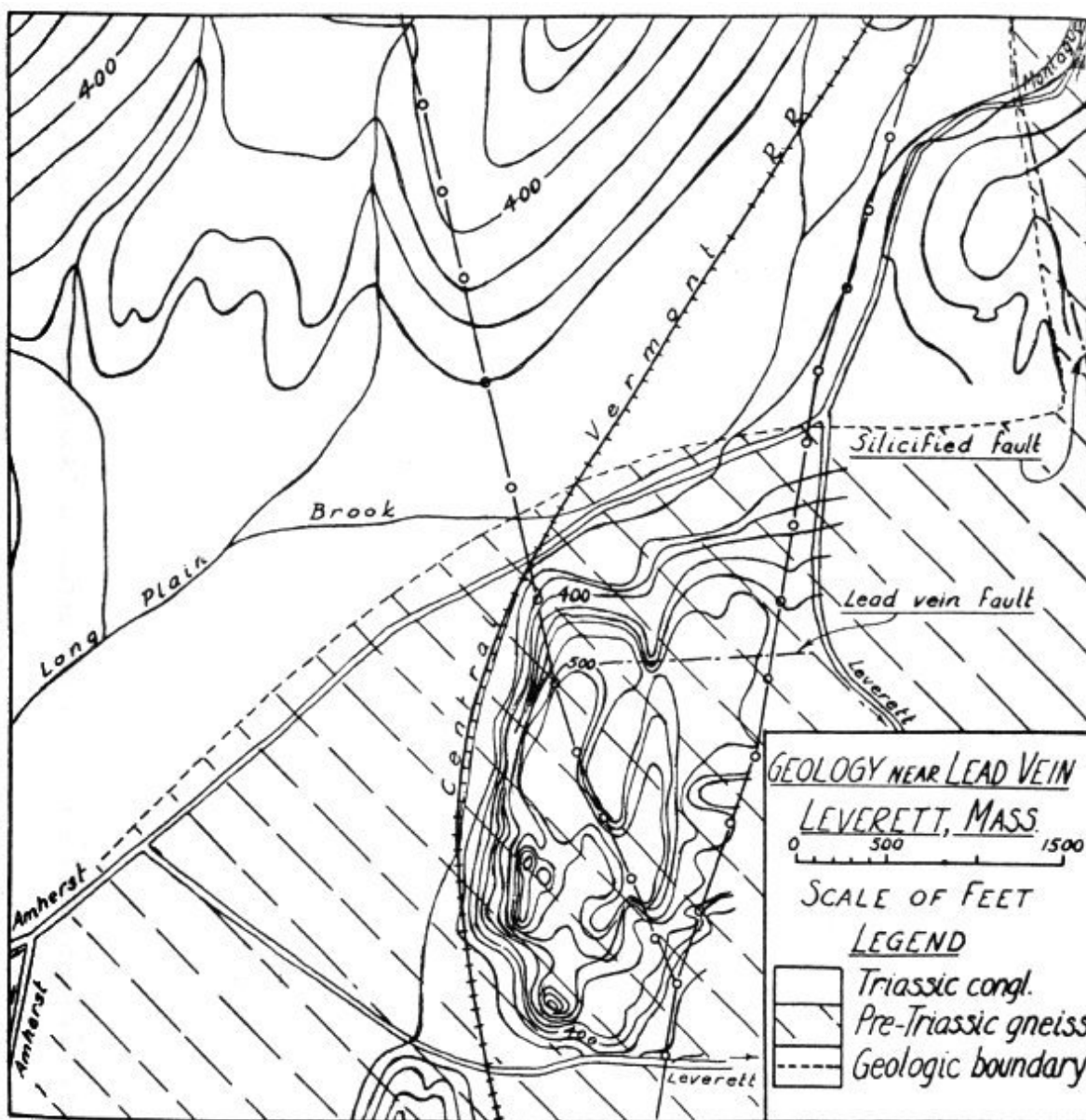


FIG. 23. *Geologic map of the region in the vicinity of the lead veins near Leverett.*

1. Only barite in these veins
2. Best mineral locality
3. Best place to see fault
4. Slickensides and tension cracks show direction of movement on fracture making opening for vein
5. Best place to see quartz replacing barite along crush zones in vein

The Leverett lead vein is the most interesting of the group because it is so well exposed that the nature of the vein system is admirably displayed. The deposit lies in a series of overlapping, nearly vertical fault fissures in gneiss. Slickensides and tension cracks on the walls of the veins indicate that the movement was nearly horizontal from northeast to southwest. Wherever a fracture begins to narrow and close up, another begins to widen and become conspicuous a few feet to the northwest of it. Several different fissures appear along the length of the mineral zone. 66

The same minerals are present as are found in the Loudville, West Farms and Hatfield veins, but barite is more abundant and quartz less so. Numerous cavities lined with crystals indicate that the vein formed close to the earth's surface. Apparently the minerals entered fractures situated near the front of a range that bordered the basin in Triassic time. A fault zone so located would lack the great thickness of rock that once lay over the gneiss and would be free from any appreciable overburden of outwash within the Triassic basins.

The Dinosaur Tracks Near Holyoke

People still write from as far away as the Rocky Mountains to ask if the dinosaur footprints beside the Connecticut River are still in place. They are. Anyone may see them in that triangular area between the Boston and Maine tracks and Federal Highway 5 about one-quarter mile north of the entrance to Mountain Park. Marvelous as their preservation from the assaults of man may seem, it is even more amazing that they should have been preserved in rock at all.



PL. 9a. *The dinosaur track preserve at Smith's Ferry near Holyoke.*



PL. 9b. *Varved clays or calendar beds on river bank south of Hadley.*

The footprint beds are shaly sandstones about thirty feet above the Granby tuff—a bed of volcanic ash formed in late Triassic time. They are inclined 15° towards the river, and even the higher strata which form the “Riffles” are footprint-bearing. The sandstones are ripple-marked, and they contain worm trails and a few casts of salt crystals. Some beds have impressions of reeds. The footprints range from half an inch to ten inches in length, and the stride of the larger animals was from five to eight feet. Most of the tracks are headed up the present slope, but a few are going in the opposite direction. 67

The sandstones were laid down as almost horizontal beds of sand which were occasionally covered and rippled by moving but rather shallow water. Rushes and reeds, which have left stray impressions in the rock, grew seasonally in the shallow waters, but in between the periodic rains and floods, the local climate seems to have been quite dry—and probably very warm. The sedimentary record suggests a lowland much like some of the tropical valleys in the West Indies, lying in the rain shadow of adjacent mountains.

The large tracks invariably have impressions of three toes. Even a careful search does not disclose the double tracks which would have been left by quadrupeds, and for years the bipedal impressions were called bird tracks. But birds have spurs which leave a mark behind the middle toe; these animals had no spurs and were not birds, but reptiles. Gregarious animals generally follow a leader, and only an occasional individual strays from the beaten path. The tracks at Holyoke suggest that these Triassic reptiles traveled in small herds.

The modern silts of the Connecticut Valley are not a good medium for the preservation of tracks because they lack coherence, and they drift with the wind as soon as they dry. Clays in a region of seasonal aridity are different. They are baked hard in the hot sun, and the water contains dissolved mineral matter which crystallizes in the clay and sand as the water evaporates, cementing the particles into a rock-like aggregate. Impressions in this sort of mud are preserved. The Connecticut Valley had the right kind of sediment and climate in Triassic time; impressions of salt crystals can be found 68 in the shales where the tracks are clearest, not only in this locality but elsewhere in the neighborhood of Holyoke and West Springfield. These precipitated salts helped hold the clays together until they were effectively buried, and afterwards a firmer cement was deposited around the particles.

Footprints are known near South Hadley, at Turners Falls, at Gill, and along the highway to the French King Bridge; but they do not portray the character of the animals, their habits and the mode of preservation of their tracks as effectively as the tracks north of Holyoke. Certainly no occurrence of tracks *in situ* is as accessible, and no geological exhibit in New England has received so many visitors.

Fossil Fishing

Many years ago men were excavating to lay a foundation for a waterwheel at what was Whittemore's Ferry, three miles north of Sunderland. They made a catch of some of the most ancient fish ever taken in New England, but the fish were petrified and did not put up a fight.

They were found in layers of black shale, in which skeletons and carbonized tissues were well preserved. Of the five genera identified, all but one were ganoids.

The shale accumulated as mud on a Triassic lake bottom, and it was covered by a coarse stream-laid gravel which has since been cemented into rock. The mud was not eroded by the stream which washed down the gravel, and the pebbles are not even impressed into the underlying shale. Apparently the fish perished as the waters evaporated and the lake became a playa flat. The limited variety of fish suggests that the connections with outside regions were restricted, and that living conditions within the basin were rigorous. The situation may have been like that found in the fresh water lakes along the western margin of the Great Basin in Nevada and eastern California.

Other lake deposits with fish remains appear at different levels from 69 Whittemore's Ferry up to the Sunderland Caves. Each is covered by a conglomerate layer, and at each place the lake clays had been partially hardened before the pebbles were washed over them. Seemingly dry alluvial plains followed transient lakes in kaleidoscopic but cyclic succession.

Calendar Beds

The lakes in which the fish lived and died date back to late Triassic time. Much younger were the lakes that followed the continental ice sheets, and in many valley localities these younger water bodies have registered their brief span of geologic life. For they, too, were settling basins for clays, which are characterized by annual depositional bands like the growth rings in a tree. These clays may be examined best in the clay pits at any of the brickyards, particularly at South Hadley Falls, or beside the high river banks rising above the Connecticut flood plain just south of Hadley.

The clays consist of alternating thin, dark, fine bands and thick, light, coarser ones. The coarser bands are sandy, and some of them have ripple-marks. The total number of pairs of beds is the number of years that glacial Lakes Springfield and Hadley inundated the valley, but it is not a simple matter to count them. Actually the lake bottom deposits are but a small fraction of the total volume of material brought to the lake. Lake shore deposits and deltas

grew outward and buried the bottom deposits after a few hundred years had passed. Thus in the pits at South Hadley Falls, the clays rest upon glacial gravels, and a scant hundred layers intervene between them and the sands above. Shore encroachment is not encountered at Hadley, but the shallow depth of the present water table hinders deep exploration, and the Fort River has removed many of the upper beds.

Long winters result in thick winter deposits, and heavy spring floods cause thick sand layers. If the winter of any year is long at Northampton, it is 70 usually long everywhere in New England; and if the Connecticut has floods, most neighboring drainage systems have them, too. In this way, similar layers, or similar successions of layers, are formed at different places at the same time; and the lake deposits at White River Junction, Deerfield, Hadley, South Hadley Falls, Chicopee and Springfield may be matched and dated with respect to each other. The complete record in the valley shows that, in the vicinity of the Holyoke Range, the lake came into existence about 18,000 years ago.

In each of the clay pits every set of lines exposed on the working faces represents a year, and the deposit as a whole is a calendar—in fact, it is also a thermograph for part of the region's post-glacial history. Some bands at South Hadley Falls and along the Hadley river bank are highly distorted, and the distorted layers are planed off smooth. Spring sand covers the distorted beds. The disturbance can be attributed to ice which froze to the lake bottom and dragged the clay layers as it expanded and contracted with changes in temperature.

Locally the clays are exceptionally hard about certain centers, forming clay stones or concretions. A willow twig or shell or some organic substance is commonly present at their cores. Groundwater has deposited calcium and iron carbonate about the adjacent clay particles and cemented them into rock.

The Holyoke Range

For years it has been a popular outdoor pastime to “walk the Range.” The distance is neither so great nor the route so rugged that it cannot be covered

in the course of an afternoon, even if ample time is allotted for stops at the many lookouts. The latter provide ever changing views of the valley from Greenfield and beyond, to Meriden, Connecticut. The buildings in Hartford are easily visible on a clear day. The trail follows the crest of the Range closely and only rarely leaves the basalt lava flow. The trip is somewhat less arduous from west to east than it is in the opposite direction, and the view from Bare Mountain is a pleasant climax for those ending their hike at the Notch. 71

At the toll booth the trail leaves the road which ascends to the Mount Holyoke Hotel and angles upward along the mountain slope. Overhead the dark basaltic lava columns rest upon red and white Triassic sandstone, and the path soon crosses the contact between the two types of rocks. A short distance above the contact the trail takes advantage of a col and climbs to the top of the ridge. Down the steep southeasterly slope Mount Holyoke College appears in the distance through a screen of oak trees.

The remainder of the climb is gentle, and soon the path enters the clearing around the hotel. The view is arresting. The Connecticut emerges from behind Mount Sugarloaf, wanders through the Hadley fields, flows through the watergap just west of Mount Holyoke, and disappears far to the south beyond Springfield. Northampton is spread out below. Automobiles on the Hockanum Road look like so many moving dots. The hills between the Range and South Hadley are made of volcanic ash and lava; many have pipe-like cores which were the necks of ancient volcanoes. Off to the east are higher points on the Range which lie on the route to be followed.

The path continues along the crest of the range and descends gradually to the toll road level at Taylor's Notch. Here it is on sandstone, and the lava-sandstone contact is exposed on both sides of the gap. Sandstone cliffs rise fifty feet high a few yards down the road; and the fine arenaceous character of the rock and its bedding are visible at some distance.

The trail climbs steeply from this col and soon skirts the edge of an abrupt cliff, in which are carved the initials of many hikers who paused on the ledge to rest and to enjoy the panorama. Eastward the path might well serve as the model for a roller coaster in an amusement park. "The Sisters" are a series of hills separated by sharp, deep valleys; and no sooner has one attained a

summit than a drop down the other side is in order. Abrupt 30-foot cliffs trending north and south form many of the valley margins; they are smooth joint surfaces where the rock is weak, and where blocks were plucked out by the great Ice Sheet. Each of “the Sisters” has a cleared lookout which affords a new picture of the Hadley-Deerfield lowland to the north. 72

The last lookout is some distance below the succession of summits, and it affords a view to the east. A cliff drops 200 feet vertically, and about one-quarter of a mile farther east other cliffs of red-weathering basalt face towards it. Almost all of the broad, low gap between the cliffs is underlain by a complicated mixture of volcanic ash, agglomerate and irregular lava flows. The cliff itself is thick columnar basalt, and at its base is a coarse sandstone. But the sandstone is thin and disappears in the depression, whereas the agglomerate and lava become very thick and extend northward to the top of “Little Tinker” and the “Tinker.” They are part of an ancient volcanic cone, buried in sandstone both to the east and to the west. Flow structures in the main sheet move away from this center, which is believed to have been one of the volcanoes on the line which supplied the basalt for the great lava field.

In the depression, the trail winds between hills of twisted lava and consolidated agglomerate. When the trees are leafed out and the surrounding hills concealed, it is easy to imagine oneself on the slope of a Pacific volcano. The trail divides at the lowest point in the depression, and the less used fork goes north to the Bay Road at the northern base of the Range. The other fork ascends Mount Hitchcock, and at a slight elevation above the low flat it crosses from the agglomerate to the Holyoke basalt sheet.

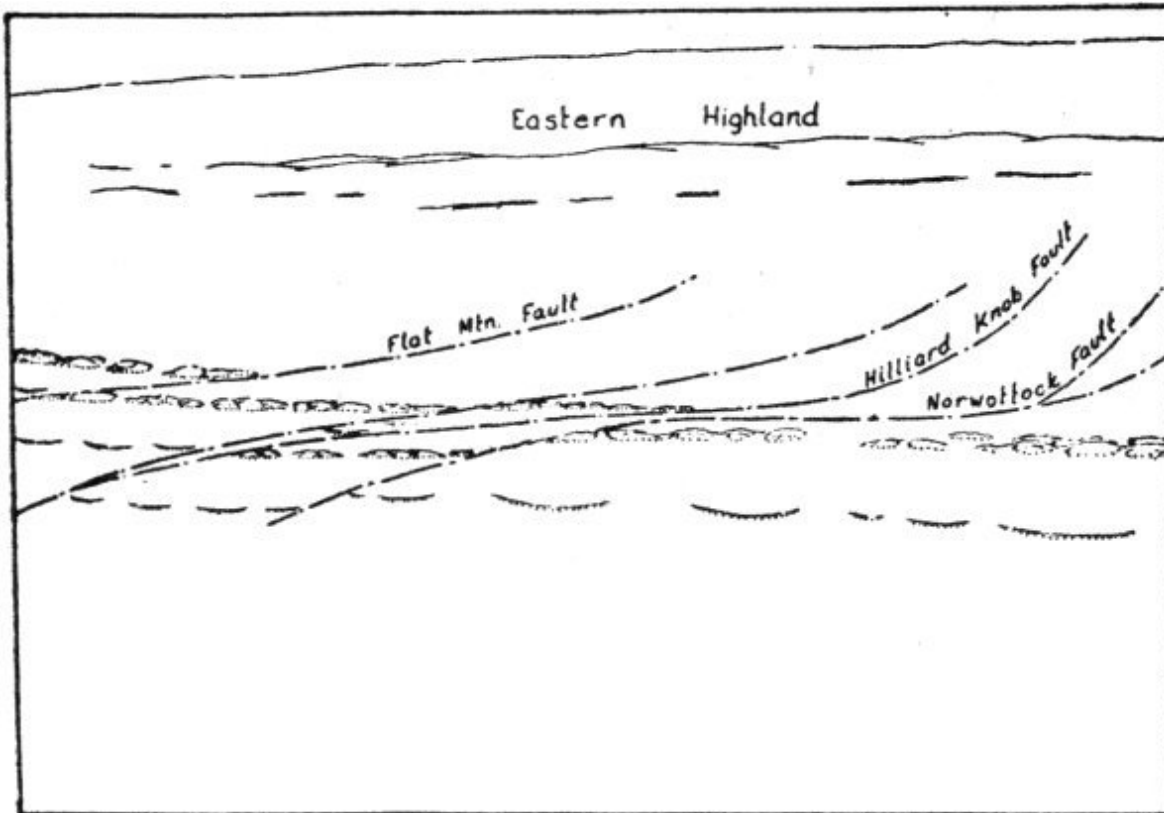
The best lookout on the Range between the Mount Holyoke Hotel and Bare Mountain is on top of Mount Hitchcock. A side trail leads out to a promontory, from which one may peer along the face of the Range, look down upon the “Tinker” and “Little Tinker,” and gaze over the lowland which the Connecticut has excavated in the New England upland 73 through the long course of geologic time.

The east slope of Mount Hitchcock descends steeply, and many a hasty hiker has made the trip in less time than he intended. The path drops to a flat

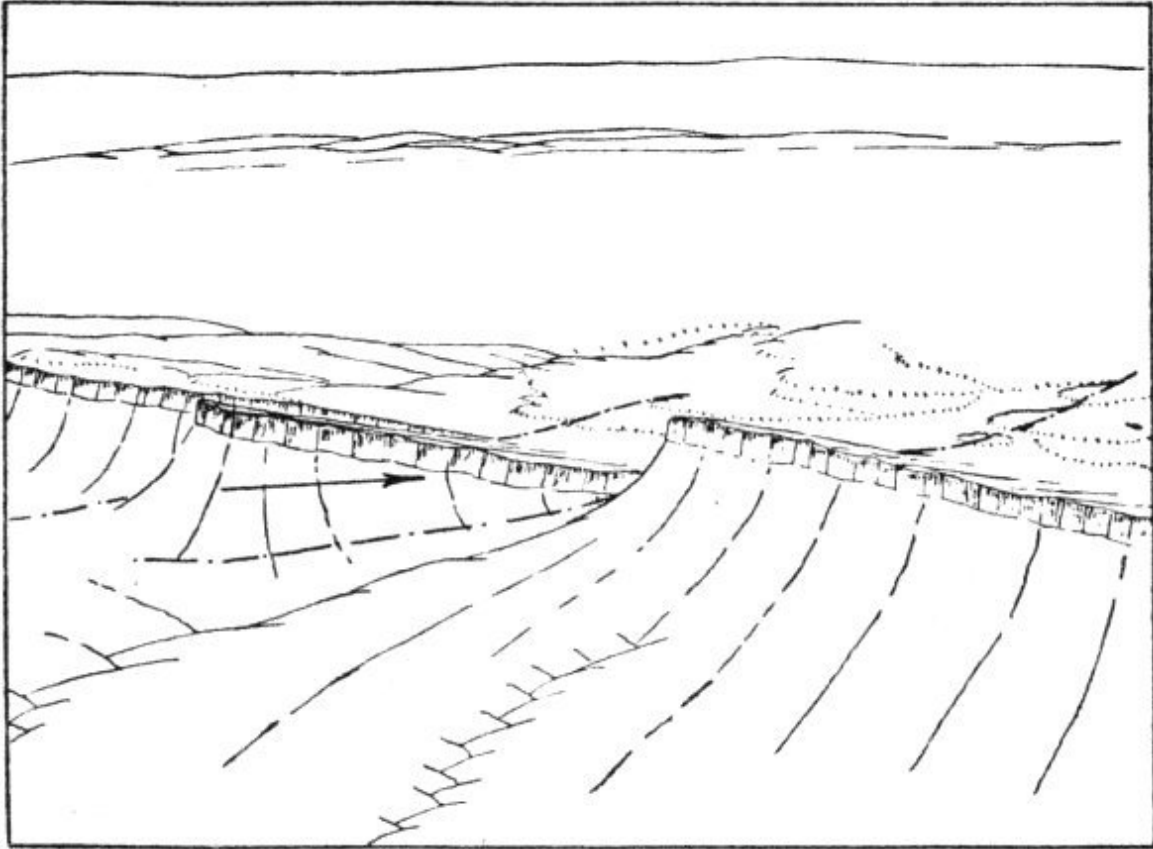
which measures about 1,000 feet across, and in which the sandstone lying below the lava sheet is sporadically exposed. Here the thick basaltic lava has been worn away; and erosion ceases both east and west at conspicuous fracture surfaces which locally become fault planes.

Beyond this low notch the trail leads irregularly upward and eventually comes out on Bare Mountain. The top is bare indeed; even scrub oak is absent from the summit. The long south slope of the Range is clearly visible, and to the west is the Mount Holyoke Hotel where the hike started. The Mount Tom Range, with the Connecticut River at its foot, is just to the left. Due south are the towers of Mount Holyoke College and the cities of Holyoke and Springfield. If the day is clear, the tall buildings of Hartford appear in the far distance. Six hundred feet directly below, the highway goes through the Notch, and across the road is the trap quarry in Notch Mountain, which supplies the crushed stone for the local highways. The face of Notch Mountain lies north of the main line of the Range because the basalt sheet has been displaced northward between fault planes that bound the eminences on either side. The notches utilized by the highway and by the power line are due to facile erosion of the crushed rock along the fault planes. Farther to the east, Mount Norwottock rises to the greatest height in the Range, and the view from its summit is at least the equal of that from Bare Mountain. The Hadley lowland stretches northward between the Pelham Hills on the east and the Berkshire Hills on the west, and protruding above its relatively flat surface are Mount Warner, Mount Sugarloaf and Mount Toby. The Deerfield gorge trenches the western upland just west of Sugarloaf, and on the skyline is Glastenbury far off in Vermont.

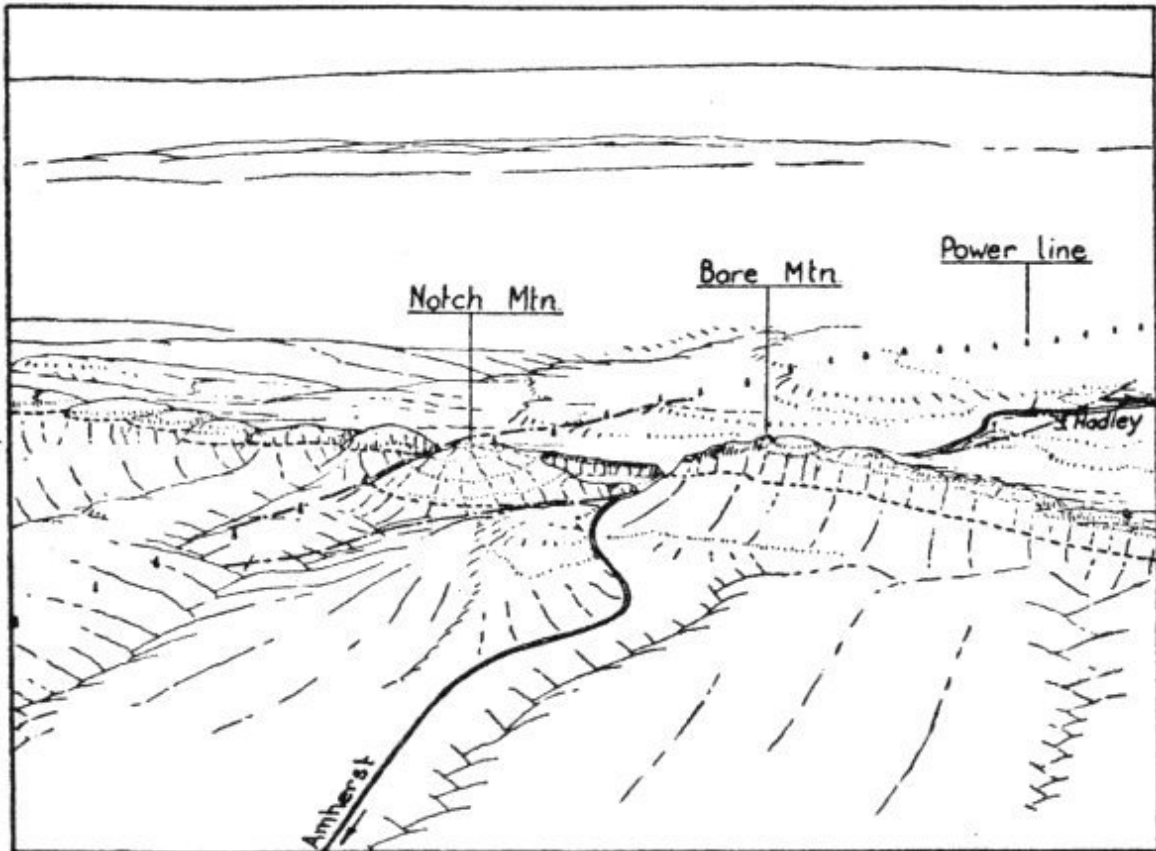
FIG. 24. *Diagrams showing the stages in development of topography in the vicinity of the Notch.*



a. *The New England peneplain stage at the Notch.*



b. The incoherent rocks are removed from the lava flow.



c. The contours of the cliffs are smoothed out.

The downward trail follows the cliff above the highway. The drop is rapid but not precipitous, and soon the western half of the trail is behind.

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The east end of the Range, especially beyond Mount Norwottock, is less commonly visited, but it offers much more of the valley's story. Here the Range is more broken than it is in the western half, and the trail winds through valleys for much of the distance. Long gentle slopes from the west lead to mountain summits, and steep eastern descents take the hiker into the valleys. Plainly the walk is much easier from west to east than in the opposite direction.

The trail leads from the crusher scales around the north base of Notch Mountain and thence up the power line to the crest of the ridge. The path lies on conglomerate below the lava sheet through most of the distance and returns to the lava only where it bends eastward along the crest line. Faulting

east of Notch Mountain has moved the base of the lava southward until it abuts on the sandstone above the lava occurring west of the fracture. Thus, the entire backslope of the Range along the power line is coarse sandstone, whereas in the woods to the east it is vesicular basaltic lava.

Many small but abrupt descents occur along the path as it follows the ridge eastward. Each of them marks the position of a minor fault, along which the eastern side has been pushed down and southward under the western side. However, the elevation of the trail increases gradually to the summit of Mount Norwottock, which is almost as high as the uplands bordering the valley. If one can momentarily overlook the lowland excavated on the incoherent Triassic sandstones, the regional surface seems to slope gently upward to the east, the north and the west. Far to the east Mount Wachusett rises above the general level, and there in the northeast is Monadnock's sharp cone. On the western skyline Mount Greylock's summit, with the fire tower at the north end, attains prominence as Massachusetts's highest peak. The long ridge of Glastenbury and the point of Bald Mountain are clearly visible in the northwest. Far to the south stretches the lowland, and on a clear day Hartford's towers stand sharp and clear against the sky. 76

The north face of the Range is a sheer 250-foot cliff. The south side is a half-mile-long, 20-degree slope. Eastward the crest terminates in a precipitous drop, and the trail winds down the corner between the north face and the cliffs at the east end. It crosses the contact between the lava flow and the red Triassic conglomerate about 150 feet below the summit. The conglomerate beds are separated by shaly sandstones, many of which have weathered out to make rock shelters; these are the so-called "Horse Sheds" and are said to have been used during Shays' Rebellion.

The great cliff at the east end of Norwottock was caused by the rapid erosion of the sandstones below the lava sheet, which has receded steadily westward as it was undermined. Recession started at a fault plane about halfway between Norwottock and Hilliard Knob, for here displacement pushed the lava down and southward on the east side until the subjacent sandstone was exposed west of the fault. Exposure led to erosion and to recession of the lava cap.

The trail passes through the “Horse Sheds” to the south base of the Range, following the contact of the lava with the overlying sandstone for about one-half mile on the way towards Hilliard Knob. This eminence lies over half a mile north of the crest of the Range, for it has been offset by faulting, much like the displacements near Mount Norwottock and Notch Mountain, and the trail passes suddenly from the conglomerate above the lava flow to the conglomerate below it. Trail markers must be observed closely through this section because many wood roads cross the path.

Eastward the way again leads upward to the lava and follows the crest of a low section of the Range, but soon another fault breaks the continuity of the ridge, and the high top of Flat Mountain stands out on the far side of a deep hollow. The hollow is underlain by sandstone below the lava sheet, and the trail follows down the steep dip slope of the beds, only to ascend again towards the reddish basalt cliffs of the mountain. At the base of the flow, the bed of a dry brook exposes a mass of frothy lava. 77

The best views from the top of Flat Mountain are those along the south slope of the Range towards Mount Tom, and northward across the Hadley lowland. The path then turns down the north face of the mountain some 200 yards along the crest from the west end and, passing over a series of conglomerate ledges underlying the lava, it continues along a wood road beside a steep-sided brook until it comes to the Bay Road at the fork to Dwight and Belchertown.

Any nature lover will find the trail very interesting. The views from the western half are unexcelled. Wild flowers and birds abound along the less frequented eastern section. Anyone wishing to see how molten lavas and earth movements in the distant past have influenced the topography of the present will find the far eastern walk a veritable revelation.

Trips from Northampton

Northampton makes an excellent base for many drives that will gratify the lover of scenery, of rocks, or of minerals. The drives range from ten to one hundred miles in length, and any one of them may be extended or shortened at the whim of the driver. To the east the hard surface of Route 9 leads through Amherst, Pelham, and Belchertown; and to the west, as the Berkshire Trail, it rises to the western upland via Williamsburg and Goshen. Side roads go to Ashfield and Conway, and permit a return by way of South Deerfield; or, via Cummington and West Chesterfield, one may come back by way of Huntington and Westhampton. Federal Highway 5 follows the Valley north to Greenfield, whence optional return routes are available through Shelburne Falls, Conway and South Deerfield in the western upland; or through Orange, Pelham and Amherst in the eastern upland. Each of these routes offers arresting views of the broad Connecticut Valley, the picturesque gorges along its margins, and the even-crested highlands with distant peaks of greater elevation. Indeed, the choice of attractive drives is bewildering, even for those who are hesitant about wandering off the surfaced highways.

In the following pages only a few of the possibilities which are available to the motorist are described. And for each one chosen, the striking views and the significant geological features are indicated, in the hope and belief that the traveler may turn explorer and, in following other byways, may reconstruct for himself many additional details of the region's geologic past.

Northampton, Amherst, Pelham

The route leaves from the Court House corner on Highway 5 and the excursion follows Route 9 eastward across the Coolidge Memorial

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^[1] Bridge (1.3), where a panorama of the floodplain with its many channel scars and terrace levels is spread out below (see pp. 1-3). Beyond Hadley (3.0) the road tops the highest river floodplain at a conspicuous

terrace (4.9) and rolls gently over the ancient bed of Lake Hadley. The shore line of this glacial lake appears as a broad flat between Orchard Street and Lincoln Avenue in Amherst (6-9). (See pp. [5-7](#).)

The route turns left at the traffic intersection (7.2) and continues to the north end of the common (7.4), where it turns right on the Pelham road. This road crosses the lake bottom from the Central Vermont Railroad tracks (7.9) to the Orient (9.9), where the delta of glacial Orient Brook made a conspicuous gravel terrace at the farthest limit of the lake. Stone fences make their appearance (see pp. [8-9](#)); rocky ledges and erratics abound at higher elevations, but perched shore lines of ice-margin lakes occur at many levels up Pelham Hill. The road to Mount Lincoln (see pp. [51-52](#)) turns right (11.2) just west of the Amherst reservoir (11.4). As the road approaches the hilltop (12.9), an opening westward through the trees reveals an unusual view of the Holyoke Range; and the broad lowland valley from Mount Tom in the south to Mount Sugarloaf in the north spreads out below. On the hilltop (13.3) the road enters the Daniel Shays Highway (13.9). Mount Wachusett (see [p. 15](#)) lies straight ahead and projects above the great expanse of the New England upland (see [p. 46](#)); Mount Monadnock rises even higher in the northeast, and everywhere, deep valleys furrow the highland and break its otherwise monotonous surface.

The Daniel Shays Highway runs north to Athol, where it joins the Mohawk Trail; but on this trip we shall turn to the right, or south, at Pelham and follow Federal Highway 202 along the valleys in the Quabbin reservoir watershed. Pelham gneiss is the most abundant rock along the highway, outcropping west of the road (14.1) in a series of eastward-dipping 80 layers that resemble sandstone. Mount Lincoln's fire tower stands high above the skyline directly west from the power line crossing (16.1). The country has a gently rolling form, which was imposed upon it by the ice sheet (see [p. 9](#)), and the miles of stone fences represent glacial debris piled up by the early settlers in an effort to bring agricultural order out of geological chaos. Blossoms on the wild cherry trees along these fences and the flowering dogwood make this a particularly attractive drive in the spring. A ledge of gneiss with large eye-shaped crystals of reddish feldspar lies east of the highway at 16.3 miles. As the road begins to descend (18.1), a panorama of the broad lowland between Belchertown and Palmer spreads out below. View succeeds view as the road drops to lower levels: At one

place it is Holyoke and Springfield; at another it is Belchertown; and finally the highway comes to the corners (21.1) where routes lead right to Amherst, left to Worcester, and straight ahead through Belchertown to Palmer, Springfield and Holyoke.

The Granby-Holyoke road (Federal Highway 202) turns right at the south end of the Belchertown common (22.0). After crossing the railroad (22.4), it passes out upon the plain of glacial Lake Springfield (25.3) where the stone fences cease to line the roads, because the lake deposits cover the glacial boulders. Rocky islands in glacial Lake Springfield surmount the flat lacustrine plain (26.9). Granby is situated on a long rolling point (28 to 31.6) that is underlain principally by flat-lying arkosic conglomerate, but more ancient crystalline rocks appear just a little farther east. In this section the lake plain is very narrow, and the drop to the Connecticut River Valley begins at 32.3 miles and continues to the junction with the South Hadley road (33.2), where varved clay (see pp. 4-7) makes its appearance in the pits to the right of the highway.

The itinerary of this excursion continues on Federal Highway 202 through Holyoke in preference to the alternate routes through South Hadley and thence either by way of Hockanum (p. 85) or via Amherst (pp. 83-84) 81 to Northampton. The Holyoke road crosses the Connecticut River (33.8 to 34.1) where the Longmeadow or youngest Triassic sandstone appears in a series of serrate ledges between the bridge and the dam at the right. Mud-cracks on some layers and ripple marks on others tell of wet and dry seasons at the time they were formed. The route turns right just south of the Holyoke post office (34.6) and right again into Federal Highway 5 (36.2), which parallels the river.

The road has been built on a terrace which was once the flat bottom of glacial Lake Springfield (37.8), but at the north end of the city it descends towards the Connecticut River, utilizing the contact between the red layers of Longmeadow sandstone and the massive, dark green Granby tuff with large volcanic bombs that are visible from the road. The twin entrances to Mountain Park (38 and 38.2) may tempt the motorist to indulge in an attractive side trip, but there is enough to occupy him on the main highway. Nearer the river (38.5), a ledge slopes from the roadway to the railroad tracks and to a series of riffles in the stream. This is the Smith's Ferry

footprint locality (pp. [66](#)-67), and the widened highway and the entrance to the ledges offer an invitation which cannot be declined (38.6).

North of the dinosaur tracks, road, railroad and river run parallel. Lateral roads are few, but there is a gateway (40.2) into the Mount Tom Reservation. The Granby tuff, which has outcropped persistently on the west side of the road, rises to a high bluff and then passes eastward beneath the river (40.6). The underlying second lava replaces it in the road cuts and is especially conspicuous along the railroad (40.9). The next dark gray bluff west of the road (41.4 to 41.6) is part of the Holyoke flow which caps the Mount Tom and Mount Holyoke ranges (see pp. [26](#)-27). Soon it, too, crosses the river to Titan's Pier (pp. [60](#)-61), and old residents say that a ledge of it outcropped in the river bed at low water before the Holyoke dam raised the water level. Directly ahead, southward-dipping beds of conglomerate outcrop on either side of the road (41.0); these beds underlie the lava forming the gentle 82 southern slopes of the ranges, and their position beneath the trap can be seen plainly on the steep northern slopes.

The road through the Mount Tom Reservation rejoins the highway (42.2) just south of the outlet from the Oxbow Lake (42.3), the upper end of which also loops 'round and abuts against the highway (42.7). (See [p. 3](#).) Annual floods inundate most of this section, and even the banked-up road and railroad periodically go under the swirling waters of the swollen river. A sign (43.5) announces that the roadway was 13.5 feet below water at the height of the 1936 flood, but it is hoped that the new dike at the southern limit of Northampton will hereafter turn the floods away from the lower sections of the city. Federal Highway 5 bears right (44.0), and the road ahead continues into the Berkshire Trail. Of passing interest is the fact that a well drilled near this junction penetrated 3,700 feet of Triassic arkose without reaching the crystalline rock floor. The road crosses the unused bed of Mill River (44.2) and comes once again to the Court House corner in Northampton (44.6).

Belchertown, Amherst, and Northampton

In our first tour we noted that a road (Route 9) turns right to Amherst at the south end of the Daniel Shays Highway (21.1), and if we will return to this junction, it will be worth our while to make the Amherst run. Just beyond the

intersection the highway traverses the level gravel plain of a nice margin lake (see [p. 7](#)) before it descends (22.0) toward the Lake Hadley plain. Erratic boulders and stone fences are abundant on the slope, and the bedrock is part of the pre-Triassic complex. One very interesting pegmatite contains inclusions of contorted schist (23.2). The road soon leaves the rocky slopes for the gravel plain of Lake Hadley, but only a short distance northward and westward lie the Belchertown Ponds, which seem to occupy a large and deep kettle hole area (see pp. [7-8](#)).

The road winds through pine-clad kame terraces, left on the margin of the ice which filled the Lake Hadley basin; and where it emerges from the woods (24.4), the line of hills making the Holyoke Range may be seen stretching westward in a series of sharp points. These are the projecting edges of the Holyoke lava flow which resisted erosion after all the softer sediment and volcanic debris flanking it were removed. 83

The road to Mount Lincoln turns right at Pansy Park (24.9), and north of this point the Amherst road follows a kame terrace between the Pelham Hills on the right, and the former ice-filled bed of glacial Lake Hadley on the left. Ultimately (27.1) the highway leaves the terrace and drops to a delta which was deposited in Lake Hadley. The view northward shows Mount Toby and Mount Sugarloaf outlined sharply, and to the east near the Orient, the sharp V-shaped notch of the north fork of Fort River cuts one of the kame terraces. The delta deposit (27.5 to 27.7) shows excellent fore-set beds in the gravel pit (27.7), and its entire surface is dotted with ponds which occupy irregular kettle holes (see pp. [7-8](#)).

The highway continues down the delta slope and crosses the Fort River (28.4). This river established a meandering course upon the bed of Lake Hadley, but its floodplain is now excavated below the level of the lake deposits, which form a terrace above the stream. The road passes through Amherst (30.2) and returns to Northampton (37.4) by the outbound route.

South Hadley, Amherst, Northampton

The route 116 north from the road junction at South Hadley Falls (33.2) also has its points of interest. After it passes over the deeply dissected deposits in

Lake Springfield, it rises above the old lake level at the Mount Holyoke campus (35.1) and continues at this higher elevation beyond the Hockanum-Amherst fork (35.9) in the center of South Hadley. Along the right fork (State Highway 116), which leads to Amherst, horizontal Longmeadow sandstone outcrops west of the road (37.1) where the slope to the valley of Bachelor Brook begins. The flat lake plain extends from Moody's 84 Corner (37.4) to the base of the Holyoke Range. A gravel road turns right from the highway (38.2) and crosses the brook one mile east, and from this locality were excavated many of the excellent dinosaur footprints in the Amherst College collection.

The lake plain ends at ledges of Granby tuff and agglomerate (38.5). The outcrops east of the road are grooved with glacial striations, and the fragmental nature of the rock is clearly revealed in the smooth surface. Lava lies on the tuff west of the road (38.7) and also at the bottom of the volcanic series at the Aldrich Lake road (39.0). Coarse conglomerates make recurrent ridges as far as the base of the Range (39.5), where the road follows a shelf cut into the Holyoke lava flow just west of the Notch fault. The conglomerate east of this fault was displaced downward; and as it disintegrates easily, a depression has been cut into the Range east of the road. The quarry situated at the top of the Range (39.8) just north of the Amherst town line, has brought to light many fault fractures that have served the mineral collector well for almost a century. The Range trail (see pp. [73-75](#)) westward leaves the highway at the town line marker, and the path eastward follows the old trolley line northeastward from the scales house.

The route begins its descent (40.2) through a cut in conglomerate, and the entire northern valley is spread out below: Sugarloaf and Toby close the eastern side of the view, and hills far up in Vermont form the background in the northwest. The road quickly reaches the flat plain of Lake Hadley (40.7), with apple orchards stretching along its gravel shore line. The Bay Road crosses the highway (41.1) and parallels the Range from end to end.

The lake deposits fail to conceal many earlier features. Two drumlins (see [p. 9](#)) rise to the east of the road (41.8) near South Amherst. South Amherst (42.7) is on an island in the old lake; erratic boulders cover the hilltop, and bare rocks mark the old wave-washed shore. The highway crosses Fort (or Freshman) River (43.8), and at the railroad tracks (44.6) it rises to the old

lake beach, which is continued in the flat land on the south side of the Amherst College campus. The route turns left at Northampton Road (Route 9) and continues to Northampton (52.2). 85

The Hockanum Road (State Highway 63), which follows the left fork at South Hadley (33.2), crosses the Lake Springfield sand plain (34.1) and rises above the lake level beyond Bachelor Brook (34.3), staying at this higher altitude beyond the junction with the Moody's Corner road (35.3). The hills directly ahead are tuff, agglomerate and lava, and are products of the last volcanic episode in this region. Dry Brook (35.6) flows on the sandstone overlying the Holyoke lava sheet, and the latter outcrops in the road cuts (35.8) and to the left in Titan's Pier (see pp. [60-61](#)). The road to the Mount Holyoke House and Titan's Piazza (see [p. 61](#)) turns right (36.0) where the highway breaks through the last of the lava mass.

The 1936 flood inundated this highway (36.5 to 37.2), and the old watermark may still be identified by debris caught in the bushes and left on pasture land. The view upstream from the floodplain (37.0) shows where the Connecticut is cutting into its eastern bank and causing it to recede (see pp. [1-2](#)). Soon it will penetrate the valley of Fort River. The road passes through a woodland on the dissected lake-shore deposits, but it soon emerges upon the lake bottom and early river silts (38.6). The Bay Road (39.8) enters from the east just south of the bridge over Fort River (39.9). The road joins the outbound route at Hadley (41) and returns to Northampton (44).

Holyoke, Easthampton, Northampton

The return from Holyoke (36.2) by way of Easthampton leaves Federal Highway 5 and rises westward across a ridge of Granby tuff. Several small lakes (36.7) occupy basins on the friable "second" sandstone between the "second" lava and Granby tuff, which lie immediately to the east, and the Holyoke lava, which lies below and to the west. The sandstone is very 86 thin, and the road shortly begins to climb up the dip slope of the Holyoke lava sheet. Sandstone crops out below the sheet at the west base of a low cliff (38.1) which continues northward to the south face of Mount Tom. The Christopher Clark road through the Mount Tom Reservation enters from the north at the summit (38.5); it follows a scenic route under the west

cliffs of the Range to its north end at Mount Nonotuck, where it drops abruptly in a series of hairpin curves to Mount Tom Junction on Federal Highway 5.

At the junction of the Easthampton and Christopher Clark roads, a turn-out offers an opportunity to view the Western Upland, within which, as it makes its way from Goshen and Williamsburg to Northampton, the Mill River has cut an impressive valley. On the long descent to the base of the mountain (39.5), the Easthampton road is cut out of coarse arkosic sandstones, but then it levels off abruptly on the flat plain of glacial Lake Hadley. The lake sediments continue into the center of Easthampton (41.3), broken only by the shallow valley of the Manhan River. From Easthampton the route utilizes the College Highway (State Highway 10) to Northampton; and its position on the lake beds affords good views of the Range and of the abnormally broad floodplain of the meandering Connecticut River in the vicinity of the Oxbow. Just north and east of the New Haven Railroad's underpass, the river has cut away the terrace followed by the road, and this low stretch, like the rest of the floodplain, is subject to frequent inundations.

The road enters Northampton east of the Smith College campus (45.5) and joins the Berkshire Trail. A right turn at the traffic light leads to the Court House corner (45.7).

Northampton, Hadley, Sunderland, Hatfield

This tour also leaves Northampton by the Coolidge Memorial Bridge, but at Hadley (3.0) it turns north on State Highway 63 and follows the river to Sunderland. Here the route recrosses the river, joining Federal 87 Highway 5 at South Deerfield, and from this point south to Northampton the road lies almost literally in the shadow of the western upland.

In Hadley (3.0) the road turns north along Center Street and then swings right at the curve in the Connecticut (3.5). The river bank is lined with riprap to resist the current and to prevent the river from washing away a substantial section of the town. Across the stream in Hatfield the Connecticut very nearly achieved the type of destruction which the residents of Hadley are

trying to escape, and the flood-channel, or “washout,” which was gouged by the swollen stream in the spring of 1936, may be seen (4.3) on the way to North Hadley. The road approaches Mount Warner, whose crystalline rocks appear at the south end of a long, low spur (4.8) on the edge of the river floodplain. Elsewhere along the base of the eminence, which scarcely merits the name “Mount,” the crystalline rocks are hidden by a terrace, but they crop out on the higher slopes. The younger red Triassic sandstones are present, too, and they may be seen dipping steeply westward in the brook bed between the bridge (5.9) and the dam (6.1) at North Hadley.

Sand dunes appear near the river on the outskirts of North Hadley (6.3) and extend north beyond Mount Warner (7.0) to the point where the road drops from the terrace to the floodplain (8.2). Here the former bed of the river is occupied by a puny brook, which enters the mainstream on the left. The terrace marking the edge of the floodplain lies close to the east side of the road for a long distance (8.2 to 9.8) and then swings a half mile eastward. The road follows a high area between two abandoned channels formerly used by the river (11.3), until it joins the Amherst-Sunderland road (11.8) at the southern edge of Sunderland. The route turns left in the center of town (12.4), crossing the river beneath Mount Sugarloaf, and it continues on to Federal Highway 5 at the traffic light in South Deerfield (14.2).

On the trip south from the junction, sand dunes appear east of the railroad between the Boston & Maine (14.6) and the New Haven crossings (14.9). The highway is situated on the flat bed of Lake Hadley from this point to Hatfield. The road to Whately, which turns west at 16.7, offers some attractions. It forks two miles beyond Whately, and the right branch leads to the Northampton reservoir and to Haydenville (see [p. 89](#)). The left branch follows West Fork Brook and comes back to Federal Highway 5 at North Hatfield (19.1). Either route provides a scenic drive over little-frequented gravel roads.

From the main highway the delta built by West Fork Brook into glacial Lake Hadley appears as a flat terrace along the western highland (18.2). The rolling fields (19.7) east of the railroad are dunes which were once raised by the wind along the old Connecticut channel. Mill River, which rises near Conway, parallels the highway for 6.1 miles and crosses it here to enter the Connecticut (20.3). The road approaches the massive gray rocks of the

western upland (20.5), and the Hatfield lead vein (see [p. 64](#)) outcrops in a bluff on the right (20.9). The view south (22.3) shows the water gap between the Mount Holyoke and Mount Tom ranges. At the State Police barracks (23.4) the Hatfield road turns left, and a short distance beyond (23.9), on the west side of the road, is the abandoned City Quarry. The granite exposed in the quarry contains a black, radio-active mineral called allanite, and each glistening black crystal is surrounded by a reddish halo caused by bombardment of the feldspar by alpha particles.

A road to Florence branches right (24.5) near the railroad crossing, and one to the Coolidge Bridge turns left across the Boston & Maine tracks (24.7). A by-pass to the Berkshire Trail (25.6) goes west, and the tour returns to the Court House corner (26.1).

Northampton, Cummington, Plainfield, and South Deerfield

This tour includes a representative section of the Connecticut Lowland, traverses rugged valleys on the western margin of the lowland, and 89 crosses a wide remnant of the New England upland. The trip is 58.6 miles long, and all of it except the last twelve miles moves through rapidly changing scenery.

The route leaves the Court House corner on State Highway 9, the Berkshire Trail, following Main and Elm Streets past Smith College. At the Cooley Dickinson Hospital the road rises from the bed of glacial Lake Hadley to the Mill River delta, which, despite some dissection, maintains the same general level through Florence (2.6) to Look Park (3.4), where a ridge dotted with glacial erratics rises through it. The road follows the delta margin past the Veteran's Hospital and shortly (4.3) climbs to the land of erratics and stone fences. The road from Whately (see [p. 88](#)) enters from the right in Haydenville (7.1), and the Trail continues up Mill River to Williamsburg (8.1). Not far beyond the center of Williamsburg the road forks left for Chesterfield and right for Cummington.

The right hand route climbs a long wooded hill with a deep valley on the right and occasional cliffs of schist intruded by reinforcing granite dikes on the left. The view back near the hilltop (12.5) offers, through a frame of

trees, a panorama of the Mount Holyoke and Mount Tom Ranges surmounting the Connecticut Lowland. The New England upland begins at the hilltop (13.1) in Goshen. Just past Goshen Pond (13.6), a road continues straight ahead to Ashfield, and the hard surface of the Berkshire Trail curves left. Ledges of flaggy Goshen schist outcrop from Goshen to Swift River (18.4); the banding of the ledges is almost horizontal at one place (14.6) and makes excellent flagging for garden walks. The west-flowing Swift River tumbles into the deeply entrenched, east-flowing Westfield River at Swift River village, and the combined streams flow due south through a “door” in a vertical wall of Goshen schist so narrow and inconspicuous that the water appears to run downhill and then up again. The Berkshire Trail follows up the north bank of the Westfield River as far as the lower bridge (19.5), at Cummington, where a road to Chesterfield turns left (19.6).

The Plainfield road branches off to the right across the river at the center of Cummington (20.5), and it climbs almost continuously from the Westfield valley to the summit of the New England upland at the Plainfield corner (24.5). Here the broad, gently rolling expanse of country offers no suggestion of the deep valley only three miles away. 90

The tour takes the road right (Route 116) to Ashfield (33.0), Conway (40.2), and South Deerfield (46.6), where it turns south on Federal Highway 5, returning to Northampton (58.6) over ground that is covered in another tour (see pp. [87](#)-88). The twenty-two miles of country between Plainfield and South Deerfield contain a succession of highland views, glimpses into youthfully incised valleys, and a final sweep of Connecticut lowland that defy description. Nearly everywhere the glaciers of the Ice Age have scraped away the soil and have exposed the underlying metamorphosed sediments. Their high structures and their metamorphism show that they are merely the roots of an ancient range that once rose majestically to summits which, were they restored, would dwarf the planed upland of today. Rugged as some of the topography may seem, prolonged erosion has greatly softened and tamed it. (For more details of the features which can be seen along this route, see pp. [94](#)-95.)

Trips from Greenfield

The most popular drive from Greenfield is westward over the Mohawk Trail, but the eastward continuation of this highway to Orange, combined with the Daniel Shays Highway to Pelham, offers almost equal attractions and should not be missed.

Mohawk Trail, Adams, Plainfield, and South Deerfield

The Mohawk Trail (State Highway 2) heads west from the center of Greenfield and crosses the Green River (0.6) before climbing out of the valley. A lookout (2.1) affords an excellent view of the north end of the Connecticut Lowland, and the observation tower on Shelburne Summit (3.2), situated on a shelf cut out of black Ordovician slate (see pp. [35](#)-37), provides a broader sweep of central New England scenery. Beyond, the upland is gently rolling, trenched by one deep valley at Shelburne Center (6.8). The descent into this valley (5.6) offers a glimpse to the south across the Deerfield River gorge, but the road soon rises again, hovering 300 feet above the sharply incised stream. The Sweetheart Teahouse (9.9) makes use of one of the ideal sites overlooking the gorge and river. The highway to Colrain (10.2) continues straight ahead, but the Trail turns left across the Deerfield River (10.6) and then right in Shelburne Falls. The road left leads to Conway and South Deerfield.

Thick, almost horizontal bands of gray granite gneiss are exposed in the road cuts (11.6) along the south bank of the Deerfield River, but the entrenched stream has left so little room for the highway that the latter soon crosses to the relatively low and more hospitable north bank (11.9). For many miles the road follows the stream so closely that spring floods occasionally cover its surface with ice cakes. The drive along this stretch to the next bridge (21.0) contains the most restful scenery on the trip, though the flat open 92 valley is hemmed in by abrupt slopes which rise for 800 feet. Nor does the flatness of the valley harmonize with the mountain-structure of the platy

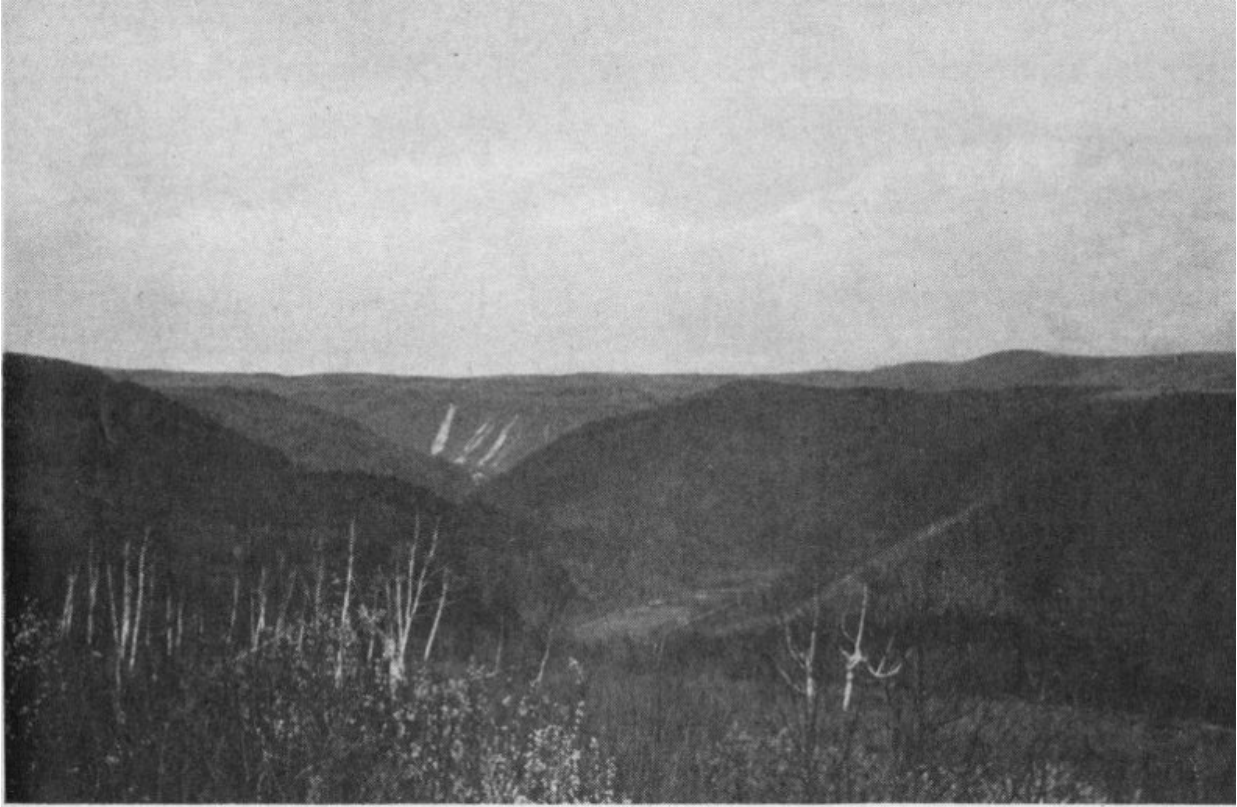
Goshen schist, which stands on edge all along the roadside. Davis Brook (18.7) crosses the route, and the road beside it leads up to the Davis Mine, which once did a thriving business extracting iron pyrites for the manufacture of sulphuric acid.

Once more the highway crosses the Deerfield River (21.0) and enters Mohawk Park, which invites the motorist to linger. A mile farther on (22.0) the road leaves the Deerfield River (22.0) and follows the narrow gorge of Cold River, which seems scarcely wide enough to accommodate it. A shady picnic ground and auto camp (23.6) lie just below the narrowest and deepest part of the gorge (24.5), where the crowding summits seem to tower high above the puny cars.

The road crosses to the north bank of Cold River (25.8) and climbs a shelf cut into green volcanic schists (25.8 to 27.6). Leaving the gorge (26.6), it ascends to the upland (29.0), while in view below is the laborious route of the Boston and Maine Railroad along the Deerfield and thence through the east portal of the Hoosac Tunnel near Zoar.

A lookout (29.4) affords a memorable view of the sharp V-shaped gorge of the Deerfield River cut deep into the highland surface, which stretches unbroken to the horizon, with only a few divides rising to greater elevations in the west and northeast. A set of broad rock benches, about 200 feet lower than the upland, forms a strath terrace (see pp. [46-47](#)) which closely follows the river's course. Great landslide scars, caused by the heavy rains accompanying the 1938 hurricane, mar the valley walls far to the north and again eastward from Zoar.

The road to Zoar (30.2) turns right a short distance east of the Whitcomb Summit (30.6), where lookout towers at an elevation of 2,240 feet enhance the excellence of the view westward across other straths to Mount Greylock, the highest point in Massachusetts. From these same vantage points, one may survey the deceptively smooth slope of the New England upland eastward down the course of the Deerfield towards the Atlantic coast.



PL. 10. View of the Deerfield gorge from the east summit of the Mohawk Trail.

The high level flat to the extreme right and extreme left is the New England peneplain. The terrace bordering the steep walls of the gorge is a strath.

The road crosses a series of strath heads, which drain into the Cold River, and ascends to the west summit (33.2). At first it seems possible to throw a stone into North Adams, so abrupt is the western slope. The city lies deep in a limestone valley, and beyond it the Taconic ranges rise steeply west of Williamstown, over six miles away. 93

On the long descent into the valley, the roadway is cut into albite-biotite schists with horizontal cleavage. Above and below the sharp hairpin turn (35.2), there is a beautiful view to the south along the strike of the limestone trench and along the route (State Highway 8) which is to be followed from North Adams (37.8) to Adams. Not far south of North Adams the road passes the west portal of the Hoosac Tunnel (39.4). Boulders on the mountainside

east of the highway are glacial erratics which were left above the level of the valley trains and above the surface of glacial Lake Bascom. The limestone which outcrops on the slopes of Mount Greylock west of the road is used for lime (41.6), and the quarries provide ideal exposures for a study of the rock. The burning plant (42.2) is at the roadside. The road branches in the center of Adams (43.6), one route (State Highway 116) continuing ahead to Savoy and Plainfield, the other veering right to Mount Greylock, Dalton, and Pittsfield.

The Savoy road follows a broad valley eastward into the hills. A perceptible steepening of the slope occurs where it crosses from the dolomitic limestone below, to the albite-biotite schist above, at a thrust fault (47.5). Hard white Cheshire quartzite (48.2) and arenaceous limestone (49.0 to 49.6) overlie the schist and outcrop by the roadside, and in places the arenaceous limestone has weathered to a white glistening sand.

The road soon drops into a wide and open valley (50.1) which seems to slope interminably southeastward; this is the head of the Westfield drainage, and it has occupied this position in the Westfield system far back in geologic time (see [p. 14](#)). The little village of Savoy (51.8) nestles near the eastern edge of the valley, and once beyond the settlement, the highway tops a divide (52.7) comprised of rolling hills. It skirts Plainfield Pond (56.0 to 56.4) and then comes out upon a panorama of the upland which embraces the entire Westfield basin (57.5). This section is underlain by the Savoy schist, which is characterized by its many large red garnets. At the hilltop (59.7) in Plainfield the road forks, the route to the right descending to the Berkshire Trail and the road ahead proceeding to Ashfield. This portion of the New England upland lies so far back from the main streams that the small tributaries have not yet cut deeply into its gently rolling surface, and no hint of hidden valleys can be detected in the peaceful landscape.

The Ashfield road traverses woodland country that is almost flat. The stream valleys are broad and are rarely more than a hundred feet below divides. Even Swift River (63.9), which crosses the road about two miles above the end of its entrenched gorge, has not deepened its valley, despite the long span of years since New England was raised to its present elevation (see [p. 47](#)). Past the alternate road to Cummington (64.0), the route continues across the flat country above Ashfield; but where the road to Goshen turns south

(66.4), deep dissection of the New England upland begins. An opening in the trees (66.6 to 67.1) discloses the valleys along the South River in the vicinity of South Ashfield, as well as the level skyline in the highlands east of the Connecticut Valley. The road drops into the South River valley at Ashfield (68.2), where a choice of routes to Greenfield is presented. The road that follows the South River to Conway (75.4) is the more interesting.

The river is joined by a tributary from the north at South Ashfield (69.9), and the streams occupy deep but open valleys. Kame terraces flank the 95 rivers and are a source of gravel for road ballast. The old dam (73.9) near Conway is a picturesque spot, and the deep, shady pool below is not neglected by anglers. Glacial erratics (see pp. [8-9](#)) dot the hill slopes, but ledges are rare and consist of the locally named Conway schist where the rock does appear at the surface. The road branches again at Conway: the left fork goes north along the dissected brink of the Deerfield gorge to Shelburne Falls, but our choice falls upon the eastward route to South Deerfield.

The highway climbs through a cut (75.8) in contorted, gneissic Conway schist, which seems to be lined with twisted white quartz veins from this point to the margin of the Connecticut Valley. The road levels off just before it reaches the New England upland, and then it drops through rolling hill country to the shaded valley of Mill Brook (77.4), which it follows to the edge of the lowland (80.1). Rocky ledges are common along this swiftly flowing stream. A good view of the Pocumtuck Hills appears on the left (79.7), with the flat plain of the old Deerfield delta stretching to their base. The road crosses this plain and enters South Deerfield (81.8).

The tour turns north on Federal Highway 5, which is built on the deposits spread in glacial Lake Hadley by the Deerfield River from its mouth eastward to the foot of the Pocumtuck Hills of Triassic conglomerate. Bloody Brook (82.2) drains this part of the plain. North of the road which goes through the notch between the Sugarloafs (82.6), the delta deposits continue as a terrace along the base of the Pocumtuck Hills as far as Cheapside. But the Deerfield has excavated its post-glacial delta, and the roadway descends to the meander-cut floodplain (84.4 to 88.1), though it rises over one of the meander scarps (85.1). Remnants of the Deerfield delta form a terrace due west across the valley and on the margin of the hills. The entire lowland north of the meander-cut terrace was inundated in 1936, and

the water level may still be identified by debris on the railroad embankment on the right. Old Deerfield (86.1) itself is on a meander-scarp terrace, 96 and the 1936 flood line is well marked along it. After the road crosses the Deerfield River (88.3), it leaves the floodplain as it climbs to the center of Greenfield (89.8).

Greenfield, Orange, Pelham, Amherst, and Deerfield

Route 2 also leads eastward from Greenfield to the French King Bridge and Millers Falls. The highway out from Greenfield turns north (0.3) along the west front of the trap ridge, near the summit of which several individual lava flows are represented by separate sets of columns superimposed one upon the other (see [p. 26](#)). The underlying bedded sandstones outcrop in the lower wooded slopes. A road branches right (1.7) to Turners Falls and crosses the lava ridge, but the main highway continues straight to a sharp curve near Falls River (3.4). Pillow-shaped masses of lava characterize the bottom of the lava flow and lie above conglomerate in the bluff to the right. The valley of Falls River is a fault zone slicing across the lava sheet, which reappears at the lookout-parking place at Turners Falls (3.5). The extent to which the waterfall has receded (see pp. [58-59](#)) may be judged from the length of the gorge.

The route continues left past the bridge entrance (3.6). Ripple-marked red shales—once Triassic muds in which stray dinosaurs left their tracks—outcrop by the roadside (4.6 to 5.7), and coarse conglomerate beds (5.9) overlie the shales and dip steeply towards the river. Somewhat farther east a broad sand plain (6.0 to 6.8) of glacial outwash (see [p. 59](#)), which ends at the French King gorge, buries the Triassic bedrock, but once again conglomerate appears and forms the west wall of the gorge. Pre-Triassic crystalline rocks (6.9) likewise outcrop on the western cliffs, and form a narrow ridge between the present course of the river and the pre-glacial channel, which lies below the glacial delta (7.0 to 7.9) of Millers River.

The road to Northfield turns left (7.9) and another (8.7) leads right to Millers Falls, but Route 2 continues east, climbing high above the river (9.9), which flows through a narrow gorge. Gneiss with horizontal banding outcrops 97 (11.2) in mesa-like hills north of the highway, which descends to a

point (12.5) that was 5.5 feet under water during the 1936 flood. The road continues near the water's edge for almost half a mile, and the narrow gorge through gneiss ends at Erving (13.8). Here the valley widens out into a hilly lowland which has been developed on schist with occasional bands of gneiss. The road follows the north bank of the river across this lowland to Orange (17.9). Route 2 continues to Athol where the Daniel Shays highway enters from the south, but an alternate route, which turns right in the center of Orange and crosses Millers River (18.0), provides a preferable short-cut to the Daniel Shays Highway (21.8). This section of road is lined with stone fences which memorialize the combined labors of the great Ice Sheet and the early settlers.

Route 32 from Petersham and Worcester enters from the left (22.7) just before the highway dips into the creek bottom at the edge of the Quabbin basin. Thence it ascends to the New England upland level, where a lookout (25.5) affords an expansive view to the east and north, with Mount Monadnock rising prominently on the distant skyline. New Salem (25.8) is on the hilltop. Hornblende schist outcrops at intervals across the broad ridge, and especially near the descent (28.4) southwestward to another stream (30.3) which empties into the Quabbin Reservoir. Once again the highway climbs rather steadily for three and one-half miles, passing the Shutesbury road (31.0) on the right, until it reaches another lookout (34.6) from which the trenched New England upland spreads out to the east. Pelham gneiss is the main rock on the broad ridge west of the Quabbin basin, especially in the vicinity of Pelham (35.2), which gave the rock its name.

The tour turns east to Amherst (41.7), following a section which has been described elsewhere (see pp. [78-79](#)). The principal sights include the panorama of the Connecticut Lowland and the ice-margin lake deposits. The drive from Amherst to Northampton (see pp. [78-79](#)) and from 98 Northampton to South Deerfield (see pp. [87-88](#)) on Federal Highway 5 has likewise been covered in other tours, but some new features may be seen along the shorter route from Amherst to Sunderland.

The Sunderland road turns right at the north end of the Amherst common. It descends, first, from Amherst Island, in glacial Lake Hadley, to the old beach at Massachusetts State College (42.6), and then from the beach to the lake bottom (43.2) north of the campus. The route takes the left fork in North

Amherst (44.2), traverses part of the old lake bed, swings west around the Long Plain delta (45.5), and crosses its entrenched brook (46.1). Most of the stream's water seeps through the delta sands and gravels, and emerges in springs at the Fish Hatchery (46.3). Gravel pits across the road furnish an excellent section of the fore-set and top-set beds of the delta. The road right (46.8) goes to the delta top east of Mount Toby, Montague and Turners Falls, but the main highway continues north.

The road turns left and then right (47.2), cutting through a beach bar in glacial Lake Hadley, and passing a sand dune area (47.6) which developed from the sandy braids in the channel of the Connecticut when it first established its course on the lake bed (see pp. [4-6](#)). The route drops down from a terrace (47.9) to the highest floodplain level of the Connecticut. Swales (48.3 and 48.5) on this flat represent former river channels, and the scalloped embankment to the east records the lateral swing and undercutting of the meandering river. The North Hadley road (48.7) enters from the south along a low ridge between two swales, and after the sharp right turn into Sunderland (49.2), the road divides, one fork going north to Montague, the other west across the Sunderland Bridge to South Deerfield.

The Sunderland Bridge (49.4 to 49.6) offers a good view downstream along the natural levees (see [p. 2](#)) and westward to the cliffs of Mount Sugarloaf (see pp. [56-58](#)). The road rises above the floodplain (49.9) and passes 99 the Sugarloaf trail (50.0) on the right. A right turn into Federal Highway 5 at South Deerfield (51.0) brings the motorist back to a section of country already described in connection with the Mohawk Trail tour (see pp. [95-96](#)), and another eight miles of driving brings him to Greenfield (59.1).

Greenfield, Turners Falls, Montague, North Amherst

A variant of the drive east from Greenfield is available in the route that turns right across the Turners Falls Bridge (3.6 to 3.8) and follows the east side of the Connecticut Valley southward. The road turns left in the center of Turners Falls (4.2) and climbs the embankment which the river excavated in the old lake beds. On the sand plain above (4.9), the left fork goes to Millers Falls and the right, to Montague. The Montague road skirts the west side of a low line of hills which terminate at a depression (8.6) on the pre-glacial

course of the Connecticut. The road goes over Saw Mill River (9.4), in the bed of which Triassic conglomerate is exposed. Conglomerate also appears in the hills directly south, but the older crystalline rocks crop out in an exhumed ridge to the southwest and in the highlands eastward. The conglomerates form the south end of a Triassic basin extending from Mount Hermon and Northfield farther up the valley (see [p. 26](#)). Beyond Montague (9.7) Triassic conglomerate appears along the roadside (10.1) as far as the forks to Sunderland and Millers Falls (10.8).

The Millers Falls road follows the foot of a terrace which rises to the old delta level, and at the next fork (11.0), the route keeps right and continues southward to North Amherst. The delta of the glacial stream buried many ice cakes which left numerous kettle holes (11.0 to 11.5) when they melted. The stratification of the deposits is displayed in the many road cuts. The route crosses the Central Vermont Railroad (11.5) and follows an old outwash plain southward past the road to Roaring Brook (13.1) (see [p. 54](#)). The tour continues through a narrow stretch in which crystalline rocks 100 predominate, as far as the Long Plain delta (15.6). Mount Toby rises steeply on the west side of the railroad. A third of the way up the mountainside can be seen (13.9) a conspicuous bench, which consists of an exhumed remnant of the ancient, sloping granite mountain front on which the Triassic sediments were laid (see pp. [20-21](#)). The bench level drops northward to the railroad at Roaring Brook, and southward it crosses the road (14.9). The conglomerate east of the road (14.9 to 15.3) fills an old mountain valley. A road east (15.3) goes to Leverett, and the lead vein is located just south of it at the hilltop.

The route skirts the margin of the crystalline rocks and crosses the railroad again (15.6). Just beyond the road to Leverett station (16.1) the motorist may exercise the option of returning to the Sunderland road (17.3) by going right across the Long Plain delta and thence to Greenfield (29.6) via South Deerfield (see [p. 95](#)). Or he may extend his trip by taking the left fork of the Mount Toby road, which follows the boundary between the Long Plain delta and the glaciated eastern highlands. Boulders and bare ledges feature in the landscape to the east, whereas the flat delta and the level beach margin (17.9) lie to the west. Beyond the limits of the delta lies a series of bare ledges of gneiss. After crossing Factory Hollow Brook (19.1), the route joins the Sunderland road (19.2) at the center of North Amherst, returning to

Greenfield (34.1) by way of Sunderland and South Deerfield, as before (see [p. 98](#)).

Greenfield, Turners Falls, Montague, Sunderland

The Sunderland road (10.8) just beyond Montague turns right and climbs the terrace along the floodplain of Saw Mill River. The plain is the delta which this stream built into Lake Hadley. A few rock ridges project above it; buried ice has melted to form kettle holes (11.4) (see pp. [7-8](#)); and post-glacial streams have cut valleys in it; yet it preserves its deltaic form to the old lake margin (11.6). Low shed-like cliffs occur east of the road (11.9); the overhanging rock is Toby conglomerate, and the excavated shelter is a 101 gray shale which was laid in a Triassic lake bed (see pp. [22](#) and [68](#)). These cliffs recede from the highway and end (12.3) at the Sunderland Caves (see [p. 55](#)). The route continues downhill and joins the river road on the floodplain of the Connecticut (14).

The road rises over a promontory (14.1) formed by the resistant Deerfield lava sheet (see [p. 26](#)) and then descends to the river floodplain and meander-cut terraces (see [p. 22](#)), which cross to the east side of the highway and continue south beyond Sunderland (15.5). In Sunderland junction is made with the longer tour through Amherst (see [p. 98](#)), and the return to Greenfield may be made by that route (25.4).

Trips from Springfield

In the vicinity of Springfield the most interesting drives are to be found on the west side of the Connecticut River, for the comparatively flat land east of the river is thickly settled and heavily industrialized, and geological phenomena are effectively masked. The country to the west offers a display of features which may be traced to the activities of the river, to the former presence of glacial Lake Springfield, to the prolonged erosion of the Triassic bedrock, and to the resistance of the pre-Triassic rocks in the western highland. Almost any trip will include this entire suite of geological phenomena. The distances which are given in the following tours have been taken from the west side of the North End Bridge.

Springfield, Holyoke, Easthampton, and Westfield

This route follows north on the floodplain along the river bank via Federal Highway 5, and the heavy retaining wall is designed to keep the river out. The sand promontory which comes to the left side of the road (0.6) is a remnant of the old lake—bottom deposits. Past the junction with the road through West Springfield (0.8), the highway utilizes the old lake beds, which form a terrace above the present floodplain; but ultimately (1.6) it drops again to the floodplain level, with its many abandoned channels, although it discreetly stays on one of the higher, and older, meander-cut terraces. A cut-off to Chicopee turns right (2.4), but Federal Highway 5 continues north across the meander terraces to the forks (4.6) which lead to the residential (left) and business (right) sections of Holyoke.

Here the main highway climbs steeply from the floodplain to the top of the lake deposits. The view north shows the Holyoke Range rising above the roofs and chimneys of Holyoke. Lake deposits form a broad flat between the Triassic volcanic ridges which lie to the west and the trench cut by the Connecticut River.

The road from Westfield and the airport (Federal Highway 202) from the left, and the road to Easthampton (7.6) turns left from the main highway, which continues north. The north route, described under the tours from Northampton (see pp. [80](#)-82), features the dinosaur tracks and the succession of Triassic rocks. The Easthampton road goes past the south end of the Mount Tom Range, and its scenic attractions have been dealt with elsewhere (see [p. 85](#)). Its most interesting sights are the line of lakes between the two volcanic series and the view from the summit of the ridge. Easthampton (12.7) lies on the old lake bottom, from which an impressive view of the palisade of massive columns comprising the tilted lava flow of Mount Tom can be obtained.

At Easthampton the tour turns south on the College Highway (State 10) towards Southampton and Westfield. The road follows the gravel plain which was spread into Lake Hadley by streams flowing out of the western highland. The plain is dissected locally by the Manhan River (15.2 and 18.0) which crosses the road twice, and one small valley near Southampton (17.1) discloses Triassic arkose buried by the sand. The road rises above the lake deposits (17.4) near the Manhan River, and at once glacial erratics become numerous. The “land of stone fences” forms a narrow divide between the Lake Hadley basin and the Lake Springfield sand plain (21.1), which extends south to the valley of the Westfield River. The Holyoke road (22.5), which enters from the left, came over the trap ridge and across the lake plain.

As the College Highway approaches the edge of the Westfield valley (23.6), it slopes steeply down to the level floor cut by the river. It crosses the Westfield (24.5) and comes to the junction (24.9) with the Jacob’s Ladder route (Federal Highway 20), which offers another interesting sidetrip into the Western Upland.

The route continues south to the center of Westfield (25.2), leaving the College Highway at the south end of the common. The Springfield road goes around the central square and starts east in the valley which the meandering Westfield River carved out of the Lake Springfield sediments. The terrace levels and the scalloped pattern of the meander scarps are conspicuous along the lowland. The highway crosses the Little Westfield River (26.1) and then the Westfield itself (27.0) just beyond the entrance to Robinson State Park.

Most mineral collectors will instantly recognize a road turning off to the left (27.8) as the way to the Westfield trap quarry. For years this locality has been as important a source of specimens to collectors as it has been of crushed rock to road-builders. Beyond the quarry road the valley narrows, and the terraces close in as the river enters the gap in the trap ridge. The black lava flow crosses the river (28.3) at the Westfield-West Springfield town line, and shortly the upper flow appears, resting on red shales in both railroad and road cuts (29.1). Actually there are two flows separated by an amygdaloidal band in the upper lava series at this place. The highway crosses the Boston and Albany tracks (29.3) and leaves the river. After passing the junction with the Holyoke road (31.0), the highway drops to the upper terrace level on the bed of glacial Lake Springfield (31.4). The upper terrace is narrow here, and the road soon descends to the meander-cut terraces of the floodplain (32.1). The road to Memorial Bridge turns right (32.3) and our route returns to the North End Bridge (32.8).

Westfield to the Westfield Marble Quarry

This is a short drive of 5.7 miles each way from Westfield, with a mile walk from the Little Westfield road to the marble quarry. The view of the Little Westfield gorge and the entire Connecticut Lowland from Meriden to Amherst makes this trip well worth taking.

The tour leaves Westfield on the Jacob's Ladder road and soon reaches the terraced margin (1.6) of the Westfield valley. The numerous benches 105 along the stream banks represent temporary flood-plain levels of the Westfield. The route turns left from the Jacob's Ladder highway (4.0) and parallels the base of the western highland to the Little Westfield road (4.9). Throughout this distance the marble quarry derrick appears on the highland skyline. Our road turns right at the next crossing and winds along the edge of the Little Westfield gorge (see pp. [61](#)-62). The narrow hill road to the marble quarry turns right (5.7), but it is inadvisable to drive. The walk is an easy one, and the view at the top is worth the moderate physical exertion.

OPTIONAL TRIPS

It must be plain, even to the casual reader, that the foregoing pages have been written with self-restraint. Many of the luring side roads were passed without so much as a pause; trips to the Cobble Mountain Reservoir west of Westfield, and to the Quabbin Reservoir east of Belchertown have not even been suggested; some of the main highways were slighted. For anyone who knows the byways and the hidden beauties that can be found in reasonably accessible places, this chapter will seem inadequate and incomplete.

But it would take a volume far beyond the scope of this brief guide to do justice to the scenery, the geography, and the geologic detail of the Connecticut Valley and its bordering uplands. The authors can merely ask the indulgence of those who would like to know more.

Mineral and Rock Collections

Travelers are inveterate collectors of mementos, and those who travel up and down and across the Connecticut Valley and who delve into its geologic history may well be interested in gathering records of its past. The best records are not in notes or printed pamphlets—not even in this volume on the subject; they are to be found imprinted in the rocks and minerals themselves. But the value of records is measured solely by their utility, and utility is achieved by systematic arrangement. So the authors will venture a few suggestions on collecting and arranging the minerals and rocks which are present in the valley and in the bordering uplands.

One mineral may come from a vein, which is the record of a fissure beneath a hot spring; another comes from a dike, which was a molten igneous rock. This specimen is a conglomerate or consolidated gravel washed into place by an ancient stream; that is a slate which was transformed from clay by intense squeezing and shearing. And if these four specimens were to constitute the nucleus of a collection, the need for classification is apparent. The first two are minerals, which are substances of limited chemical composition and well defined physical properties. The last two are rocks, which are aggregates of minerals or of mineral grains. And the minerals may be further classified according to their separate modes of origin. So, too, with the rocks. Their mineral composition indicates some of the conditions which existed where the minerals originated; the shapes of the mineral grains reveal the process which moved them to their present site; and the arrangement of grains discloses the conditions existing during aggregation at this new locality. Mineral make-up, size, shape, and arrangement of the grains provide means of recognizing major rock varieties—namely, sedimentary, igneous and metamorphic types,—and also of reading each rock's history.

The vein minerals, which are deposited in conduits for hot spring water, commonly possess attractive crystal forms; they include barite, quartz and amethyst, fluorite, calcite, datolite, galena, sphalerite, pyrite and others almost too numerous to list. Almost equally attractive crystals may be obtained from some metamorphic rocks, in which they have formed as heat and pressure abetted the growth of certain minerals at the expense of their less favored fellows; this group contains garnet, kyanite, chlorite, amphibole, epidote and many others. Less spectacular are the minerals resulting from the decay of rocks by percolating surface water, such as kaolin, limonite, some calcite, and the bright-colored copper carbonates. Two additional types of minerals are formed as the result of normal sedimentary and igneous processes, which will be described at length in connection with these two kinds of rocks. So, after the rock specimens are sorted from the minerals, the latter may profitably be arranged into five groups:

1. THE VEIN MINERALS.
2. THE MINERALS OF PEGMATITES AND IGNEOUS ROCKS.
3. THE MINERALS OF METAMORPHIC ROCKS.
4. THE MINERALS OF SOILS AND ROCK DECAY.
5. THE MINERALS OF SEDIMENTARY ROCKS.

The Vein Minerals

The mineral list which follows is far from complete; it contains only those minerals which are most commonly found in casual visits to the localities discussed in connection with the local tours of the Connecticut Valley. Additional species are listed and described in any textbook on mineralogy.

QUARTZ is a hard, white or colorless mineral which will scratch glass easily. In the technical language of the crystallographer, crystals are hexagonal or six-sided prisms, terminated by hexagonal pyramids; and the six flat faces which make the sides, together with the six triangular faces which form the apex, are readily recognized. Massive forms break with a curved or conchoidal fracture and were used by the Indians to make arrow-heads. The mineral is very abundant in all the lead veins and trap quarries; and in some of the latter, specimens of the purple variety of quartz, amethyst, are common. A black, smoky variety has been

discovered in the pegmatite dikes of the highlands. Chemically quartz is the dioxide of silicon (SiO_2).

CALCITE breaks along three smooth surfaces or cleavage planes. Each surface is rhomb-shaped, and the six rhombic faces fit together into a characteristic rhombohedral form. A knife will scratch the mineral easily. Calcite is abundant in the white veins of the trap quarries and is the principal constituent of the crystalline limestones in the Hoosac Valley between North Adams and Pittsfield. Calcite is a carbonate of lime (CaCO_3).

BARITE resembles calcite because it can be scratched with a knife and has three smooth cleavage planes. It differs in having one cleavage perpendicular to the other two, which intersect at angles of 78° . The mineral is more than four times the weight of an equal volume of water, and it feels heavy. It is found in the lead veins at West Farms, Hatfield and Leverett. In large quantities it has commercial value as a source of the element barium, for it is the sulphate of barium (BaSO_4).

GALENA is the chief metallic mineral in the veins at Leverett, Hatfield and Loudville. It is very heavy and has a metallic gray color; it breaks into perfect cubes. A knife scratches it easily and crumbles it to a black powder. The mineral is a lead sulphide (PbS).

SPHALERITE is a lustrous, resinous brown mineral in these same veins. It cleaves into multi-faced fragments and is softer than a knife. Chemically it is the sulphide of zinc (ZnS).

PYRITE is the deceptive golden-colored, metal-like mineral which has earned the name of "fool's gold." It will scratch glass, and it crushes to a black powder. The materials in it are iron and sulphur (FeS_2).

CHALCOPYRITE resembles pyrite but will not scratch glass and has a greenish yellow color. It is a compound of copper, iron and sulphur (CuFeS_2).

The veins in the Connecticut Valley region contain many other minerals, among which must be mentioned datolite, natrolite, apophyllite, thomsonite,

fluorite and babbingtonite in the lavas; and siderite, rhodochrosite, rhodonite, wulfenite and pyromorphite in the older veins of the highlands.

Minerals of Pegmatites and Igneous Rocks

The minerals found in pegmatites are legion. More than thirty can be collected on any trip to Collins Hill near Portland, Connecticut, or to the Ruggles Mine near Grafton Center, New Hampshire. Only the minerals appearing most commonly in pegmatites are described, but a list of others is appended as an aid in consulting a textbook. Igneous rocks contain practically the same suite of minerals as pegmatites, but in smaller grains.

MICROCLINE is a white to flesh-colored feldspar with two almost perpendicular cleavages. It will scratch glass or a knife. One cleavage face shows a grid of translucent and transparent lines intersecting at 90° .

ALBITE is the second most abundant feldspar. It is white and may generally be recognized by its two cleavage surfaces at 86° . Its growth may be likened to piling a series of plates with their surfaces parallel to one of the cleavages; during growth the plates are laid alternately face up and face down, so that the 86° cleavage edges zigzag in and out, forming a surface which, on the average, is perpendicular to the growth cleavage surface. The separate plates can usually be detected as fine bands or striations. The mineral scratches either glass or a knife.

MUSCOVITE is the white mica found in tabular crystals that can be cleaved into flexible and elastic sheets. It can be scratched and cut easily with a knife or shears.

BIOTITE is an amber-colored to black mica. Like muscovite it is flexible and elastic, but it is slightly more brittle.

TOURMALINE crystals occur in triangular prisms with the corners bevelled so as to give them a rounded appearance. They lack cleavage, are very brittle, and will scratch glass. Black is their usual color, but red and green varieties are present in many pegmatites.

SPODUMENE crystals are white to pale rose in color, and they occur as flattened prisms with bevelled corners. They cleave parallel to the surfaces bevelling the corners. The mineral is much harder than a knife, and the cleavage surfaces have a lustrous, slightly satiny appearance.

RADIOACTIVE MINERALS occur in many pegmatites and metamorphic rocks of this region. The species which have been formed as a result of recent alteration are brilliant golden or green encrustations in cracks or on a pitchy-black nucleus. The most abundant ones are uranite, autunite and torbernite. Older primary or source minerals are pitchy-black and are surrounded by a narrow rusty red zone or “halo,” $\frac{1}{16}$ to $\frac{1}{8}$ inch wide; an elongate species resembling a rusty hand-made nail is allanite; the more pitchy, irregular-shaped mineral is usually uraninite or pitchblende. 110

Other minerals found in pegmatites in the Connecticut Valley region include beryl, apatite, zircon, garnet, fluorite and lepidolite.

Most of the minerals in normal igneous rocks are too minute to be recognized easily, but a few have distinctive characteristics which serve to identify them. QUARTZ is a hard, dark, glassy-looking mineral without cleavage. ORTHOCLASE and MICROCLINE feldspar are hard, flesh-colored (occasionally white) minerals with flat cleavage surfaces. The minerals making the white lathlike mosaic on the weathered surface of the Range at the Mount Holyoke House are LABRADORITE feldspar. They are about $\frac{1}{4}$ inch long and $\frac{1}{50}$ inch thick—too small to permit testing by ordinary physical methods, although unweathered pieces have essentially the same physical properties as orthoclase and microcline. The MICAS are flaky and reflect light like minute pieces of tinfoil; muscovite is white, and biotite is amber-colored to black. CHLORITE resembles mica but is less lustrous and is dark green.

Some minerals of igneous rocks do not appear in pegmatites. Among them is OLIVINE, which has almost the same color as chlorite but is harder than a knife and is massive or granular. It is commonly associated with massive green SERPENTINE, which is softer than a knife. These three minerals are especially abundant in rocks found in the vicinity of Blandford, Massachusetts, and Dover and Chester, Vermont.

AUGITE is a dark brown to black pyroxene which occurs between the mosaic of whitish labradorite feldspar prisms in the weathered diabase near the Mount Holyoke House.

AMPHIBOLE crystals are dark green to black, “match-shaped” crystals. They have almost the same hardness as a knife and are characterized by two cleavages parallel to their length and intersecting at 56° . The mineral is also abundant in metamorphic rocks and is frequently reported as a “fossil fern” from ledges at Charlemont and Shelburne Falls.

Minerals of Metamorphic Rocks

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The principal minerals of metamorphic rocks include many which are likewise present in pegmatites and igneous rocks, such as microcline, albite, quartz, muscovite, biotite, amphibole, serpentine and tourmaline. But there are others which are more exclusively metamorphic:

GARNET occurs in twelve- or twenty-four-sided red crystals. It is much harder than a knife. The geometric form is diagnostic, and crystals up to an inch thick are obtainable in Plainfield, Massachusetts, and at Grafton, Chester, and Gassetts in Vermont. They occur in a muscovite schist, in which the muscovite flakes are wrapped around the individual crystals.

TALC is a white to pale-green mineral found around the margins of intrusive rocks that are rich in olivine and serpentine. It is foliated and is so soft that even solid masses will rub off on cloth. It is present in the green marble quarry near Westfield.

KYANITE is a sky-blue, bladed mineral, with two excellent cleavages at nearly 90° , and a good smooth fracture at almost 90° to both. One face is harder than a knife and the other two are softer. It is very abundant in the country rock southeast of the Westfield marble quarry.

The Minerals of Soils and Rock Decay

Aluminous minerals decay to KAOLINITE, and those with a high iron content alter to LIMONITE. Both these products of decomposition form a

sticky paste in their original forms. Kaolinite is white to yellow, and limonite or ochre is yellow to orange. Limonite also appears in orange-colored or brown balls, in icicle-like masses, and in thin beds. Specimens have approximately the hardness of a knife. Quartz does not decay easily and remains behind in solid granules.

The Minerals of Sedimentary Rocks

Most sedimentary rocks are formed by the cementation of deposits of transported waste, derived from older materials. They may contain 112 anything. The minerals which undergo rapid decay break down to limonite, kaolinite and quartz, leaving only the more resistant varieties, which include, in order of decreasing resistance, quartz, microcline, orthoclase, albite and muscovite. Less abundant constituents are garnet, tourmaline, zircon and magnetite.

Certain kinds of sedimentary rocks may be formed through other agencies—for example, limestone, which is composed of calcite, initially precipitated by lime-secreting organisms or by the evaporation of lime-charged waters. The effects of organic activity may be seen in the limestone near Bernardston, but most of the calcite now present in the rocks of western Massachusetts is of vein or metamorphic derivation. Salt (halite) and gypsum are formed by the evaporation of saline waters, but only the vacated casts of salt crystals have been detected in the Triassic sediments of the valley.

THE ROCKS

Rocks record three distinct methods which nature employs in the aggregation of minerals. The sedimentary rocks register the work of wind, water and ice. Deposits left by wind and water are generally stratified or bedded, and they, together with glacial deposits, are composed of fragments which touch one another and are cemented at the points of contact. Igneous rocks record the solidification of hot liquids which injected themselves into older rocks or filled crevices, and which, upon cooling, formed masses of closely fitting crystals. The third group includes types which are crystalline

like the igneous rocks, and which may be laminated somewhat like the sediments; they show effects of heating and squeezing until their original forms and even their minerals were changed. These are the metamorphic rocks.

Anyone who wants an orderly record of geologic history will arrange his rocks into these three groups—the sedimentary, the igneous, and the metamorphic. In the Connecticut Valley the metamorphic rocks reveal the ancient phases of earth history, and the sediments contain the details of younger or later geological episodes. The igneous rocks have a wider historical range; and, like the other types, they record a long period of violence and upheaval which seems out of harmony with the placid countryside for which they now provide a solid foundation.

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The Sedimentary Rocks

The sedimentary rocks are built from the disintegrated wreckage of older ones. The products of rock decay are picked up and dragged, or carried in suspension or solution, by wind, running water, or moving ice. They are deposited when and where the transporting agent can no longer function. Such rocks are usually layered because the transporting power of the carrying agent fluctuates. Bands of one kind of material, separated by dissimilar materials above and below, are called beds.

The bedded or stratified rocks of the Connecticut Valley vary greatly, from the coarse bouldery deposits in Mount Toby to the fine-textured, red and black laminated beds at Whittemore's Ferry. Conglomerate, arkose, graywacke, shale and even limestone are represented, but there is little true sandstone. Sandstone is an even-textured, granular rock, most commonly composed of cemented quartz grains. Its uniformity of grain-size and composition reflects prolonged weathering of the original rock and good sorting of the fragments as they were transported to their new resting place. The sequence of exposure, transportation and deposition was too rapid in the ancient Connecticut Valley to permit appreciable decay and sorting; hence sandstones are absent. Limestones and salt beds are likewise rare, but the metamorphosed limestones which are found in the western highlands and in

the Berkshire valley demonstrate that limestone-forming processes played a significant, if intermittent, part in the history of the region.

CONGLOMERATE is consolidated gravel. Pebbles and boulders of all sizes are packed together by the stream which was moving them, and the spaces between the larger fragments are filled with the sand that settled in from the stream bed. The entire mass is cemented by silica, limonite, carbonates or some other substance deposited by percolating ground-water. The Devil's football near the Mount Holyoke House is a famous piece which was dislodged from the hillside above; and excellent specimens may be collected on Mount Toby, on Mount Sugarloaf, and in the cut at Mount Tom Junction. 114

ARKOSE resembles conglomerate, but the individual grains consist of mineral fragments, among which reddish feldspar is prominent. Quartz and mica may be present, too; and all the pieces are characteristically angular, commonly ranging from $\frac{1}{16}$ to $\frac{1}{8}$ inch in size. The rock is red and crumbles easily. Beds of arkose alternate with conglomerate on the steep sides of Mount Sugarloaf.

GRAYWACKE is light to dark gray in color, and the fragments composing it are sand size pieces of older rocks. A few mineral grains, such as quartz, may be present, but mica is rare. Graywacke occurs interbedded with arkose in some parts of the valley.

SHALE is a thinly laminated sediment composed of microscopic quartz, feldspar, mica and kaolinite grains. Most shales in the Connecticut valley were deposited as muds in old lake beds. Some are red and record ephemeral pools, but others show from their black color, their coal layers, and their fish skeletons, that the water bodies in which they accumulated remained in existence for a comparatively long time.

LIMESTONE is a rock composed of calcium carbonate, and it consists essentially of an aggregate of calcite crystals or calcite fragments. It will give off gas bubbles in a very dilute solution of hydrochloric acid, and it exhibits other properties peculiar to the mineral calcite. A thin, sandy limestone bed has been identified in several sections of Holyoke.

The Igneous Rocks

Igneous rocks were once molten, and in this hot fluid state some were extruded at the surface as lava flows. Congealed flows reveal the motion, which brought them to their present resting places, in the banding and streaks that are so evident in the patterns of steam holes and minerals; but their massive structure bears witness to stagnation as they hardened. Other molten masses insinuated themselves into underground openings, where they solidified as intrusives, varying in size from small dikes less than an 115 inch wide, to huge masses that can be measured in miles in any direction. Most of the igneous rocks in the highlands of western and central Massachusetts are massive intrusive types; light-colored varieties predominate, but some dark-colored dikes cut the older rocks both east and west of the valley. Dark-colored, massive and banded lavas are conspicuous in the ranges within the valley.

Igneous rocks may be divided into three general groups for practical classification, and each major group may be further subdivided. Rather conveniently each of the major groups may be recognized by the prevalent color of its rocks—whether dark, medium-colored, or light. And within each major classification there may be flows, characterized by banded structures and fine textures; small intrusives composed of well formed crystals in a fine-grained groundmass; and large intrusives consisting of good sized, equigranular crystals. Not all of these types can be found in central Massachusetts, but the variety of igneous rocks is surprising and offers some excellent possibilities for the collector.

The Dark Rocks

The dark rocks owe their color to iron-bearing minerals like olivine, pyroxene (augite), amphibole and biotite. All of these minerals weather to a rusty red surface, which is typical of their outcrops at many places.

BASALT is a black rock, dense in some places but perforated with bubble holes or vesicles, at others. It occurs throughout the length of the Holyoke, Tom and Pocumtuck Ranges; and fragments of basalt are abundant in the Granby tuff and agglomerate.

DIABASE resembles basalt but is distinguished by the thin, short crystals embedded in it. These crystals of labradorite feldspar resemble pieces of clipped thread, and they sparkle in reflected light. Almost all dark-colored dikes and the slowly cooled central portions of thick lava flows consist of diabase.

PERIDOTITE is a dark green, coarse, granular rock composed of olivine and subordinate amounts of pyroxene. It occurs near Westfield and Blandford, and at many places in Vermont. 116

The Medium-Colored Rocks

The medium-colored rocks contain approximately the same proportions of light- and dark-colored minerals. The dark iron-bearing minerals are relatively stable, but the light-gray feldspars decompose to kaolin and give the weathered rock a chalky white surface. Surface flows of this group are unknown in central Massachusetts, but the coarsely granular intrusives are well represented.

GRANODIORITE PORPHYRY is a greenish-gray rock occurring in many dikes in the western highlands. It has rectangular crystals of andesine feldspar up to $\frac{1}{8}$ inch across, and these have a dull porcellaneous luster. A few dark-green amphibole crystals are only slightly smaller. Both feldspars and amphiboles are embedded in a very fine-textured, pale greenish groundmass.

GRANODIORITE is a gray equigranular rock containing flesh-colored microcline feldspar, white andesine feldspar, greenish flakes of chlorite, needles of amphibole and sparse grains of brown biotite. All crystals are about $\frac{1}{32}$ inch thick and commonly display a parallel arrangement. This rock forms huge irregular masses at Williamsburg, Whately and Belchertown.

The Light-Colored Rocks

The light-colored rocks are well represented by dikes and large masses but not by recognizable surface flows in central Massachusetts. Their exposures

have rarely weathered much, because the predominant minerals are quartz, microcline, orthoclase and albite, which resist decay.

QUARTZ PORPHYRY is a light gray rock that is found in dikes. It has porcelain-white cleavable feldspars up to $\frac{1}{8}$ inch thick, and dark glassy quartz of equal size in a granular mass of very fine-grained crystals. Intrusives of this type are numerous in the vicinity of Whately.

GRAY ALBITE GRANITE occurs in many dikes and small irregular masses throughout the highlands. All crystals have approximately 117 the same size and rarely exceed $\frac{1}{32}$ inch in thickness. They consist of white orthoclase and albite, dark sugary quartz, and brown to black biotite.

RED MICROCLINE GRANITE is found in very large, irregular intrusives in the highlands. The crystals are $\frac{1}{16}$ inch or more in thickness. The red color is due to the flesh-colored microcline. Quartz is dark and glassy, and muscovite is the typical mica.

The Metamorphic Rocks

Metamorphic rocks were once sedimentary or igneous rocks which have been changed by intense pressure, by heat, or by solutions moving through them. Pressure usually produces a sheeted or foliated structure along which the rock exhibits a tendency to part—somewhat like the pages in a book that was bound before the ink was dry. Percolating solutions may produce chemical alterations in the original materials and even crystallize new substances along the foliated surfaces within the rock, much as water circulating through cooled soil may solidify to ice and cause heaving. Many of the rocks in the highlands bordering the Connecticut Valley are highly foliated or banded in consequence of the mechanical deformation they suffered when the ancient upland mountain system was created. They include the slates, schists and gneisses. A few massive types, like marble, serpentine and soapstone, owe their origins chiefly to the effects of heat or of the hot, chemically charged solutions which permeated them.

SLATES are fine-grained rocks characterized by flat, parallel cleavage surfaces which usually cross the original sedimentary structure. They were

formed from shales, by shearing and compression during ancient mountain-making movements. Slates crop out beside the station platform at Brattleboro, Vermont, and at many places southward along Federal Highway 5 to Greenfield.

SCHIST is foliated, too, but it is composed largely of cleavable minerals, such as chlorite, muscovite, biotite and amphibole, which are distributed along the cleavage surfaces. These minerals result from the chemical activity of hot solutions circulating along a slaty cleavage, re-crystallizing old materials, and bringing in new to make these coarse mineral flakes. The schist receives its specific name (biotite schist, chlorite schist, etc.) from the mineral which accentuates its cleavage structure. 118

A few schists contain large crystals which bulge the schistose surfaces outward around them. Garnet is characteristic in this role, and a muscovite schist with garnets in it is called a GARNETIFEROUS (or garnet-bearing) MUSCOVITE SCHIST. Other minerals with occurrences similar to the garnet are microcline, albite, staurolite, amphibole, tourmaline, pyrite and magnetite.

GNEISS is a banded rock containing cleavable minerals, but it lacks the cleavage structure of schist. The cleavable minerals (biotite, muscovite, amphibole, etc.) may give the gneiss its specific name, but as often as not, the name is derived from the whole mineral assemblage, or from an assumed origin, as in the case of granite gneiss. As in the igneous rocks, the mineral ensemble is held together by interlocked quartz and feldspar grains. Black-banded biotite gneiss and hornblende gneiss are the most abundant varieties in the neighborhood of the metropolitan reservoir east of Pelham.

MARBLE is a granular rock composed of calcite crystals. It is formed when heat volatilizes the bituminous coloring agents of ordinary limestone and simultaneously causes enlargement of the calcite grains. It is the principal rock in the Berkshire Valley in which North Adams, Adams and Pittsfield are located.

OPHICALCITE is a lime-silicate rock. It is formed by the chemical reactions of hot solutions on limestone or marble at considerable depth

within the earth. The original calcite is converted into diopside, garnet, vesuvianite and tremolite, forming a rock that may be massive, or which may preserve some of the original bedded structure. It is found in association with the crystalline limestone and magnetite at the old iron mine, located one mile north of Bernardston.

SERPENTINE is a dark-green rock made almost exclusively of the mineral serpentine. It results from the reaction of hot solutions on olivine and pyroxene rocks (peridotites). Serpentinite is present in the Westfield marble quarry and at Zoar on the south side of the Deerfield River.

SOAPSTONE is composed principally of talc. It, too, results from the chemical activity of hot solutions ascending through serpentine and causing the mineral transformation. Bodies of this material are associated with the serpentinite at Westfield and Zoar, and northward in 119 sections of Vermont. It is mined for talc, but in colonial days it found many uses. The colonists used cross-cut saws to make blocks for foot warmers in their sleighs, to control the heat in the old wood-fired ovens and to make water pipes before iron and lead were available in adequate quantities. Many soapstone articles may be seen—and purchased—in Wiggins Country Store and in other good antique shops through the valley. One of the most primitive Indian cultures in this region utilized soapstone pots, and exhibits are on display at both the Springfield Museum of Natural History and the Amherst College Museum.

Conclusion

To anyone who has had the patience to read through the preceding pages and to reach these concluding remarks, it must be obvious that geology is not merely a pastime for specialists. It does not take half a dozen college and university degrees to collect rocks and minerals, and to understand what they mean; or to appreciate not alone the beauty, but also the long and involved, yet logical, origin of scenery; or to comprehend from a rock-cut or cliff the vast changes which have occurred in the course of geologic time; or to grasp the current significance, as well as the historical importance, of such rock and mineral products as the trap, the limestone, the pyrite, the lead veins, the soapstone, the varved clay, the gravel banks.

Whether one's interests are practical, historical, acquisitive, esthetic, philosophical or scientific, the geological features of the Connecticut Valley possess the variety to gratify them all. One must indeed be blind if he cannot find something of interest—a hobby—even a profession in the geological display spread before him in central Massachusetts. Let it not be thought that this little volume tells the whole story. On the contrary, its authors expect to have a difficult time justifying their sins of omission, more particularly because many of the omissions have been conscious and deliberate. But they trust they have left for the reader a wealth of features which he can make his own by right of discovery. For it will not take him very long to penetrate the fourth dimension of geologic time more deeply and intimately than is possible in the pages of a book.

Footnotes

- ^[1] The figures denote the distance in miles from the starting point to the feature mentioned.

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